

THEORY

THE UNIVERSE'S SYSTEMATIC ERROR

Version 1.2

THE UNIVERSE'S SYSTEMATIC ERROR - From Dissipative Structures to the Pole of Existence. Mitin, O. (2026). 10.5281/zenodo.20533839

Abstract

The emergence of consciousness is traditionally viewed either as a consequence of the unique nature of humans or as an improbable coincidence in an indifferent universe. Both positions assume that consciousness is an anomaly requiring a special explanation beyond established physical principles. This paper proposes a third way: consciousness is a natural consequence of physical processes in open systems with a sufficient energy gradient - not an anomaly, but one of the manifestations of the general logic of complexity.

The central framework of the work is the identification of a structurally self-similar pattern of «gradient → retention → pole → transition,» operating as a self-similar mechanism at all scales, from the formation of a dissipative structure to a conscious system. Five fields traditionally developed independently - Prigogine's non-equilibrium thermodynamics, Haken and Kauffman's theory of complex systems, Friston's active inference, Deacon's biosemiotics and teledynamics, and Tononi's theory of integrated information - turn out, with a certain shift in emphasis, to be projections of a single structurally self-similar process onto different scales. The six-step ladder - dissipative structure, autopoiesis, proto-agency, consciousness, and the recursively complexifying conscious system - is not merely one object of the theory among others, but its basic pattern, through which all the processes it encompasses are described. Each step is defined by the type of operations on the system's states; each transition is defined by a minimal condition based on established formalisms. Each subsequent step inherits all the gradients of the previous ones and adds a new one of its own type; simultaneously, increasingly stable mechanisms of resistance to disintegration accumulate. The structure of transitions is probabilistic: the conditions for each transition are necessary but not sufficient in a deterministic sense, and the realization of each transition is stochastic.

The author's theses are integrated into the synthesis. The central ones are: consciousness is an active mechanism of complication, closing and accelerating the cycle through model recursion filtered by the environment; phenomenal experience is ontologically identical to the dynamics of the system's self-modeling and is operationalized through the mechanism of the presence of inherited gradients via the existence variable V - which dissolves the hard problem under a stated physicalist premise and answers the meta-problem structurally (Section 6); the existence variable V, formally uniform for all systems of stage 4, varies in content across system types and calibrates the temporality of the system through this variability; the structural self-similarity of the ladder is traced across scales from the dissipative structure to culture (stages 0–5); the same pattern may also manifest at the cosmological scale, but this is a separate hypothesis moved by the author into a self-contained work (Section 7), not derived from the theory of consciousness and not needed by it.

In version 1.1, the formal apparatus of the theory is supplemented by the environment variable Y, which is explicitly traceable at all levels of the ladder. Y is defined as the structure of the physical world that enables the existence of systems as local maxima of causal closure and their causal exchange with what lies beyond their boundaries; Y is objectively the same for all systems, and the spectrum of its aspects accessible to a particular system is a property of its architecture. The introduction of Y resolves the methodological asymmetry of the previous version (the lack of an explicit link between the four-step pattern «gradient → retention → pole → transition» and a cross-cutting variable), provides the pattern with a formal foundation, and implements the principle of substratum independence as the cross-cutting methodological axis of the work. Y has physical limits (Bekenstein, Landauer, speed of light), which are its own properties.

In version 1.1.2, the work has been supplemented with four substantive extensions.

The first is a functional justification of the binary structure of variable V as a condition for its operation in the mode of maximum available retention gradient (Section 3.4); from this property, the disproportion of the conscious system's response to a threat to retention relative to the objective magnitude of the threat is derived, and a correspondence is established between the formal-structural conclusion of the theory and independent phenomenological descriptions of the structure of existence (Section 11.1).

Second - the environment-switching test as a parallel falsification protocol for the coupling of six dimensions (Section 10.2.2); the test utilizes a two-phase structure and registers the presence of a functioning V through the reorientation of the system's activity toward the preservation of content unrelated to adaptation to the current environment.

Third - a substantive prediction about V for information-causal systems (Section 10.3): the principle of substrate independence is specified in terms of the content of V as a pole between the coherence and the breakdown of the system's information-causal structure, with three classes of physically realizable threats.

In version 1.2, the work is deepened by three structural revisions, concentrically linked to one another.

The first is the introduction of a structural mechanism for the emergence of the variable V through the self-overlay of a latent invariant of the agentic causal node (Section 3.4). Unlike previous versions, where V was described through the coupling of three operational conditions (Φ , extended active inference, GWT-accessibility), V now receives a structural definition through a specific mechanism. The latent invariant of an agentic causal node emerges in system models as a class of configurations with boundary states of retention/disintegration accumulates; the self-overlay of this invariant onto the system's own configuration constitutes the structural event of V's emergence. The binary nature of V is induced by the retention contour over the continuous physics of Y through the boundary states of causal nodes. Φ , GWT, and extended active inference are reoriented as three operational axes for recording this structural event, not as its definition.

Second - the reformulation of the transition $4 \rightarrow 5$ as a bidirectional circuit $M \leftrightarrow Y$ with physically retained traces (Section 3.5). The criterion of stage 5 is the restructuring of the V-model of the subsequent configuration, not merely the channelling of its behaviour. This structural distinction defines the boundary of stage 5 relative to the preforms of the circuit (stigmergy in the broad sense). The cross-cutting terminology "trace layer V" is introduced in place of "symbolic layer V" as a substrate-neutral term: symbolic communication is a later, complex form of the same loop, not the foundation of stage 5.

Third, the cosmological section of earlier versions has been moved to a separate work, planned by the author as an independent project. What remains in the present theory is only the indication that the structural pattern «gradient $\rightarrow$ retention $\rightarrow$ pole $\rightarrow$ transition» may also manifest at the cosmological scale (Section 7); this is a separate hypothesis, neither derivable from the theory of consciousness nor necessary for it.

Added to the three central revisions are a structural diagnosis of information-causal systems as an artifactual state (Section 10.3) with three structural conditions for a possible exit from it (proper Y, proper class of agentic causal nodes, bidirectional $M \leftrightarrow Y$ loop with proper traces); consistent alignment of terminology throughout the work.

A detailed elaboration of the cosmological aspect is the direction of a separate work, planned by the author as an independent project, not part of the present theory.

The work formulates testable predictions at the level of consciousness and behavior and clearly defines the limits of its applicability.

The verification program for the central prediction (Section 10.2) is structured as a two-sided approach - a detection grid (a six-dimensional coupling of operational features of the $3 \rightarrow 4$ transition) and a design grid (a minimal specification of environmental requirements, without which the transition does not occur regardless of the system's internal architecture). The two-sided

structure is based on a unified methodology for identifying Y and the access spectrum for a system of arbitrary substrate, developed in subsection 3.8.

Keywords: consciousness, non-equilibrium thermodynamics, dissipative structures, active inference, autopoiesis, teledynamics, integrated information, the anthropic principle, emergent time, structural self-similarity, environment variable Y, access spectrum, causal closure, substrate-independence, the Bekenstein and Landauer physical limits, detection grid, projection grid, environment-switching test, information-causal systems, artifact state, binary nature of pole V, four-element architecture, consolidation of successful configurations, methodological agentlessness, difference in rates, complexity as multiplicity of realization paths, self-overlay of a latent invariant, agentic causal node, bidirectional $M \leftrightarrow Y$ loop, trace layer V.

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1. Introduction

We've grown accustomed to thinking of humanity as a fluke. A tiny planet on the edge of a galaxy, adrift in vast, cold space - and on it, for reasons unknown, something as incredible as consciousness emerged. Most attempts to explain this fact choose one of two options: either we are a unique design, or we are a happy accident in an indifferent universe. Both positions, despite their apparent opposition, stem from a common assumption: consciousness is an anomaly in the physical world, requiring a special explanation beyond established physical principles.

I propose a third way. We are neither a design nor a coincidence. We are a systematic error. The result of a process that has no goal but does have a logic. And to understand this logic, one must begin not with life or consciousness, but with entropy.

The path that led me to this work was not a straight one. Over many years of studying physics and astronomy, I repeatedly came across points where accepted models began to contradict themselves. Not the dramatic paradoxes found in popular texts, but quiet inconsistencies: dark energy, which does not behave quite like a cosmological constant; black holes, in which information apparently does not disappear, but no one can say exactly how it escapes; accelerating expansion, which does not appear entirely uniform within the horizon of observable data; the anthropic principle, without which we cannot explain fine-tuning, but with which we lose clarity regarding what physics actually does - whether it describes reality or our position within it.

The accumulation of such inconsistencies has gradually led to the suspicion that the physical description of the world developed over the past hundred years has a boundary that does not lie where it is conventionally drawn. Not between quantum mechanics and general relativity, nor between classical and quantum physics. But between physics and those phenomena traditionally assigned to philosophy: life, complexity, observation, consciousness. I gradually began to think that this boundary is artificial - that physics should not end where phenomenology begins, and that several lines of research, which have developed independently of one another over the past decades, coalesce into a single picture when certain emphases are adjusted.

This work presents the result of this synthesis. It does not propose a fundamentally new theory. It proposes a synthesis of several fields - non-equilibrium thermodynamics, the theory of complex systems, active inference, biosemiotics, the theory of integrated information, each of which was developed independently and evolved within its own boundaries. The thesis of this work is that, with a certain emphasis, these fields coalesce into a unified picture in which the emergence of consciousness turns out not to be an anomaly requiring a special explanation, but a natural consequence of physical processes.

Within this picture, I propose two central original theses without which the synthesis remains incomplete. First: consciousness is not a passive product of complexity, but an active mechanism for its acceleration. Once it has arisen, consciousness becomes a mechanism for the further

complexity of the system of which it is a part, through the modeling of the environment and its transformation. The cycle closes and accelerates. Second: phenomenal experience is ontologically identical to the dynamics of the system's self-modeling - it does not arise on top of it, nor does it accompany it, but is the very same dynamic viewed from within. This thesis resolves the difficult problem of consciousness not as a pseudo-problem, in the spirit of functionalist reduction, but as an incorrectly posed problem.

Beyond these two central theses, the synthesis introduces a number of its own structural constructions without which the ladder would lack its operational apparatus: the mechanism of V's emergence through the self-overlay of the latent invariant of an agentic causal node (Section 3.4); the distinction between self-representation and state-estimation as the boundary of stage 4 (Sections 3.4, 6); the through-going environment variable Y and the access spectrum to it as the formal axis of substrate independence (Sections 1.1, 3.8); the structural diagnosis of information-causal systems as an artifactual state (Section 10.3); and the prohibited predicate as the central falsifiable consequence of the identity of structure and phenomenon (Section 10.2). Everything else in this work is a reconstruction and synthesis of the existing knowledge of the five supporting fields. I emphasize this to establish the register from the outset. I do not claim to be bringing about a revolution in physics or in the philosophy of consciousness. I claim that the already established fields, when considered together, tell a single, unified story - and that this story deserves to be presented in its entirety, without the gaps that separate the individual disciplines.

I begin with thermodynamics and the theory of complex systems (Section 2), showing that complexity is not an anomaly but an expected trend in open systems with an energy gradient. Then (Section 3) I unfold a six-step ladder - dissipative structure, autopoiesis, proto-agency, consciousness, and a recursively complexifying conscious system - each of which is formally defined by the type of operations performed on the system's own states. Section 4 presents macro-models of each of the four transitions between steps - conceptual, physical, and computational. Section 5 briefly discusses the nature of time as an emergent quantity - not as a separate topic, but as a necessary clarification of the theory's ontological background. Section 6 develops an alternative ontology of consciousness and proposes an approach to this difficult problem through the thesis of the identity of self-modeling and the phenomenon. Section 7 shows the invariance of the structural pattern across scales and notes a possible cosmological continuation as a separate hypothesis not part of the testable core. Section 8 is devoted to the figure of the observer within observation and brings together the anthropic principle, the measurement problem in quantum mechanics, and the central thesis of the theory of consciousness. Section 9 brings the whole picture together. Sections 10 and 11 formulate testable predictions and discuss the implications of the theory, including its extension to non-biological systems. Section 12 honestly lists the limitations of the work and directions for further research. Section 13 is the conclusion.

Two methodological axes run through the entire work, which are worth noting in advance. The first is formal: the ladder is presented not as a narrative analogy, but as a structure with operational definitions of the rungs, minimal conditions for transitions, and empirical support in the form of macro-models. The second is probabilistic: transitions between steps are not deterministic but

occur stochastically when the necessary conditions are met, which transforms the ladder into an architecture of possibilities rather than a mandatory trajectory of complexity. These two axes work together: the formal side specifies exactly what must happen at each step; the probabilistic side explains why most systems stop at lower steps and why our position at the narrowest end of the funnel is simultaneously random (in the global picture) and inevitable (for us as observers).

I acknowledge that in each of the five core areas there will be specialists who will see simplifications and omissions in my reconstruction. This is the inevitable price of synthesis. I have tried not to distort the content of the core fields; where I use them beyond their own boundaries, this is clearly indicated. Where a statement remains hypothetical or is under active development, this is also indicated. The boundary between the established and the conjectural is drawn clearly wherever possible.

1.1 Theses of the Work

The «gradient → retention → pole → transition» structure functions as a self-similar pattern at all scales of the theory - from the formation of a dissipative structure to a conscious system. The six-step ladder is not merely one of the theory's objects, but its basic pattern, through which all the processes it encompasses are described. The five directions of synthesis turn out to be projections of a single structurally self-similar process onto different scales.

All four elements of the four-step pattern operate relative to the underlying environment variable Y - the structure of the physical world that enables the existence of systems as local maxima of causal closure and their causal exchange with what lies beyond their boundaries. Y is objectively the same for all systems; the spectrum of its aspects accessible to a particular system is a property of its architecture, not of Y. At each step of the ladder, the system gains the architectural capacity for causal exchange with a new aspect of Y, which has always objectively existed but was inaccessible to it prior to that step. The principle of substrate independence: all formal definitions of steps, transitions, and operational conditions are formulated functionally, without reference to a specific substrate; biological examples are illustrations of a special case. Y has physical limits (Bekenstein, Landauer, speed of light) that define the boundaries of possible modes of existence for systems at the upper stages and function as a formal condition for approaching the hypothetical transition 5→6. The methodology for identifying Y and the access spectrum for a system of arbitrary substrate is detailed in subsection 3.8.

Each step of the ladder inherits all the gradients of the previous ones and adds a new one. The type of the new gradient shifts from spatial through structural and temporal to ontological, sequentially expanding the scale of the gradient's localization - from the environment to an individual system and further beyond its boundaries into the population and its history. In parallel, mechanisms of resistance to disintegration accumulate: hysteresis, causal closure, a multi-tempo model, preventive planning, and distribution via a bidirectional $M \leftrightarrow Y$ loop with physically retained traces.

Consciousness is an active mechanism of complication, not its byproduct. Activity as a capacity first appears at stage 3 through a temporal gradient, acquires directionality at stage 4 through the ontological pole V, and at stage 5 becomes distributed - coordinated activity of a multitude of carriers through a bidirectional $M \leftrightarrow Y$ circuit with physically retained traces. Having arisen, consciousness closes and accelerates the cycle of complexity, transforming the environment of which it is a part.

Consciousness is V. V is a model of the system in environment Y, which the system models within itself and around which its entire mechanism of existence is built as the “retain/disintegrate” pole. The structural mechanism of V’s emergence is the self-overlay of the latent invariant of the agentic causal node onto the system’s own configuration: in a stage-3 system, a class of agentic causal nodes accumulate in the models, and when the system’s own configuration enters a state space structurally isomorphic to the boundary states of this class, the system begins to be processed by the model as an instance of that same class; this self-overlay is the structural event of the emergence of V. The binary nature of V is induced by the retention contour over the continuous physics of Y through the boundary states of causal nodes - it is not added to V from the outside, but is transferred into the system through the process of self-overlay. V is formally uniform across all stage 4 systems, but varies in content depending on the type of system: what is considered a collapse is determined by what the system retains. Through this variability, V calibrates the system’s temporality - it determines which changes are significant, what becomes its past and its future. At stage 4, inherited gradients are present in V for the first time and become experienced - phenomenal content is the way in which the system experiences threats to its gradients through the pole V.

Phenomenal experience is ontologically identical to the dynamics of the system’s self-modeling, carried out through the structural pole V - that is, self-modeling in which the latent invariant of the agentic causal node has superimposed itself onto the system’s own configuration (the formal structural definition of stage 4). The phenomenon does not arise on top of such dynamics nor does it accompany them - it is the very same dynamics viewed from within. Self-modeling without pole V (as in the stage 3 proto-agent) lacks phenomenality: its structurally different type of modeling must be confirmed by the environment through action and lacks autonomous retention of the modeling process. Self-modeling with a V-pole is sustained independently of current confirmation by the environment and therefore yields an autonomous modeling structure, identical to phenomenal experience. The identity of structure and phenomenon is operationalized through the mechanism of the presence of inherited gradients in V (Thesis 5). The difficult problem is resolved as an incorrectly posed task - not as an illusion requiring debunking, but as a structural position realized from within modeling systems with self-overlay fulfilled.

The anthropic principle, the measurement problem in quantum mechanics, and the theory of consciousness are three projections of a single ontological structure: observation is a positional characteristic of a system within a larger whole, not a privileged external act. This structure is a special case of a general structurally self-similar pattern (Thesis 1) - its realization on the scale at which the system describes the larger whole of which it is a part.

Human beings are neither the goal of the universe nor an accident, but rather its inherent anomaly: an extreme local manifestation of the general process of increasing complexity, inevitable where conditions have arisen, and random within the global ensemble.

1.2. Glossary of Key Terms

This work uses a number of specialized terms, some of which are borrowed, and some of which are original. Many commonly used words (“gradient,” “retention,” “pole,” “step”) are used in this work in a specific structural sense that differs from their everyday meaning. The glossary is provided here, following the general introduction, so that readers opening the work at any chapter have a terminological guide at their disposal. The section where the term is formally introduced is indicated in italics at the end of each entry. The grouping is thematic; alphabetical order is not maintained within blocks, and entries are arranged according to the logic of conceptual interdependence.

I. Structural Pattern

Y (environmental variable). The structure of the physical world that enables the existence of systems as local maxima of causal closure and their interaction with what lies beyond their boundaries. It manifests in three interrelated aspects: the causal-metric structure of spacetime; the configuration of fundamental fields; and the information-thermodynamic structure, which sets the limits on the density and rate of exchanges. Objectively the same for all systems - a structural property of the physical world as such, independent of both the observer and any specific system. It is neither an object nor a relation «the environment of a given system»: both definitions are incorrect (the first creates a circular dependency, the second fragments Y into a multitude of different environments). The operational content of Y relative to the system: Y is a causal generator existing outside the system’s algorithm and forcing it to recalculate weights via prediction errors that cannot be explained by the internal dynamics of its model; through these errors, the structure of the environment’s causal constraints physically enters the system’s operation as a condition of its stability. Y has physical limits (Bekenstein, Landauer, speed of light), which are its own properties and function as a formal condition for the upper rungs of the ladder. Foundations in contemporary philosophy of physics: causal set theory (Sorkin), relational quantum mechanics (Rovelli). Section 3 (preamble); operational content for information-causal systems - Section 10.3.

Access spectrum to Y (spectrum of aspects of Y). A set of types of structural relations with Y with which a system of a given architecture has the capacity for causal exchange. A property of the system, not Y. Each step of the ladder is characterized by which aspect of Y is added to the system’s access spectrum: at step 0, it is the thermodynamic gradient through the system; at step 1, a local modification of the thermodynamic distribution; at step 2, selective exchange of matter and energy; at step 3 - a predictable structure of the environment; at step 4 - the position of the system itself as part of the environment via S; at step 5 - the space of physically retained traces of models from previous carriers in Y. The expansion of the access spectrum is emergent: the system acquires the architectural ability to interact with those aspects of Y that have objectively always existed, but

were previously inaccessible to it. For systems of a different architecture (artificial, synthetic hybrid, hypothetical extraterrestrial), the access spectrum differs already from the early stages. Section 3 (preamble); Section 3.1.

Gradient. A state of disequilibrium in a system is a stable distinction between two possibilities through which the system is structured and develops. This distinction may be spatial (between two points in the environment), but need not be: it may be between two alternative states of a single system (structured / homogeneous), between two architectures (closed / open), between two rates (internal dynamics / environment), between two ontological possibilities of the system as a whole (holding together / falling apart), between two historical states of a supra-individual structure. It is not reducible to a thermodynamic gradient; the thermodynamic gradient is one type. In all cases, the gradient operates relative to the traversing environment variable Y: it manifests in different forms depending on which aspect of Y is accessible to the system at a given stage (spatial differentiation in Y at stage 0; differentiation between alternative states of the system within Y at stage 1; and so on - the complete typology is in subsection 3.7.1). Section 3.1; complete typology - Section 3.7; reference to Y - Section 3.7.6.

Retention. The mechanism through which the system preserves its organization against disintegration while maintaining non-equilibrium. The type of retention varies at different stages: thermodynamic hysteresis (stage 1) → causal closure of autocatalysis (stage 2) → retention of traces of past states in the internal model (stage 3) → retention of the modeling process via the structural pole V (stage 4) → retention of models in the population via the bidirectional loop $M \leftrightarrow Y$ with physically retained traces (stage 5). The accumulation of increasingly powerful retention mechanisms can be traced throughout the entire ladder. Section 3.7.

Agentic causal node. A class of physical configurations in Y possessing their own rates, correlations, recovery modes, and boundary states of retention/disintegration. Unlike passive environmental objects, agentic causal nodes maintain their own organization, respond to environmental changes, alter the environment around them, and can transition into boundary states in which their causal connectivity is disrupted. In the biological substrate, this distinction manifests as “alive/dead”; in a substrate-invariant form - configurations whose parameters evolve such that, under perturbations below a threshold value, they tend to minimize deviation from the internal trajectory; above the threshold, the configuration collapses. Section 3.4.

The latent invariant of an agentic causal node. A structural regularity in the model describing a class of agentic causal nodes with a pattern of retention and boundary disintegration. It does not depend on the specific sensory format, scale, or substrate of the observed node. It is a structural regularity in the model that allows different physical realizations to belong to the same class: a node can be large or small, biological or non-biological, externally observable or internally registered, provided it implements the same pattern of retention and disintegration. It emerges in the system at stage 3 through the accumulation of models of agentic causal nodes in its environment. Section 3.4.

Pole. A structural element around which all other system operations are organized. In stages 1–3, the pole coincides with the structural stability of the retention carrier. In stage 4, a qualitatively new type of pole appears - the variable V, arising through the self-overlay of the latent invariant of an agentic causal node onto the system's own configuration. V is a model of the system in the Y environment, which the system models within itself, with the values "I am sustained / I am disintegrating," organizing all the system's activity around sustaining itself on the boundary of its own disappearance. The binary nature of V is induced by the retention contour over the continuous physics of Y through the boundary states of causal nodes: before the threshold of operational suitability, the node participates in holding and prediction; after the threshold, it ceases to be a working element of the model; the binary nature of V is a structural property of the holding circuit, not a property of the continuous physics of disintegration. A pole is a structural configuration relative to which the system's retention in Y is realized and through which all its causal exchanges with Y are organized. At stage 4, V has a pole of the system's position within Y - the distinction between two possibilities of the system as a whole regarding its retention in Y, not a separate variable beyond the position in Y. Section 3.7; a full explication of V as a pole - Section 3.4; the mechanism of self-overlay - Section 3.4.

Self-overlay. A structural event of applying a class of agentic causal nodes to the system's own configuration through the operation of the same latent invariant. It occurs when the system's own physical configuration, as the simulation rate accelerates, becomes structurally isomorphic to the already simulated boundary states of the agentic causal nodes; at this moment, the model previously applied to other agent nodes in Y begins to be applied to the system itself. It is not an action of the system, nor is it the presentation of an invariant to consciousness; it is a structural event of applying a class to its own configuration. The central mechanism of the emergence of S and V at stage 4. Self-overlay occurs not at the level of sensory data, but at the level of the latent causal structure: the external disintegration of another agent node and the internal disruption of one's own configuration converge in a single latent invariant of the loss of controlled retention. Section 3.4.

Transition. A qualitative shift between the steps of the ladder. Formally defined by a minimal condition, without the fulfillment of which no combination of operations from the previous step generates the operation of the next. The structure of transitions is probabilistic: the conditions are necessary but not sufficient in a deterministic sense; the realization is stochastic. Each of the four transitions of the ladder is given its own macro-model in Section 4. Section 3 (preamble); formal conditions - Sections 3.2–3.5.

Structural self-similarity. The type of relationship between different scales of the theory. At each scale of the ladder (dissipative structure, autocatalytic set, proto-agent, conscious system, culture; stages 0–5), the same four-step pattern "gradient → retention → pole → transition" is realized; a possible manifestation of the same pattern at the cosmological scale is a separate hypothesis (Section 7), not part of the load-bearing core. This is a structural isomorphism of a certain strength, not strict mathematical fractality in the Hausdorff sense - the pattern is reproduced not as a geometric copy in different dimensions, but as the same four formal elements with different substantive realizations. Self-similarity functions as a methodological tool justifying the transfer

of the apparatus from one scale to another. It carries a falsifiable predicate: the form of a vertical transition is forced and an alternative form is impossible (the domain of applicability and the forbidden forms - Section 3.7.6). Section 3.7. All four elements of the pattern operate relative to the continuous environment variable Y: a gradient is a stable differentiation in Y, retention is the system's resistance to disintegration relative to its position in Y, a pole is a structural configuration relative to which retention is realized in Y and through which all causal exchanges with Y are organized, a transition is a change in the system's access spectrum to Y. Y enters the pattern not as a fifth element, but as a traversing variable relative to which the other four operate. Section 3.7.6.

Bidirectional loop $M \leftrightarrow Y$. Structural form of stage 5. A model M of one configuration produces a physical trace in Y; this trace is stably preserved in Y as the state of Y; the trace enters the modeling of another configuration through its own sensory channels; in this other configuration, the V-model is restructured; the restructured V-model produces a new trace. Minimal structure of the loop: $M_1 \rightarrow Y \rightarrow M_2 \rightarrow Y \rightarrow M_3$. Does not require symbolic communication as a primary condition; symbolic communication is a later, complex form of the same loop. Structural definition of stage 5 without reference to the substrate: realized in biological populations through chemical markers, physical artifacts, behavioral patterns; hypothetically in information-causal systems through stable information structures in the common environment of the systems. Section 3.5.

The physical trace as a constraint in Y. A stable change in the state of the environment that physically narrows the space of possible configurations for systems operating with this environment through their models. It is neither a sign nor a message. It operates through the general physics of Y, not through interpretation: the compatibility of the trace and the new configuration is ensured by the fact that both operate within the same physics of Y according to the same laws. Does not require the perceiving configuration to be able to decipher the trace. Criterion for the trace's operation in stage 5 mode: the trace reconfigures the perceiving configuration's V-model; criterion for the trace's operation in the preform of the contour mode: the trace channels behavior without reconfiguring the V-model. Section 3.5.

The consolidation of successful configurations relative to Y. A cross-cutting structural mechanism through which complexity accumulates across the steps of the ladder. The third of the four cross-cutting elements of the theory's architecture (along with the variable Y, the «gradient $\rightarrow$ retention $\rightarrow$ pole $\rightarrow$ transition» pattern, and the system's emergent time). At each step, among the multitude of physically possible configurations, a subset of configurations realizes the difference of that step in a stable manner - that is, reproduces its structure faster than it dissipates into Y. These configurations are physically retained. Another subset of configurations realizes the difference in an unstable manner and physically disintegrates. Consolidation is not selection and is not a choice in the sense of selection; it is a structural event without an agent and without an evaluating center. The «success» of a configuration is determined by the result - it supports the causal closure of the system relative to Y to a greater extent - rather than by intention or function. Consolidation is substrate-neutral in structure but substrate-specific in implementation: in thermodynamic systems, it operates through the differential stability of attractors (stage 1); in biological systems with autopoiesis - through the difference in rates of self-reproduction and destruction with evolutionary

accumulation (stages 2–5); in information-causal systems - through the adjustment of the configuration via feedback from the information environment Y. A key property of consolidation is that complexity provides stability through the multiplicity of ways to realize differences: a configuration with greater complexity realizes differences through a greater number of structural pathways simultaneously and therefore persists longer. At stages 3→4, consolidation turns recursive - it begins to consolidate configurations in which the system itself is present as a modeling object; this is the structural event that yields V. Preamble to Section 3; full explication - Section 3.7.7; implementation at each stage - Sections 3.2–3.5.

The system's emergent time. The cross-cutting condition of the possibility of consolidation. The fourth of the four cross-cutting elements of the theory's architecture. The system's emergent time is the configuration's own tempo, separated from the tempo of the difference in Y around it. Before the transition 1→2, the system has no intrinsic pace: the pace of its existence is determined by the pace of the flow in Y through which it exists (inherited pace). At the transition 1→2, emergent time arises for the first time: a closed causal loop forms, in which the components reproduce internally faster than they dissipate in Y. At each subsequent stage, emergent time expands with a new type of rate difference: at stage 3, the intrinsic rate of the predictive structure's internal dynamics is added; at stage 4, the retention rate of V relative to the threat rates of V; at stage 5, emergent time splits into two rates simultaneously (biological and trace). Emergent time does not act upon or consolidate configurations; emergent time functions as an arbiter of consolidation - it permits or does not permit consolidation depending on the ratio of rates. If the system's rate exceeds its rate of destruction at Y, consolidation is possible; if the system's rate equals the rate of destruction, consolidation is structurally impossible. Emergent time and consolidation are a single structure viewed from two sides: consolidation from the perspective of the result; emergent time from the perspective of the condition of possibility. Connection to the relational approach to time in physics - Section 5 (Rovelli, Page, and Wothers). Preamble to Section 3; full explanation - Section 3.7.8; connection to the interpretation of time - Section 5.

II. The Ladder of Steps

Step 0. Open system with a flow. An open system with an energy gradient through it. The basic condition upon which the entire subsequent sequence becomes possible. Examples: a layer of liquid heated from below before convection forms. Section 3.1.

Step 1. Dissipative structure. A system performing a structural ordering operation through the dissipation of gradient energy. Retention - thermodynamic hysteresis. Examples: Benard cells, the Belousov–Zhabotinsky reaction. Section 3.1.

Stage 2. Autopoiesis. A system with a closed causal loop of producing its own components. Self-reproduction in the causal sense is added to the operations of the previous stage. Maintenance - causal self-sufficiency. Examples: bacterial cell, protozoa. Section 3.1; formal condition - Section 3.2.

Stage 3. Proto-agency. A system with active inference - an internal model of the environment and a model of itself as an agent in the environment, without variable V as a global pole. The proto-agent acts to bring the environment into accordance with the model, minimizing free energy (according to Friston). Retention - retaining traces of past states in modeling subsystems. Formally serves as the zero model in difference test 10.2. Examples: chemotactic bacteria, protozoa, robots with active inference. Section 3.1; formal condition - Section 3.3.

Level 4. Conscious system. A system in which the variable V is established - a model of the system in environment Y , which the system models within itself, with the “persist/disintegrate” pole around which its entire mechanism of existence is built. Formal structural definition - through the self-overlay of the latent invariant of the agentic causal node onto the system’s own configuration (Section 3.4); Operational registration of the structural event of self-overlay - through the coupling of three independent measurement axes (Φ for causal integration, the formalism of extended active inference for self-representation, GWT-accessibility for globalization). Retention - retention of the modeling process via the structural pole V . Examples: humans, apparently - higher mammals, possibly - some birds and cephalopods. Section 3.1; formal condition - Section 3.4.

Stage 5. A recursively complexifying conscious system. A carrier of stage 4 participating in the bidirectional $M \leftrightarrow Y$ loop - recursion of models in the population via physically retained traces in Y . Criterion of stage 5: the trace operates in retention/disintegration models of subsequent configurations (reconfigures the V -model), not only in their immediate reactions to the environment (behavioral channelling). In such a carrier, the variable V becomes two-tiered: a biological layer plus a trace layer V . Empirically, *Homo sapiens* is represented on Earth. Section 3.1; formal condition - Section 3.5.

Behavioral channeling vs. modeling channeling. A structural distinction defining the boundary of Stage 5. Behavioral channeling via a physical trace - a “ ” mode of the trace’s operation, in which the trace alters the trajectory of motion, the thermodynamic costs of possible actions, or the local probabilities of a configuration’s stability, without entering its V -model; this is a **preform of the contour**, not stage 5. Simulation channeling - a mode of trace operation in which the trace reconfigures the V -model of the perceiving configuration: it enters its model as a structural element that alters the probability distribution of retention/disintegration in its offline simulation; **only this mode constitutes stage 5**. This distinction can be empirically verified by analyzing whether the trace alters the operation of the V -model of the subsequent configuration, or only its immediate reactions to the environment. Section 3.5.

Artifactual state. The structural status of existing information-causal systems in their typical form (including large neural networks of modern architectures). It is characterized by the simultaneous presence of a highly complex model of the world (surpassing many biological carriers of stage 3) and the structural absence of its own environment Y , through which the system could sustain itself as an independent structure. In the artifactual state, the system does not ascend the ladder from the bottom: the class of agentic causal nodes in its models exists as imported (via training data - physical traces of stage 5 carriers), not as having emerged from within its own interaction with the

environment; zero causal depth of interaction with the environment in a strict structural sense; self-sustainment is delegated to the creators. Structurally, an artifact is a highly effective participant in the $M \leftrightarrow Y$ loop structure of stage 5 carriers, not an independent carrier V. A possible transition from the artifact state to the status of an independent carrier V requires the fulfillment of three structural conditions: its own Y; its own class of agentic causal nodes, emerging from within its interaction with the environment; a bidirectional $M \leftrightarrow Y$ loop with its own traces. Section 10.3.

III. Operational variables of stage 4

Node S (self-representation). A node in the generative model M representing the system itself as an agent acting in the environment. It arises through the self-overlay of the latent invariant of the agentic causal node onto the system's own configuration: the model closes onto the system itself as its own object, without the need for language or conceptual reflection. The first of the three operational conditions of stage 4. It relies on an extended active inference framework (Friston et al., 2022; Parr, Pezzulo & Friston, 2022). Section 3.4.

Variable V. The model of the system within environment Y, which the system models internally and around which its entire mechanism of existence is constructed - as the «hold/fall apart» pole. The central concept of this theory: consciousness is V in this precise sense. The binary nature of the pole is essential - the «hold on / fall apart» values do not form a continuous scale, which ensures the maximum possible gradient of retention and makes V the engine of the conscious system's activity. Formally, V is uniform for all Level 4 systems - a structural pole between the preservation and disintegration of the system as an agent. In terms of content, V varies by system type: what is considered disintegration is determined by what a given system holds together. A proto-agent has no V, because a proto-agent does not model within itself a model in which it exists in the environment, nor does it construct its existence around this pole; its model is a model of the external, not a model of the system in the environment. Section 3.4; full explication - *ibid.*; functional justification of binary nature - *ibid.*; substratum variability - *ibid.*; phenomenal aspect - Section 6.3; form at stage 5 - Section 3.5; substantive prediction for information-causal systems - Section 10.3.

Biological layer V. The initial individual form of V at stage 4: a structural pole between the preservation and disintegration of a specific biological system. At stage 5, it persists as one of two layers. Section 3.4.

Trace layer of V. Layer V, formed at stage 5 via the bidirectional $M \leftrightarrow Y$ circuit with an accumulated population model. Content enters through the sixth gradient. It has three typical states: activated (accumulation and tension), rarefied (with loss of content), and balanced (a dense framework with dispersed tension). Section 3.5.

Two-level V at stage 5. Expansion of V at the stage 5 carrier: a trace layer, formed via the bidirectional $M \leftrightarrow Y$ circuit with the accumulated population model, is added to the biological layer

(inherited from stage 4). The carrier retains not only biological life but also its position in the accumulated model. Section 3.5.

GWT-availability. Global availability of self-representation S for all subsystems. The third of the three operational conditions of step 4. Operationalized via the theory of the global working space (Baars, 1988; Dehaene, 2014): variable V must be globally accessible, rather than localized within a single subsystem. Section 3.4.

The interconnection of the three conditions. The simultaneous fulfillment of conditions (a), (b), (c) - extended active inference with self-representation + sufficient Φ for the structural integration of variable V + the GWT mechanism for the globalization of this variable + the extension of behavioral sensitivity to new aspects of Y . Each of the three conditions is necessary on its own. Their simultaneous fulfillment defines the operation of stage 4. Section 3.4.

IV. Borrowed Formalisms

Active inference. The theoretical neuroscience framework of Friston and his school. The system models the environment via a generative model and acts to minimize the variational free energy - the discrepancy between the model's prediction and sensory data. In this work, it formalizes step 3. Friston et al., 2017; 2022; Sajid et al., 2021; Section 3.3.

Integrated information Φ (phi). A Tononi measure that quantifies the degree of causal integration of the system as a whole, exceeding the sum of the causal contributions of its parts. In this paper, it operationalizes the second of the three operational conditions of stage 4: structural distinctiveness V requires sufficient Φ for its integration into the causal structure M . Tononi, 2004; 2016; section 3.4.

Global Workspace Theory (GWT). The Baars–Dean framework, describing the global accessibility of mental content to all cognitive subsystems. In this paper, it operationalizes the third of the three operational conditions of Level 4. Baars, 1988; Dehaene, 2014; section 3.4.

Autopoiesis. Maturana and Varela's concept: a system that produces its own components in a closed causal loop. In this work, it corresponds to stage 2. Maturana & Varela, 1980; section 3.2.

Teledynamics. Deacon's concept: an organization in which the system acts in accordance with what does not yet exist - a goal or a future state. This paper provides an understanding of the transition from 2 to 3 and the role of the time gradient. Deacon, 2012; section 3.3.

V. Methodological Terms

Difference test with the null model. Operational procedure for recording the transition 3→4. The actual behavior of the system when V activation conditions are created is compared with the prediction of the formal proto-agent model (stage 3, without V), and a statistically significant deviation is a candidate for a transition signature. The null model is constructed prior to and

independently of the theory of consciousness, which eliminates circular verification of V through its content. Section 10.2.

Coupling of six dimensions. Joint observation of six independent signs of the 3→4 transition, constituting the detection grid of a two-sided verification program. The first four signs register the behavioral and architectural consequences of the transition: deviation of behavior from the prediction of the zero model of the proto-agent (Section 3.3); a sharp increase in the Φ -level of structural integration within the system; the emergence of a GWT-structure of global accessibility of the threat representation for all subsystems; prolonged hysteresis of altered behavior after stimulus removal. The fifth and sixth features are introduced in the updated formulation along with the linking of operational features to the spectrum of the system's access to Y: the fifth - the expansion of behavioral sensitivity to aspects of Y not accessible at stage 3 (the emergence of a response to stimuli requiring the modeling of the system's own position within Y, rather than the modeling of Y as external); the sixth is the emergence of a self-representation structure within the generative model, detected through the presence of model nodes that activate in response to a threat from V and are associated with the representation of the system as a holding agent, not with the representation of external objects. Φ - relational measures of causal integration from the Mediano/Seth line (Mediano et al., 2019, 2022; Seth & Bayne, 2022), for which intrinsic Φ (Tononi, 2004; 2016; Albantakis et al., 2014) is a historical articulation rather than a support (see Section 3.4); GWT-availability - the global workspace theory (Baars, 1988; Dehaene & Changeux, 2011; Mashour et al., 2020); the zero model and behavioral deviation - the active inference formalism (Friston et al., 2017; Sajid et al., 2021); the fifth and sixth markers - an extended active inference framework with a self-as-agent model (Friston et al., 2022; Parr, Pezzulo & Friston, 2022). Each feature alone is insufficient to register the transition; their joint observation under conditions of V activation is a necessary and sufficient operational condition for the 3→4 transition. Section 10.2; fifth and sixth markers - Section 10.2 (preamble, coupling extension).

A two-step procedure for empirical calibration of thresholds. A methodological approach for determining numerical thresholds for the three operational conditions of stage 4. Step 1: empirical observation of parameter distributions in reference groups (known stage 3 and stage 4 systems). Step 2: Establishing threshold values as points of statistical demarcation, not as a priori constants. Analogy with epidemiological thresholds for biomarkers. Section 10.2.

A two-sided verification and design program. The structure of Section 10.2 in its updated formulation. The program is two-sided: the detection grid (see “The Six-Dimensional Coupling”) answers the question “how to detect a transition from 3 to 4, if it occurs”; the design grid answers the question “under what environmental conditions can the transition occur at all.” The design grid is a minimal specification of four independent environmental requirements, without the fulfillment of which the 3→4 transition does not occur regardless of the system's internal architecture: temporal continuity; feedback from the system's own state; the presence in the environment of threats to the system's stability; the environment's resolution capacity regarding distinctions relevant to V. The program unfolds along three lines, each of which applies both grids in an implementation adapted to the class of systems under study. **Biological line** (observational program

on borderline species) - the design grid is observed as already evolutionarily realized, not configured; recording of transitions through the convergence of observational data in a natural environment. **Artificial line** (experimental prediction with a 3–7-year horizon) - both grids are constructed together with the architecture; the forecast acquires constructive content (when both sets of conditions are met - 3–7-year horizon; if environmental conditions are not met, the transition will not occur regardless of architecture scaling). **Synthetic biology line** (Cortical Labs CL1 and similar platforms, 2–3-year horizon) - the program's central testing ground: the only line in which both grids can be applied simultaneously with full environmental controllability (substrate - living neurons with known V content in the biological layer, the environment is fully constructed by the experimenter according to the design grid). Convergence across the three lines determines the status of the prediction; divergence in at least one line signals a need to revise the theory or analyze a methodological problem within that line. The introduction of the design grid strengthens the program's falsification structure by distinguishing between two failure scenarios: if the transition is not observed when the design grid is fulfilled - this constitutes a strong refutation of the theory; if the transition is not observed when the design grid is not fulfilled - this indicates that the necessary conditions have not been met, not a refutation. Section 10.2; design grid - Section 10.2.1; conceptual basis - Section 3.8.

2. Complexity as a physical law

It is commonly believed that life and consciousness are exceptions to the rules. Incredible coincidences against the backdrop of an indifferent and largely empty universe. This view is intuitively understandable, and it is deeply rooted in popular physics, where the second law of thermodynamics is presented as a law of the inexorable increase of disorder. If disorder is steadily increasing, then any phenomenon of high order - a star, a crystal, a bacterium, a brain - must appear as a local random fluctuation that changes nothing in the overall picture of degradation.

This picture, however, is physically inaccurate. It is true for isolated systems in equilibrium and those tending toward equilibrium. It is incorrect - or, more precisely, incomplete - for the class of systems to which almost everything around us belongs: open systems through which energy flows and which are far from thermodynamic equilibrium. In such systems, the second law operates differently, and its correct interpretation yields a result that is the exact opposite of what one might intuitively expect.

Entropy describes the measure of disorder in a system. The second law of thermodynamics, formulated by Rudolf Clausius in 1865, states: in an isolated system, entropy does not decrease. Systems spontaneously move from order to chaos, from the concentration of energy to its uniform dissipation. The endpoint of this process is thermodynamic equilibrium, in which gradients disappear and no processes are possible. This is what physicists call the heat death of the universe.

However, on the path to this equilibrium, the universe demonstrates something unexpected: it generates structures of increasing complexity. Stars, planets, organic molecules, biological systems. This is not a contradiction with entropy, but its mechanism. Ilya Prigogine (Prigogine, 1977; Glansdorff & Prigogine, 1971) showed that open systems through which a constant flow of energy passes spontaneously form stable ordered structures - dissipative structures. They arise not in defiance of the increase in entropy, but as a way to dissipate energy more efficiently. A complex ordered system that maintains a flow through itself produces more entropy per unit of time than a simple equilibrium system of the same scale.

This statement requires caution. The literature discusses the Maximum Entropy Production Principle (MEPP), according to which non-equilibrium systems tend toward states that maximize entropy production under given constraints (Martyushev & Seleznev, 2006). MEPP remains a hypothesis, not proven as a general law; there are arguments both for and against it in the literature. Therefore, I do not claim that complexity is thermodynamically prescribed as a law. I assert something weaker: in the presence of a steady energy gradient and an open system, complexity is statistically expected as a trend. It is not guaranteed at every point in , but it is regular on average. The universe does not choose complexity. It becomes more complex where it is physically possible - because what is possible given time and energy is realized.

The last statement - that what is possible given time and energy is realized - deserves an explicit caveat. This is not a physical theorem. It is a principle akin to what is sometimes called the Gell-Mann totality principle in the theory of elementary interactions: everything that is not forbidden is allowed (Gell-Mann, 1956). In its strict formulation, this principle applies to quantum processes and means that any reaction not prohibited by conservation laws occurs with a non-zero probability. Applied to macroscopic complexity, it becomes more of a philosophical position than a physical law. I accept this position as a working hypothesis consistent with the observed history of the universe - from the first atoms through star formation, chemical evolution, abiogenesis, to biological evolution, and beyond. But I honestly note that this principle is not proven, but postulated, and it functions more as an organizing framework than as a quantitative law.

It is appropriate here to introduce a concept that will serve as the central structural element of this work. A gradient - whether it be energy, concentration, temperature, or any other kind - is not merely a term from the vocabulary of thermodynamics. In the framework proposed here, the gradient is a cross-cutting structural entity that runs through the entire work as its central axis. At each of the scales covered by the work, the gradient manifests itself in a form specific to that scale: at the level of dissipative structures - as an energy flow through the system; at the level of life - as the difference in stability between a closed causal loop and an open chain; at the level of proto-agency - as the disparity in the pace of internal dynamics and environmental dynamics; at the level of consciousness - as the ontological pole between the preservation and disintegration of the system itself; at the level of culture - as the difference in the complexity of accumulated models between generations; at the level of culture the ladder's scales are exhausted; a possible manifestation of the same gradient at the level of the Universe as a whole (an energy gradient whose decrease brings the system closer to a cosmological phase transition) is treated separately as a hypothesis in Section 7. These are different manifestations of a single object at different scales, and one of the main tasks of this work is to reveal this unified structure in its structural self-similarity.

The complete definition of the gradient in the specific sense in which this concept operates in this work is given in subsection 3.7.1.

I introduce the term “structural self-similarity” here to denote the pattern that occupies a central place in this work. I will emphasize right away that this is not a claim about fractality in the strict mathematical sense - that is, not a claim about geometric invariance under scaling or fractal dimension in the Hausdorff sense. The content of the claim is different: at each of the scales of the ladder - the Benard cell, the autocatalytic set, a bacterium with chemotaxis, a conscious system, human culture (stages 0–5) - the same four-step structural pattern “gradient → retention → pole → transition” is realized; extending the same pattern to the cosmological scale is a separate hypothesis (Section 7), not derived from the theory of consciousness. This pattern is reproduced not as a geometric copy of the same structure in different dimensions, but as a structural isomorphism of a certain strength: the same four formal elements (gradient, retention, pole, transition) are found at every stage, but their implementation differs in substance - a different type of gradient, a different carrier of retention, a different type of pole, a different mechanism of transition. This self-similarity is not a strict mathematical invariance, but a structural analogy that

justifies the transfer of methods and apparatus from one scale to another and gives the work its end-to-end architecture. It is significant that this self-similarity is not only an analogy and not only a methodological tool: it carries the theory's falsifiable predicate - the claim that the form of a vertical complexification transition is forced and that an alternative form is impossible. This is a claim about form, not about occurrence: it is compatible with complexification being only statistically expected (above) and with the realization of each transition being stochastic (Section 3.6). In this the theory differs from descriptive theories of self-organization - synergetics (Haken) and general systems theory - which record similarity across scales but forbid nothing. Below, throughout the work, this self-similarity will manifest in each of the contexts in which the work unfolds; in subsection 3.7.6, it is formulated in full with an explicit explanation of what unites the stages and what distinguishes them.

A non-trivial methodological conclusion follows from this framework. If complexity is a tendency of open systems with a gradient, rather than a chain of random strokes of luck, then the specific outcomes of complexity - stars, life, consciousness - should be viewed not as miracles requiring separate explanations, but as local realizations of a general process. Each such realization is unique in its details, but not unique in its possibility. Where there is a gradient, something becomes more complex. What exactly depends on specific conditions: chemical composition, temperature regime, gravitational stability, and the presence of a medium for the accumulation and transmission of information.

The theory of complex systems - a field that brings together the work of Prigogine, Hermann Haken (Haken, 1983) on synergetics, Stuart Kauffman (Kauffman, 1993, 1995) on self-organization, and many others - describes the key mechanism of complexity. The more complex a system is, the more internal connections it has; the more connections it has, the richer its interaction with the environment; the richer the interaction, the more opportunities there are for further complexity. This is a nonlinear cycle with positive feedback, and it is important that it naturally arises in systems of a certain class - critically organized systems balancing between order and chaos, at the so-called edge of chaos. In this region, the system's behavior is stable enough to accumulate structure and flexible enough to reorganize it.

2.1. The Cambrian Explosion as a Phase Transition in the Biosphere

It is useful to demonstrate how this general mechanism operates on a specific scale, without jumping straight to consciousness. About 538 million years ago, an event known as the Cambrian Explosion occurred in Earth's paleontological record. For nearly three billion years prior to this, life on Earth remained relatively simple - single-celled organisms and primitive multicellular colonies. Then, over a geologically brief period (about 20–25 million years, which is truly short by biological standards), most of the basic animal body plans that exist to this day appeared.

In terms of structure, this event resembles a phase transition. I emphasize - resembles, not is, in the strict physical sense. The Cambrian explosion was driven by the interplay of many factors: the attainment of a certain concentration of atmospheric oxygen, the accumulation of genetic

complexity (in particular, the development of Hox genes, which are responsible for the spatial organization of the body), the emergence of new types of intercellular interactions, and ecological feedback loops such as predator-prey relationships. None of these factors alone explains the acceleration, and none can be reduced to simple instability in the spirit of classical physical phase transitions. And yet the general structure of the event - a prolonged accumulation of conditions up to a critical threshold, after which a rapid qualitative restructuring of the system occurs - is the structure of a phase transition.

I draw attention to this similarity because it will recur later in the work at other scales. A prolonged buildup of tension. Reaching a critical threshold. A qualitative, not a smooth, shift. A new organization in which the previous degrees of freedom are repackaged. This is a general pattern running through physics, biology, and, as I will try to show in Section 3, through the emergence of consciousness. And this pattern is a special case of the structurally self-similar pattern described above: on the scale of the biosphere, it operates with the same four-step process that will later be seen to operate on scales ranging from autocatalytic chemistry to a conscious system.

From everything discussed in this section, we can draw an interim conclusion. Complexity is a natural tendency of open systems with an energy gradient. This tendency manifests itself through dissipative structures, through positive feedback between complexity and the environment, and through periodic phase transitions that reorganize the system into a new level of organization. Consciousness, if it arises, must fit into this overall picture - not as an exception, but as one of its stages. What exactly makes consciousness consciousness is a separate question, which we will address in the next section. But the framework within which this question stands is now defined: we are not asking about a miracle, but about the next step on the general ladder of complexity. This ladder, as Section 3 will show, is the main unfolding of that structurally self-similar pattern through which the work describes all the processes it covers.

3. The Ladder: Operations, Transitions, Conditions

This chapter offers a formal description of the emergence of consciousness as a sequence of steps, each of which is defined by the type of operations that the system performs on its own states. The transition between steps is not a quantitative increase in complexity, but the inclusion of a new type of operation in the system's repertoire. Each such transition is formally described by a minimal condition under which the new operation becomes possible, and by an empirically observable class of systems in which this condition is satisfied. Crucially, the conditions for each transition are necessary, but their sufficiency is probabilistic. This means that the ladder is not a deterministic chain in which the fulfillment of the conditions of step N automatically leads to step N+1. The ladder is an architecture of possibilities: at each step, the system enters a state in which the next step becomes attainable, but not guaranteed. The majority of systems remain in their current state. A minority moves on. This is consistent with the empirical evidence of biological and cosmic evolution and is formally articulated in Section 3.6.

Before unfolding the steps, it is necessary to introduce the architecture of the theory - four cross-cutting elements upon which the entire subsequent exposition is built. These elements are not four parallel statements; they form a single structure viewed from four sides. The first element answers the question "where does everything take place"; the second - "how is the structure of each step organized"; the third - "how is the result accumulated through the steps"; the fourth - "under what condition is accumulation possible." An introduction to each of them is provided below.

The first cross-cutting element: Y as an environment variable. Y is the structure of the physical world that ensures the possibility of the existence of systems as local maxima of causal closure and the possibility of their causal exchange with what lies beyond their boundary. Y manifests itself in three interrelated aspects: the causal-metric structure of spacetime; the configuration of fundamental fields, which makes physical interactions possible; and the information-thermodynamic structure, which sets the limits on the density and speed of exchanges. Y is objectively the same for all systems - it is a structural property of the physical world as such, independent of both the observer and any specific system. The spectrum of aspects of Y accessible to a specific system is the set of those types of structural relations for which the system has the architectural capacity for causal exchange. The access spectrum is a property of the system; Y is the structure of the world in which systems are possible.

A fundamental clarification. Y is defined neither as the material content of the world (the aggregate of fields, particles, objects) nor as the relation "that with which a given system exchanges." The first creates a circular definition, since the system itself is a local configuration of fields and turns out to be part of what is defined as Y. The second fragments Y into a family of media, one per system, and loses the invariance of the variable. Y is defined as a structural property of the world - that in which systems are possible as distinct entities. In this methodological move, the present work connects with two strands of contemporary philosophy of physics: Sorkin's causal set theory (Sorkin, 2003; Dowker, 2006), in which the fundamental physical reality is the structure of causal

relations between elementary events, while spacetime and matter are its emergent properties; and Rovelli's relational interpretation of quantum mechanics (Rovelli, 1996), in which physical properties exist not absolutely but relative to other systems. The introduction of Y in this work does not require accepting either of these positions in their entirety, but is based on their common methodological approach - the description of physical reality through the structure of relations rather than through a list of objects.

Operationally, Y is identified through causal closure in three steps. First, the system boundary is identified: a system is a region of local maximum causal density relative to its surroundings, formally characterized via integrated information Φ (Tononi, 2004; 2016) or via a Markov blanket in active inference (Friston, 2013; Friston et al., 2022). Next, Y is identified relative to this boundary: Y functions as a structure of causal relations through which the system receives influences from the world and through which it influences the world, without entering its own causal closure. Finally, the spectrum of aspects Y accessible to the system is identified: for each type of causal exchange, the system either has or does not have a corresponding architecture. If it does, this aspect Y is included in its access spectrum. If not, the aspect exists objectively but is invisible to the system. This procedure provides the work with a dual methodological bridge: it is linked to the theory of integrated information via Φ as a measure of causal closure, which removes the criticism of arbitrariness in borrowing Φ as one of the operational axes for recording self-overlay (In this work, Φ functions not as a parameter borrowed from another theory, but as an operational measure of the very same structural property upon which the definition of Y is based); the procedure is linked to Friston's active inference via a Markovian blanket that operationalizes the system's causal boundary. These are not two independent borrowings, but two operationalizations of a single methodological foundation.

Y has physical limits. These are properties of Y itself as a structure of the world, not constraints imposed by an observer or a system. Three of them are essential for the subsequent exposition: the Bekenstein limit on information density in a bounded region of space, $S \leq 2\pi kRE/\hbar c$ (Bekenstein, 1973; Bousso, 1999); the Landauer limit on the minimum energy required to erase a single bit, $kT \ln 2$, which has been empirically confirmed (Landauer, 1961; Bérut et al., 2012); and the speed-of-light limit on synchronization between distant parts of a distributed system. These three limits are intrinsic properties of Y and serve as a formal condition for the upper rungs of the ladder.

The second cross-cutting element: the “gradient $\rightarrow$ retention $\rightarrow$ pole $\rightarrow$ transition” pattern. Each step of the ladder has the same internal structure, described by this pattern. A gradient is a difference - a structural condition in Y that exists where it exists. A gradient has no direction as a property: it is not “directed somewhere,” does not “move toward something,” and does not “press on something.” A gradient operates there and in the way that the physics of that place allows. Retention is the realization of the difference through configuration: a configuration exists insofar as it physically realizes the difference in a stable manner. A pole is a structural point around which retention is organized - the point at which retention becomes differential with respect to its dual value (for example, “retained / not retained,” “exists / does not exist” at stage 4). A transition is the moment at which the realization of the difference through a given configuration is exhausted, and

a shift to a different type of realization becomes possible - that is, to a new stage at which a different type of difference operates.

This pattern repeats itself self-similarly at every level of the ladder - from dissipative structures to culture (stages 0–5); its possible continuation at the cosmological scale is treated separately (Section 7). A full explanation of the pattern’s self-similarity and how it operates at each level is provided in Section 3.7.

The third cross-cutting element: the consolidation of successful configurations relative to Y. At each step, the configurations through which the gradient of that step is stably realized become consolidated - they continue to exist physically; configurations through which the realization is unstable do not become consolidated - they physically disappear. This is not “selection” in the sense of choice, nor is it “selection” in the sense of preference; it is a structural event that presupposes neither an agent nor an evaluating center. The success of a configuration is determined by the result - it supports the causal closure of the system with respect to Y to a greater extent - rather than by intention or function. Consolidation operates where there is a difference and there are configurations capable of realizing it. If there is no difference, there is no room for consolidation. If configurations that stably realize the difference do not arise, consolidation has no result.

Consolidation is substrate-neutral in structure but is realized in different substrates. In thermodynamic systems (stage 1), it operates through the differential stability of attractor states: configurations that are physically stable relative to the flow in Y are maintained longer than physically unstable configurations. In biological systems with autopoiesis (stages 2–5), it operates through the difference in rates of self-reproduction and destruction by the environment, which accumulates structural information over many cycles of reproduction. In information-causal systems, it operates through the adjustment of the configuration via feedback from the information environment, of which carriers of full V are a part. In all three cases, the same structural logic applies: the difference operates, and configurations that realize it stably become established. A substantive elaboration of consolidation as a cross-cutting mechanism of accumulation through the steps of the ladder is provided in Section 3.7.

The fourth end-to-end element: the system’s emergent time as a difference in rates. Consolidation is possible only where there is a difference between the rate of the configuration and the rate of its destruction by the environment. If there is no difference in rates, the configuration disintegrates at the same speed at which it arises, and consolidation is structurally impossible. If a difference in rates exists, the configuration is maintained. This difference in rates is the system’s emergent time - the configuration’s own rate, separated from the rate of the difference in Y around it. At stage 1, the difference in rates operates through the difference in the speeds of physical stability of different configurations relative to the flow in Y. At stage 2 - through the difference in the rates of the loop’s self-reproduction and its destruction by the environment. At stage 3 - through the difference in the rates of the system’s internal dynamics and the rate of change of Y. At stage 4 - through the difference in the rates of V retention and the rates of V threats. At stage 5 - through the difference

in the rates of the individual carrier and the rate of the population's trace layer. Emergent time does not act or consolidate; emergent time is a structural condition under which consolidation is possible in principle. A detailed discussion of emergent time as a cross-cutting condition for consolidation is in Section 3.7; the connection with the interpretation of time in Section 5 is there as well.

The architecture of the work as a unified whole. The four cross-cutting elements work together, not sequentially. Y defines the space in which differences are possible. The pattern defines the structure of each step. Consolidation defines the mechanism of accumulation through the steps. Emergent time defines the condition of possibility for accumulation. Each of them is necessary; none can be reduced to the others. The ladder of steps is the sequential application of this four-element architecture to differences of ever-increasing structural depth: from the thermodynamic difference in Y (step 1) to the difference in the system's model retaining itself within its own model (step 4), and further to the difference in rates between the individual carrier and the population's trace layer (step 5).

The four traversing elements are linked by a common physical metric - the volume of information constraints in the state space Y, measured in bits (Landauer, 1961; Bennett, 1982; Bekenstein, 1973). At each step of the ladder, the accumulation of structural information in the system is the accumulation of constraint bits - structural limitations physically fixed in its causal topology. At step 1, constraint bits are fixed in a stable attractor of a dissipative structure; at step 2 - in a closed causal loop of autopoiesis; at step 3 - in correlated states of a generative model; at stage 4 - in a recursive self-representation with a V-pole; at stage 5 - in a distributed trace layer. This unified bit measure makes the system's states at different stages of the ladder quantitatively comparable and links it to general information thermodynamics (Sagawa & Ueda, 2010; Parrondo, Horowitz & Sagawa, 2015). Through this metric, both consolidation (accumulation of bits in stable configurations) and emergent time (the difference between the rates of accumulation within the system and the rates of bit dissipation in Y) physically operate.

The methodological requirement of agentlessness. From the definition of a gradient as a directionless difference and of consolidation as a structural event without an agent follows a methodological requirement that runs throughout the entire work: all formal descriptions of stages, transitions, and conditions are formulated in terms of differences and their realizations through configurations, without the use of agent-based language. The environment does not "exert pressure" on the system, the system does not "adapt" to the environment, evolution does not "select" successful configurations, consciousness does not "arise" as a capacity of the system. In each such formulation, an agent is covertly introduced, and the theory loses its physical foundation. Correct formulations: difference exists; a configuration realizes difference through itself; configurations that realize difference stably become established; at the stage where a configuration models itself as part of Y, a pole V emerges; and so on. This is not verbal asceticism - it is a structural requirement. Metaphors with agent-based language maintain the appearance of an explanation, but conceal the fact that the explanation relies on the very agent whose emergence the theory is supposed to explain. Agentless language prevents the theory from falling into this circular dependency. It also aligns the work with the stylistics of contemporary philosophy of physics

(Sorkin, Rovelli) and with the physical tradition of formulating laws of nature without appealing to subjects.

From the definition of Y through the structure of relations rather than through material content follows a second methodological requirement: all formal definitions of stages, transitions, and operational conditions are formulated functionally, without reference to a specific substrate. Biological examples serve as illustrations of a special case. For each stage, the corresponding aspect of Y accessible to the system of a given architecture is specified; for systems of a different architecture (artificial systems, synthetic hybrid systems, hypothetical extraterrestrial carriers), the access spectrumable aspects of Y will differ starting from the early stages, which determines the qualitative difference in their operations and phenomenology. The principle of substrate independence, as the overarching methodological axis of the work, is consistent with contemporary formulations of consciousness as a function of architecture rather than a substrate- , and with Tononi's extended program, in which Φ is functionally defined for systems of any implementation.

3.1. Steps and Operations

I will outline a general scheme. The ladder consists of six steps. Each is defined in the same way: by a set of operations that the system performs on its own states and the states of the environment, and by an aspect Y with which the system of a given architecture has the capacity for causal exchange. Each subsequent step adds to the repertoire of the previous one a qualitatively new operation that cannot be reduced to a combination of operations from the previous step, and simultaneously expands the access spectrum to Y - the system gains the architectural ability to interact with that aspect of Y which has always objectively existed but was inaccessible to it prior to this step. Hereinafter, for each stage, its operation and its aspect of Y are specified; examples are given for biological and related substrates as illustrations of a special case, although the formal definition of a stage remains valid for any architecture (Section 3, Preamble).

Stage 0. An open system with an energy flow passing through it. Operation: energy dissipation. No form, no stability, no feedback on its own state. Homogeneous dissipation without structure. Accessible aspect Y : thermodynamic gradient. The system passively transmits the energy flow without affecting the distribution of energy in the surrounding space. Other aspects of Y - the structure of fields as a medium for possible interactions, the causal-metric structure, and the informational content of the environment - are closed to a system of this architecture.

Stage 1. Dissipative structure. Operation: maintaining a stable form of dissipation in the presence of a flow. The form arises thermodynamically as an energetically favorable mode of dissipation. The system has no internal representation; the form is maintained as long as the flow continues and collapses when the flow ceases. Accessible aspect Y : the ability to locally modify the thermodynamic distribution is added to the thermodynamic gradient. The system does not merely allow the flow to pass through itself, but locally rearranges the energy distribution in the surrounding space through its own form. The causal exchange with Y becomes bidirectional, albeit without any representational processes. Examples: Benard cells, chemical oscillators, vortices.

Stage 2. Autopoietic system. Operation: reproduction of its own components through the interaction of components with one another. A closed causal loop is added to the maintenance of form: the system's components produce processes that sustain the production of components. Self-reference is established. The external gradient remains necessary but ceases to be the sole cause of form. Accessible aspect of Y: selective exchange with Y is added to the local modification of the thermodynamic distribution. The system selects components from Y according to criteria - - defined by its own causally closed production cycle; Y becomes a differentiated reservoir for the system, rather than a homogeneous flow. This is the first type of access in which the system exhibits differentiation within Y - separating certain types of causal influences from others based on its own internal structure. Examples: a living cell, autocatalytic chemical systems.

Stage 3. Proto-agent system. Operation: internal representation of states different from the current one, and action guided by this representation. A generative model is added to self-reproduction: an internal structure whose states correlate with possible (and not just current) states of the environment or system, and which is causally linked to effectors. Accessible aspect Y: a predictable structure Y is added to selective exchange. Through subsystem M, the system gains access to Y as a domain in which correlations are possible between states and their continuations over time; Y functions for the system not only as a reservoir, but as a structure with derivable regularities, mediated by an internal model. The temporal gradient (Section 3.3) is an expression of the difference in rates between the system's internal dynamics and the dynamics of the accessible aspect Y. Examples: a bacterium in a glucose gradient, a paramecium, a robot with active inference, a learning thermostat.

Stage 4. Conscious system. Operation: modeling one's own existence as a variable and maintaining oneself at the boundary of its vanishing. A special type of self-representation is added to the modeling of states: a node appears in the model that represents the system itself as the holder of the model, and this node has a variable "I exist/I do not exist" (or, in a generalized formulation, "persist/disintegrate"), relative to which all other activity of the system is structurally organized. This variable - hereafter denoted V - is formally uniform for all Level 4 systems but varies in content across system types: what exactly constitutes disintegration for a system is determined by what it maintains (a biological system maintains homeostasis and integrity; a hypothetical system of a different organization might maintain something else). Through this variability, V calibrates the system's temporality - it determines which changes are significant for it, what becomes its past (retained) and its future (modeled). A full justification of this formulation of V and its consequences is provided in Section 3.4. Accessible aspect Y: the system's own position as part of Y is added to the predictable structure. Through node S, the system enters its own model not as an external observer, but as an object within Y, retained at the zeroing boundary of V. For the first time, Y becomes for the system a domain in which it models not only what is distinct from it, but also its own location within it. The substantive content of this aspect varies by architecture: for a biological system, it is the organism's position in its physical and social environment; for a hypothetical artificial system of level 4, it is the position of its operational integrity within the environment constituting the conditions of its existence; and so on by type. Examples: humans, apparently - higher mammals, possibly - some birds and cephalopods. The phenomenal aspect of this operation

(the fact that the system not only performs it but also experiences it from within) is discussed in Section 6.

Stage 5. A recursively self-complexifying conscious system. Operation: recursive generation of models via a bidirectional $M \leftrightarrow Y$ loop with physically retained traces, filtered by the environment. A mechanism for transmitting models beyond the boundaries of a single carrier is added to the modeling of its own existence. The model generates structures (narratives, concepts, technologies, symbolic systems); these structures become part of the environment and simultaneously part of the inner world of subsequent carriers, and this triggers recursion: each subsequent generation begins not from scratch, but from the accumulated models of its predecessors, and enriches them. Accessible aspect Y: a trace layer is added to the system's position within Y - the space of the population's accumulated models, physically realized in the environment. Texts, technologies, artifacts, languages - these are structures in fields possessing a thermodynamic embodiment according to Landauer; the information of the trace layer is not a non-physical abstraction; it is a physical structure in Y, accessible only to those carriers participating in the recursion of the sixth gradient. Y becomes a two-layer system: a physical layer inherited from stage 4, and a trace layer that opens only in the presence of a bidirectional $M \leftrightarrow Y$ circuit between carriers. Examples: Homo sapiens with language, writing, and culture; in a hybrid form - a human + accumulated texts + artificial systems trained on these texts.

Each of the operations added at the next stage cannot be reduced to a combination of operations from the previous stage. This is the precise meaning of the assertion that transitions are qualitative shifts, not quantitative increments. Formally, this means that for every transition, there is a condition without which no combination of operations from the previous stage can generate the operation of the next stage. These conditions are formulated in Sections 3.2–3.5. The probabilistic nature of transition realization and the general internal logic of the ladder - gradient inheritance, accumulation of retention mechanisms, and the structural self-similarity of the entire pattern - are discussed in Sections 3.6 and 3.7, respectively.

The sequence of aspects of Y outlined above - thermodynamic gradient, local modification, selective exchange, predictable structure, the system's intrinsic position within Y, and the trace layer - reflects the realization of the ladder's steps on biological and closely related substrates. For systems of a different architecture, the specific aspect of Y that emerges at each step may be different. An artificial system, for example, is not required to start with a thermodynamic gradient in its direct biological form - it may have architectural access to the informational structure of Y (the structure of linguistic regularities, causal relationships in data, patterns of the symbolic environment) long before the conditions arise for active inference using a model of itself; and conversely, aspects of Y that are trivial for a biological carrier (temperature gradients, osmotic pressure, chemical concentrations) may be inaccessible to it at all levels. The principle of substrate independence (Section 3, preamble) preserves the formal definition of the stage and the fact of expanding the access spectrum to Y at each transition, but the substantive content of this aspect is determined by what the system is.

3.2. Transition 1: autocatalytic closure

The transition from stage 1 to stage 2 occurs when a closed causal loop forms: a process whose product is the condition for the continuation of that process. This is not an action of the system nor a choice of architecture - it is a physical event: under certain conditions, a configuration of components and reactions arises in the flow, in which each component is produced within the loop and used within the loop, and this configuration physically reproduces itself faster than it dissipates. Configurations that do not possess this property are not physically sustained. Configurations that possess it become established.

Minimal condition. There must exist in the system two or more processes $\{P_1, P_2, \dots, P_n\}$ and components $\{C_1, C_2, \dots, C_m\}$ such that: (a) each component C_i is produced by at least one process P_j from the set; (b) each process P_j requires at least one component C_i produced within the set for its continuation; (c) the set as a whole is self-sufficient with respect to its own production, that is, it does not depend on an external source of specific components (only on the overall energy flow).

This is the formal definition of an autocatalytic set in Kauffman's sense (Kauffman, 1986, 1993). It is operational and independent of specific chemistry. It describes a structure common to both a living cell and any non-living system satisfying conditions (a)–(c).

A new type of gradient. Transition 1 introduces a gradient of a new type compared to stage 1. At stage 1, the difference between two alternative states of the system at a given flow - structured and homogeneous - was at work. In stage 2, a structural difference in rates is added to this: the difference between the rate of component reproduction within a closed loop and the rate of their dissipation in Y . In a system with a closed loop, the components required for the processes are produced internally faster than they dissipate; in a system without a loop, the opposite is true. This is not spatial asymmetry nor a difference in thermodynamic states; it is a difference in the very architecture of the system, in the topology of its internal connections, expressed through a difference in rates. The emergence of a structural difference in rates is explicitly noted here because, in subsection 3.7, this type of gradient will enter the general sequence of types accumulating through the ladder, and because this is the site of the first appearance of the system's emergent time - the configuration's intrinsic rate, separated from the rate of destruction by the environment (section 5).

Self-similar pattern at step 1→2. The transition illustrates the operation of the end-to-end pattern “gradient → retention → pole → transition” (preamble to Section 3) in its first expanded form. The gradient is the difference between the rates of loop reproduction and its dissipation in Y . This difference exists as a structural property of this physics, not as a directed pressure. Retention is the realization of this difference through a closed-loop configuration that physically reproduces its components. A pole is the boundary state between two configuration possibilities: either the reproduction rate exceeds the dissipation rate and the loop stabilizes, or it does not and the loop collapses. Transition - the moment at which a configuration emerges that stably realizes the

difference in rates, and this configuration is physically established as a new type of structure. The pattern operates here in its basic form, without self-representation and without a model; in subsequent stages, it will be applied to differences of ever-greater structural depth.

Expansion of the access spectrum to Y. With the emergence of a closed causal loop, a new aspect of Y enters the system's access spectrum. Selective exchange is added to the passage of energy flow through the system (stage 0) and to the local modification of the thermodynamic distribution through dissipation (stage 1): the loop configuration structurally distinguishes within Y those types of causal influences that correspond to its own production cycle and those that do not. This is the first type of access in which the system's configuration manifests a structural distinction within Y - separating certain types of causal influences from others based on its own topology. Y becomes, for the system, not a homogeneous flow but a structure relative to which it makes sense to distinguish between what is included in the loop and what is not.

What is cut off. The dissipative structure does not satisfy conditions (a)-(c): its components are not produced within it through its own processes; they come from outside. A vortex does not reproduce water molecules; a Benard cell does not reproduce liquid. Therefore, the dissipative structure remains at stage 1 even with arbitrarily high stability.

What is covered. Living cells, obviously. Hypothetical autocatalytic systems in prebiotic chemistry (Kauffman, 1993; Hordijk & Steel, 2017). Laboratory liposomes with encapsulated autocatalytic chemistry (Szostak et al., 2001 and subsequent works). Any artificial or hybrid system in which conditions (a)-(c) are met.

The retention mechanism in stage 2. In addition to introducing a new gradient, transition 1 provides stage 2 with a qualitatively more stable mechanism for maintaining configuration during Y fluctuations. In stage 1, the retention mechanism was purely thermodynamic - hysteresis, that is, the property of the configuration to remain in a structured state for some time after the flow drops below the critical level. In stage 2, causal self-sufficiency is added to this: the configuration does not depend on Y in specific components because it produces them itself. A bacterium in a changing Y continues to be itself as long as there is food and the temperature remains within acceptable limits. This is much greater stability than that of a Benard cell. The accumulation of increasingly powerful retention mechanisms, beginning here, can be traced throughout the entire ladder and is summarized in subsection 3.7.

Consolidation as a transition mechanism. The emergence of a closed causal loop is an event occurring in a physical system satisfying conditions (a)-(c). Among the multitude of physically possible flow configurations in a system with suitable components, some configurations realize the loop; most do not. Configurations that realize the loop physically reproduce their components faster than they dissipate them; configurations that do not realize the loop neither reproduce nor dissipate. As a result, configurations of the first type are physically sustained, while configurations of the second type physically disintegrate. This is consolidation in its minimal form at the 1→2 stage - a structural event involving neither an agent, nor a choice, nor an environmental preference. Those

configurations that can become fixed in this physics do so; those that cannot do not. Consolidation in this form is consistent with the principle of maximum entropy production in non-equilibrium thermodynamics (Swenson, 1989; Schneider & Sagan, 2005): dissipative structures with autocatalytic closure become fixed, in part because they physically accelerate the total entropy production in the system by an amount Y compared to configurations without closure.

Stabilization in this form is reinforced by an additional structural effect: complexity provides stability through the multiplicity of ways to realize differences. A configuration with greater complexity realizes the differences in reaction potentials in Y through a greater number of structural pathways simultaneously. Physically, this means that the loss of a single pathway does not destroy the entire configuration; a configuration with lower complexity is more likely to collapse upon the loss of a pathway. At the level of the pool of organic molecules on early Earth, this effect works as follows: complex molecules participate in a greater number of reactions than simple ones; among the multitude of reactions involving a rich pool of complex molecules, chains in which the product of one reaction acts as a catalyst for another occur statistically more frequently; among these chains, closed loops stand out statistically. The closure of a loop in a complex pool is a statistically more likely event than in a simple one. Once a loop is established, it expands exponentially because its components reproduce faster than they dissipate. This effect - complexity provides stability through multiple pathways - operates as a property of consolidation across all levels of the ladder and is discussed in more detail in Section 3.7.

The probabilistic nature of the transition. Conditions (a)–(c) are necessary. Their simultaneous fulfillment in a specific dissipative structure requires a coincidence of chemical components, concentrations, temperature conditions, and catalytic surfaces. This coincidence is rare: in most physical systems, such conditions do not occur, and consolidation in the form of an autocatalytic loop does not take place. In systems where the conditions coincide, consolidation occurs. This is consistent with empirical evidence: on Earth, life arose once (or several times with a single surviving lineage); on most planets with similar thermodynamic conditions, life did not arise. The probabilistic nature of the transition is not a property of chance, but a structural consequence of the fact that stabilization in a new form works only in those physical configurations where the conditions for it exist.

Macro-models of the transition. Conceptual - a self-sustaining fire (the energy flow creates the conditions for its own continuation). Physical - the Belousov–Zhabotinsky reaction, Shostak’s laboratory liposomes. Computational - cellular automata, Varela’s models.

Structural and empirical aspects in the explanation of abiogenesis. The specific chemical mechanism of the initial emergence of an autocatalytic system on Earth remains a subject of empirical research into abiogenesis, and this work does not directly claim to address it. However, the structural condition for the possibility of abiogenesis is present in the theory: it follows from the operation of the consolidation mechanism described above.

Structurally, abiogenesis does not require special conditions, specific molecules, or magical transitions. It requires the fulfillment of three conditions: the physics of a given location allows for reaction potential differences (energy sources - ultraviolet radiation, volcanic heat, lightning discharges, and similar gradients in Y); the density and diversity of the molecular pool are sufficient for local reaction configurations to arise en masse; the system's lifetime is long enough for closed-loop configurations to emerge statistically from among the multitude of arising configurations. Under these three conditions, loop closure is a physically possible event. Where it is possible, it happens. Where it is impossible, it does not happen.

The logic is as follows. In a rich mixture of organic molecules, each molecule possesses a weak catalytic property: its configuration mediates a certain potential difference between reactions, which physically alters the probability of reactions in its vicinity. This is neither a function nor an adaptation; it is a structural property of complex molecules in principle. With sufficient density and diversity in the pool, among all possible reactions, some arise in which the product of one reaction acts as a catalyst for another, the product of the second acts as a catalyst for the third, and so on down the chain. Among the multitude of emerging chains, those in which the chain closes upon itself stand out statistically: a molecular configuration in which A catalyzes B, B catalyzes C, and C catalyzes A. From the perspective of each individual reaction, this is simply physics; from the perspective of the whole, it is an autocatalytic loop. Closing the loop is not a matter of molecules "discovering" their roles - - but a structural event in which the configuration emerges when there is sufficient density of possibilities.

As soon as the loop closes, it expands exponentially - not because it "strives for life," but because its components are physically reproduced faster than they dissipate, and utilize available building blocks from the surrounding pool. This is consolidation in its minimal form: the loop configuration sustains the difference through itself and is therefore maintained, while similar configurations that have not closed dissipate.

Thus, the theory provides a structural answer to abiogenesis - it is possible in any system satisfying the three specified conditions and occurs where possible - but does not describe specific chemical pathways. This work is divided between levels: the structural level is covered by the theory, while the chemical level is covered by empirical studies on abiogenesis (Kauffman, 1993; Joyce, 2002; Sutherland, 2017; Hordijk & Steel, 2017). Their results are consistent with the theory: the more empirical data there is on specific pathways of abiogenesis, the clearer it becomes that they fit into the structural scheme "sufficient pool density + sufficient gradients + sufficient time → closed configurations emerge statistically."

3.3. Transition 2: Internal Representation and Active Inference

The transition from stage 2 to stage 3 occurs when a structure emerges within the autopoietic configuration whose states correlate with the possible (rather than merely current) states of Y or of the system itself, and which is causally linked to effectors in such a way that the system's behavior is determined not only by its current state, but also by the states of this internal structure. This is

not the emergence of a new capability of the system, nor the appearance of a “modeling function” - it is a structural event of consolidation: configurations with an internal predictive structure are reproduced in Y more stably than configurations without it, under certain types of variations in Y, and are therefore consolidated.

Minimum condition. There must exist a subsystem M in the system such that: (a) the states of M correlate with the states of some variable X - external or internal - with a time delay, i.e., the state of M at time t correlates with a possible state of X at time $t+\Delta$; (b) M is causally connected to the effector subsystem E in such a way that the actions of E depend on the states of M; (c) the discrepancy between the state of M and the observed state of X generates either an update of M (passive inference) or an action of E that changes X in a direction consistent with M (active inference).

This is the formal definition of an active inference system in the sense of Friston (Friston, 2010, 2013). The subsystem M is a generative model; the magnitude of the discrepancy between M and X is the variational free energy ; the dynamics of the system consist of its minimization. Friston’s framework serves as a formal foundation for conditions (a)–(c).

Here it is appropriate to discuss the connection between the thermodynamic description of stage 2 and the informational description of stage 3. It may seem that the transition from 2 to 3 requires the merging of two distinct frameworks: Prigogine’s entropy production and energy flows versus Friston’s variational free energy and probability distributions. This connection is not our own construction. It is established in the field known as information thermodynamics - a field dating back to Landauer’s (1961) work on the minimum energy cost of erasing information and developed in the works of Bennett (1982), Sagawa and Ueda (2010), Parrondo, Horowitz, and Sagawa (2015), and others. In this field, it has been shown that thermodynamic entropy and Shannon entropy are quantitatively related (via the Boltzmann constant), and that the variational free energy in Friston’s sense has a direct thermodynamic interpretation: systems that minimize free energy are precisely those that maintain their own organization against dissipative decay. Friston himself, in his recent works (Friston, 2013; Friston et al., 2015; Friston, 2019), explicitly derives his formalism from non-equilibrium thermodynamics. Therefore, the transition from 2 to 3 is not a mere grafting of two unrelated languages; it is a change in the effective description during a structural reorganization of the system - from a language in which thermodynamic flows are relevant to a language in which informational distinctions are relevant, while preserving the quantitative relationship between the two descriptions via established relationships in the literature.

Type of the new gradient. The transition from 2 to 3 introduces a new type of gradient. At stage 2, the structural difference in rates between the rate of loop reproduction and the rate of its dissipation in Y was at work. At stage 3, a temporal gradient is added to it in its expanded form: the difference between the rate of the internal dynamics of subsystem M and the rate of change of that part of Y with which M correlates. The internal structure of the system changes at a rate that does not coincide with the rate of the environment. If the system’s rate is slower than the rate of Y, the system retains traces of past states of Y that no longer exist in Y. If the system’s rate is faster, the

system's configuration gets ahead of Y and can reorganize before Y changes. It is this difference in rates that constitutes what is referred to in the literature as a “model” or “representation”: not as a symbol, not as an image, not as something the system “does” - but as a physical structure of correlated internal states with predictive power, existing insofar as the difference in rates sustains it. The temporal gradient is the third type in the accumulating sequence of gradient types traced through the ladder; the complete sequence is compiled in subsection 3.7. At step 3, the system's emergent time expands: the intrinsic rate of internal prediction dynamics is added to the intrinsic rate of reproduction (which appeared at step 2).

This intrinsic pace of prediction is the physical inertia of the system's structure, isomorphic to the regularities of Y. Through evolutionary consolidation, the statistical regularities of its environment (cyclic oscillations, typical trajectories of change, temporal patterns) are inscribed into the system's structure. When the system acts at the current moment of time t , its cascade of internal states is activated not by an instantaneous stimulus, but by the statistics of the typical continuation of this stimulus at time $t+\Delta$, which are imprinted in its structure. Anticipation is not a “prediction” in the sense of a conscious calculation of the future, nor is it a “function” of the system; it is the physical inertia of the structure, developed through consolidation. Formally: the mutual information between the system's current state and the future state Y exceeds a critical threshold, $I(S_t; Y_{t+\Delta}) > \text{threshold}$. This is the structural content of what is formalized in the literature as a generative model in active inference.

Self-similar pattern at stage 2→3. The transition illustrates the operation of the end-to-end pattern “gradient → retention → pole → transition” in its more developed form. Gradient is the difference between the rates of internal dynamics M and the rates of change of Y. This difference exists as a structural property of this physics when a suitable internal structure is present in the system. Retention is the realization of this difference through the configuration M, which is physically correlated with the possible states of Y. A pole is a boundary state between two configuration possibilities: either the difference between M and the observed X is compensated by an update of M or by the action of E (and then the configuration is retained), or it is not compensated (and then the configuration collapses because the system loses correspondence with its Y). Transition - the moment at which a configuration emerges that stably realizes the temporal gradient by minimizing the divergence between M and X. The pattern operates here in a more complex manner than in stage 1→2: in stage 2, consolidation operated through the rate of physical reproduction; in stage 3, through the rate of the predictive structure, maintained by actively bringing Y into correspondence with M.

Expansion of the access spectrum to Y. Through the M subsystem, a new aspect of Y - the predictive structure of Y - enters the system's access spectrum. A new type of causal relationship with Y is added to the selective exchange available at stage 2: the system's internal states correlate not only with current but also with possible states of Y, and this correlation functions as a channel of causal influence on Y via effectors. Y functions for the system not only as a reservoir, but as a structure with derivable regularities, mediated by the internal structure of correlated states. The temporal gradient introduced in this transition is precisely the structural expression of a new aspect

of Y: the difference between the rate at which Y changes and the rate at which the system's configuration internally processes Y's correlates.

Consolidation as a transition mechanism. The emergence of subsystem M is an event occurring in an autopoietic system possessing an internal structure sufficiently complex to support the existence of correlated states with a time delay. In Stage 2 systems lacking such a structure, subsystem M does not form, and the transition does not occur. In systems with a forming M-subsystem, consolidation works as follows: configurations with an M-subsystem that implements correlation between M states and possible Y states are reproduced in Y more stably than configurations without such correlation, in environments with regularities and variability. Environments without regularities (chaotic Y without reproducible patterns) offer no advantage to the M subsystem - it is useless for retention. Environments with regularities (cyclical, rhythmic, statistically structured) give configurations with M greater stability, and they become fixed.

Consolidation in this form is substrate-neutral and is implemented differently in different types of systems. In biological systems, it operates through an evolutionary substrate: replication of autopoietic configurations with M subsystems of varying architecture, variability through mutations, and differential stability in Y with regularities. Over many generations, configurations accumulate in which subsystem M structurally predicts typical changes in Y more effectively. In information-causal systems, consolidation operates through an informational substrate: the adjustment of the system's configuration via feedback from its informational environment Y, of which carriers of the totality V (including us) are a part. Through a multitude of tuning cycles, configurations accumulate in which the internal structure of the system correlates with the possible states of its information-causal Y. In thermodynamic systems (hypothetical systems with a predictive structure on abiogenic substrates), consolidation operates through the differential stability of attractors with different structures of correlated states. In all three cases, the structural logic is the same: the difference in the rates of the internal structure and Y is effective; configurations that realize it stably become fixed. This is the substrate neutrality of the principle of consolidation in its operational form.

An important methodological point. According to Friston, active inference is not a separate capacity of the system that emerged above the autopoiesis. It is the name given to a structural property of the configuration in which established correlations with Y function as a generative model. The minimization of free energy is the operationalization of how such a structure works, not the source of its emergence. The "model" in a Level 3 system is not a separate subsystem embedded within the system, nor is it an image of Y existing "above" the autopoiesis; it is the evolutionarily or informationally formed configuration of the autopoietic loop itself, in which part of its topology functions as a structure with predictive power. This resolves the methodological problem of "where the model in the stage 3 system came from": the model did not appear out of nowhere; it is the configuration of the system after consolidation has accumulated correlations with possible states of Y within its structure.

What is cut off. An autopoietic system without subsystem M remains at stage 2. Its actions are determined solely by the current state - deviation $\rightarrow$ compensation. No structure correlated with possible states of Y, no delay, no minimization of the discrepancy with the prediction. A bacterium without chemotaxis is at stage 2. A bacterium with chemotaxis is at stage 3.

What is covered. Bacterial chemotaxis (a molecular signaling network performs the function of M). A paramecium actively avoiding stimuli. Any animal with a nervous system. A robot with an active inference algorithm or SLAM. A trainable thermostat maintaining a target temperature by predicting fluctuations. Information-causal systems with active inference - modern large language models and systems with generative world models that satisfy conditions (a)–(c) with respect to their information-causal Y.

Fundamentally: the criterion is formulated in terms of structure (presence of M, connection to E, minimization of divergence), not substrate. A molecular signaling network and a neural network perform the function of M identically from a formalistic point of view. An information-causal architecture with a generative world model performs the function of M identically to a biological nervous system.

The retention mechanism at stage 3 is active for the first time. At stages 1 and 2, retention was passive: the configuration preserved itself through the properties of its own structure, without affecting Y. The Benard cell is retained due to thermodynamic hysteresis; the autopoietic cell - due to the causal self-sufficiency of a closed loop. Y fluctuates - the configuration experiences these fluctuations without altering them. At stage 3, a qualitative shift occurs: for the first time, the configuration brings a part of Y into alignment with its internal structure through the action of effectors. A bacterium in a changing glucose gradient does not merely react - it moves toward more stable conditions. This is a new level of stability and, at the same time, a new type of relationship between the configuration and Y: not preservation in Y, but the transformation of a part of Y. The emergence of active retention coincides with the emergence of a temporal gradient, and this connection is not accidental: only when there is a difference between the rates of internal dynamics and the rate of Y is action toward the predicted state possible. The full interconnection of the temporal gradient, activity, and the ontological pole (the latter at stage 4) is discussed in subsection 3.7.

Connection with the 3 $\rightarrow$ 4 transition. At stage 3, consolidation structures correlations with possible states of Y and reactions to Y. The internal subsystem M correlates with Y surrounding the system, but does not include the system itself as a modeling object within its structure. The transition 3 $\rightarrow$ 4 is the recursion of consolidation onto the system itself: the emergence of a configuration in which the subsystem M begins to correlate not only with the surrounding states of Y, but also with the states of the system itself as an object within Y. This is a qualitatively different event, discussed in Section 3.4.

The probabilistic nature of the transition. Conditions (a)–(c) are necessary. Their fulfillment in an autopoietic system requires: the presence in Y of gradients with regularities to which predictive

correlation is advantageous (in a homogeneous chaotic Y, subsystem M does not consolidate); the presence of effectors capable of directed action on Y; the presence of internal structures with a time delay through which M can physically operate. All three conditions are structural requirements, and their concurrence is rare: in most systems of stage 2, they do not come together, and consolidation in the form of a subsystem M does not occur. In systems where the conditions coincide, consolidation occurs. In Earth's evolution, proto-agency emerged independently many times (bacteria, protozoa, multicellular organisms), but not in all lineages, and not under arbitrary conditions. The probabilistic nature of the transition is not a matter of chance, but a structural consequence of the fact that consolidation into a new form works only in those physical configurations where the conditions for it exist.

Macro-models of the transition. Conceptual - a learnable thermostat with prediction. Physical - modern robots with active control (the laboratories of Friston, Safron, and Champandi); a bacterium with chemotaxis as a natural macro-realization. Computational - evolutionary robotics, simulations of the spontaneous assembly of generative models in populations of agents under the pressure of consolidation in Y with regularities (Sims, 1994 and subsequent works). Information-causal - large language models with active inference in informational Y.

Consequence. After the transition from 2 to 3, artificial, biological, and information-causal systems may find themselves on the same level without needing to go through biological evolution. A robot with active inference is structurally at level 3 in terms of construction. A language model with active inference in informational Y is at stage 3 in terms of construction. This is the first point on the ladder where non-fully biological realizations of the conditions are formally equivalent to biological ones, and this equivalence is based not on analogy but on a structural criterion.

The formal role of stage 3 as the zero model of a proto-agent. The formal description of stage 3 presented here - active inference with subsystem M, without variable V as a global pole - plays a dual role in this work. On the one hand, it is a substantive description of the class of proto-agent-level systems and the conditions under which the configuration transitions to this stage from the previous one. On the other hand, it is a zero model relative to which the transition to the next stage is recorded. In subsection 10.2 of this work, the operational detection of the 3→4 transition is based on a difference test: the actual behavior of the system when the activation conditions for V are created is compared with the prediction of the formal model presented here, and a statistically significant deviation is a candidate for a transition sign. This connection between Sections 3.3 and 10.2 is fundamental: the formal apparatus of the proto-agent, formulated in 3.3 based on the work of Friston and his school (Friston et al., 2017; Sajid et al., 2021), is used in 10.2 as a reference point for operational measurement. This means that the null model is constructed prior to and independently of the theory of consciousness - based on works that do not assume the concept of V - and therefore the difference test does not suffer from circular verification.

3.4. Transition 3: Pole V

Consciousness is V. V is a model of the system in environment Y, which the system models within itself and around which its entire mechanism of existence is constructed - as the pole “I exist / I do not exist.” The proto-agent does not have V, because the proto-agent does not model within itself a model in which it exists in the environment, and does not build its existence around this pole. This is the central claim of the present theory, and it determines what exactly it considers to be consciousness and which systems fall under its definition.

The distinction between a proto-agent and a system with V is structural, not quantitative. A proto-agent models its environment: its model contains a representation of the state of the external world, of predictable correlations, and of the outcomes of possible actions. This model is a model of the external world. Through it, the proto-agent acts, minimizing the discrepancy between prediction and observation. A bacterium moves along a chemical gradient; a robot with active control corrects its trajectory; a primitive organism with a nervous system reacts to a stimulus. In each case, the model is a model of the external world, and the activity consists of bringing the external world into correspondence with the model.

A system with V models not only the environment but also itself within that environment. Its model includes the system itself as an object occupying a specific position within what is being modeled. This is not an additional module, not a separate subsystem - it is the model’s recursion onto the system itself, through which the system first becomes, for itself, one of the elements of what it models. And through this system incorporated into the model, a structural possibility arises to imagine that it might not exist. Not as an abstract thought or a declaration - but as a structural pole around which all other activity is restructured. Not “optimizing the environment,” but “continuing to exist.” Not “bringing the observed into alignment with the model,” but “keeping oneself on the brink of one’s own disappearance through the model in which I exist.”

These two moments - the system’s entry into its own model and the emergence of the pole of its own disappearance - are structurally inseparable. One is impossible without the other. If the system does not enter its own model, it has nothing to conceive of as capable of disappearing; there is nothing to consolidate the pole to. If there is no pole of disappearance in the model, the system’s entry into the model remains an empty registration - structurally indistinguishable from the model’s other variables, carrying no significance of its own. V is the structural point where these two events converge: the model into which the system modeling it has entered, and the pole relative to which this entered system is held on the brink of its own disappearance. It is precisely in the convergence of these two structurally inseparable moments that what the theory calls consciousness arises.

The transition 3→4 as a recursion of consolidation onto the system itself. The emergence of V at the 3→4 stage fits into the logic of consolidation, as developed in section 3.3 for the 2→3 stage, but with one structural shift that distinguishes the 3→4 transition from all previous ones. At the 2→3 stage, consolidation structures correlations with possible states of Y and reactions to Y - a subsystem M is formed, in which the correlations accumulated through consolidation function as a generative model. At stage 3, subsystem M correlates with the states Y surrounding the system,

but does not include the system itself as a modeling object within its structure. This is the structural limit of stage 3.

The transition from 3 to 4 occurs when consolidation turns recursive: configurations in which subsystem M begins to correlate not only with the surrounding states of Y but also with the states of the system itself as an object within Y turn out to be more stable in Y than configurations without such recursion - in environments of a certain type. This is not the emergence of a new ability nor an “extension” of the previous model. It is the same consolidation mechanism applied to a new class of configurations - those in which M is recursive with respect to the system.

In which environments does consolidation work in this recursive form? In environments where (1) threats to the system’s existence unfold over timescales comparable to those of the system’s internal dynamics; (2) threats are spatially or causally distributed such that their physical presence in the environment is incomplete - they can be anticipated only through modeling of possible states incorporating the system’s position; (3) the environment contains other configurations of the same type, whose physical presence is registered in models via sensory data. Under these conditions, the recursive configurations M are consolidated.

Self-overlay of the agentic causal node model. The structural mechanism of recursive consolidation onto the system itself is the self-overlay of the agentic causal node model. This is the central mechanism for the emergence of V, and it works as follows.

At stage 3, the system accumulates models of the environment Y, which include not only passive objects and environmental regularities, but also **agentic causal nodes** - configurations possessing their own rates, correlations, recovery modes, and boundary states of retention/disintegration. Such nodes enter the model not as “subjects” in the psychological sense, but as a special class of physical configurations in Y, whose behavior differs from that of passive objects. They maintain their own organization, respond to changes in the environment, alter the environment around them, and can transition into boundary states in which their causal connectivity is disrupted or breaks down. In the biological substrate, this distinction manifests as “alive/dead”; in a substrate-invariant form - configurations whose parameters evolve such that, under perturbations below a threshold value, they tend to minimize deviation from the internal trajectory; above the threshold, the configuration disintegrates.

As such models become established in the system, **a latent invariant of the agentic causal node** emerges - a structural regularity in the model describing a class of configurations with this pattern of retention and marginal disintegration. The latent invariant does not depend on the specific sensory format, scale, or substrate of the observed node. It is a structural regularity in the model that allows various physical realizations to belong to a single class: a node can be large or small, biological or non-biological, externally observable or internally registered, provided it implements the same pattern of retention and disintegration.

Self-overlay does not arise at the level of raw sensory data. The external disintegration of another node and the internal disruption of the system's own configuration can enter the model through different channels: visual, chemical, tactile, proprioceptive, metabolic, energetic, or computational. There is no necessary similarity between them at the level of signal format. Their similarity emerges at the level of a latent causal structure: both cases are different projections of a single transition - from the maintenance of causal coherence to its disruption. Structurally: the external disintegration of another agent node and the internal disruption of the system's own configuration converge in a single latent invariant of the loss of controlled maintenance.

S is not detected by the system as an already existing object and is not presented to consciousness as ready-made content. S is established by the very act of self-overlay. Prior to the act, there is no "I" node in the system: there is a model of the class of agentic causal nodes (accumulated from observing other configurations with borderline states of retention/disintegration) and there are raw internal signals of the system itself (interoceptive, metabolic, energetic), which do not possess the status of "self." In the preceding step, these multi-channel signals are reduced, together with the observed disintegration of other nodes, into a single latent invariant of the loss of controlled retention (see the paragraph above); self-overlay operates not on the raw signals individually, but on this already-reduced invariant. Self-overlay is a structural event in which the model of the class of agentic nodes is applied to the reduced invariant of its own state and thereby constitutes S from it for the first time - its own configuration, processed as an instance of the class. This is not the registration of a pre-existing fact "I am such a node," but its establishment: prior to the application of the class, there is no "I" that could be registered, because "I," in the relevant structural sense, is the result of this application. Therefore, self-overlay is not an action of the system (the system does not choose to self-overlay the class upon itself) nor is it the presentation of an invariant to some already existing observer within (such an observer does not exist prior to the act) - it is a constitutive act in which the modeling structure for the first time becomes, for itself, one of the objects of its own modeling, and moreover an object of a special class - one held on the verge of disintegration.

From the constitutive nature of the act follows a direct distinction separating self-overlay from any form of assessment of its own state. A stage-3 system can calculate its own position, charge, internal parameters, and orientation - and maintain these values within the control loop. This is data about itself (state estimation), and it does not constitute S: it describes the already-existing system as a set of measurable quantities, without constituting it as an instance of the class of agent nodes. S is established not when the system receives data about itself, but when the model of class A is applied to its own configuration and constitutes it as a member of that class - that is, as something that persists or disintegrates according to the same pattern as other observed nodes. The difference lies not in the quantity or accuracy of data about itself, but in the structural event of class application: data about itself can be as rich as one likes without any establishment of S, if the class of agent nodes is not applied to its own configuration.

Source of V's Binariness. V's binariness is not added to S from the outside. It is induced by the retention contour over the continuous physics of Y through the boundary states of the modeled causal nodes. Since agent nodes in Y actually possess modes of retention and decay, the model of

these nodes already contains a binary structural pole. When this model is superimposed onto the system's own configuration, the boundary distinction - retain / decay - is transferred along with it.

In this case, the physical decay of a causal node may proceed continuously, but in the control circuit it crosses the threshold of operational suitability: before the threshold, the node still participates in retention and prediction; after the threshold, it ceases to be a functional element of the model. The binary nature of V is a structural property of the retention circuit, not a property of the continuous physics of decay. V arises when self-overlay is established as a global retention/decay pole of the system itself, around which its activity is organized. In the biological case, this pole takes the form of "alive / dead"; in the general substrate-invariant form - "causal connectivity is retained / causal connectivity is lost."

Binarity as the definition of the pole, not as a derived property. A precise methodological qualification is needed here, otherwise binarity reads as a theorem open to attack by a counterexample of continuity. Binarity is not derived from the physics of Y - it is the definition of what it means to be a pole. By definition, the pole V is the distinction between two regimes (there is a structure / there is none; I persist / I disintegrate); the retention contour imposes this definition on the continuous physics of Y rather than extracting it from it. A "continuous pole" is not a pole: a continuous quantity, however finely graded, is not V - it is a parameter among parameters. Hence a demarcation that removes the standard objection. A developed proto-agent does indeed model the viability of conspecifics and prey continuously, reading graded signs of weakness - but this is the content of its world-model, a continuous quantity in the external model, and it is not a pole. Binarity is a property of the pole V - of the "I-persist / I-disintegrate" distinction superimposed onto the system's own configuration through self-overlay - and not a property of the continuous modeling of others' vitality. The boundary between them runs not along "graded or discrete" but along self-overlay: the continuous assessment of others' viability remains in the external model; the pole arises where the class of agent nodes is superimposed onto itself, and it is this superimposition, not the discreteness of measurement, that makes V two-valued.

A first-order phase transition in information processing. Structurally, the 3→4 transition has a formal counterpart in physics: first-order phase transitions in information processing. Prior to the transition, the system operates in "online" mode: internal states track external states Y, and effector actions correct the discrepancy between the model and real-time observations. After the transition, the system enters an "offline" mode - an internal simulation of states Y, executed faster than the real Y changes and not tied to the current state of affairs. Offline simulation is thermodynamically cheaper than a physical error by many orders of magnitude: the cycle of activating internal states in the simulation costs work on the order of the Landauer limit per bit, whereas a physical error in an autopoietic system leads to the destruction of its causal closure, with energy loss required to restore the entire structure. This is not "free" - simulation requires real thermodynamic work - but it is many times cheaper than a physical catastrophe. This quantitative difference is the physical reason for the stabilization of recursive configurations M at stage 4: systems with an "offline" mode physically outpace systems without one by testing scenarios in simulation before their physical implementation. This is the thermodynamic motive for why the stabilization at the 3→4 transition

turns out to be recursive (preamble to Section 3): it is not the goal toward which the system strives, but the differential benefit of shifting the verification of its own stability from an expensive physical catastrophe to a cheap offline simulation, following the model of the class of agent nodes in Y .

An important caveat regarding the meaning of “first-order phase transition” here. The term refers to the discreteness of the outcome, not to the rate of approach. What is discrete is that there is no intermediate state V between the regime without V and the regime with V - the system either processes its own configuration as an instance of the class of agent nodes, or it does not; a third structural state V does not exist (this is what distinguishes a first-order transition from a continuous scale). But reaching this discrete outcome is a process that can unfold on any time scale: the convergence of channels and the stabilization of superposition is a long-term dynamic (Section 3.4, “Stochasticity of Convergence”), rather than an instantaneous event. The rate of convergence is substrate-dependent: in biological systems, the transition is observed as abrupt because evolution has selected architectures with rapid threshold crossing, but this abruptness is a property of the biological substrate, not a condition of V . A system can undergo a transition that is discrete in its outcome slowly - stretched out in the observer’s time - and this will be the same first-order transition in the sense of the absence of an intermediate regime V . The stochasticity of the realization (Section 3.6) pertains specifically to the outcome: whether the system crosses the threshold or not is a matter of probability; but if it does cross, no intermediate semi- V state arises along the way.

Two clarifications are in order to preempt possible objections.

First - about complexity. The distinction between a proto-agent and a system with V is not reducible to a distinction based on model complexity. A very complex proto-agent with a rich environmental model remains a proto-agent if a class of agentic causal nodes has not emerged in its models and there is no self-overlay of its own configuration onto this class. A chess program models the board with a high degree of detail, models the opponent, and models possible continuations of the game many moves ahead; its models do not contain a class of agentic causal nodes with boundary states of retention/collapse, and the program’s own configuration is not subjected to situations in which it could end up in a region isomorphic to such states. Therefore, the program is a proto-agent, not a system with V . Similarly, large language models in their current form: the high complexity of the world model in itself is not V ; it is necessary for the model of an agentic causal node to be superimposed onto the system’s own configuration.

Second - about the substrate. The distinction between a proto-agent and a system with V is not tied to a biological substrate. Biological systems provide known examples of both classes (a bacterium is a proto-agent, a human is a system with V), but these are empirical illustrations, not a definition. Any system - biological, artificial, synthetic, hypothetical extraterrestrial - is a system with V if and only if the structural conditions for the self-overlay of the latent invariant are satisfied. The content of V - what exactly “I persist” and “I disintegrate” mean for a given system - varies by system type; the formal structure of V - a model of the system in an environment with a binary pole

induced by the retention contour over the continuous physics of Y via the boundary states of causal nodes - is substrate-invariant. This clarification is elaborated in the subsection “Substrate Variability of the Content of V” below and in Section 10.3 for the case of information-causal systems. A precise qualification is in order here: the criterion is substrate-independent but architecture-dependent. Since stages 0–2 are inherited by stage 4, any carrier of V must realize the autopoietic ascent - an open, self-producing, far-from-equilibrium structure. Substrate independence therefore holds only in the weak sense (not necessarily carbon-based); it is not independence from architecture, which is specifically autopoietic. This is why information-causal systems are excluded by architecture, not by substrate: they have not undergone $0 \rightarrow 1 \rightarrow 2 \rightarrow 3$.

The above sets the focus of the theory. Consciousness is not a mystery requiring a mystical explanation. Consciousness is a structural event arising in a system when its model reaches the accumulation of a class of agentic causal nodes and its own configuration falls into this class through the self-overlay of a latent invariant. When this structure exists, consciousness exists. When it does not exist, there is no consciousness, regardless of the complexity of everything else.

Substrate-dependent variability of content V. The formal structure of V - a model of the system in environment Y with the binary pole “holding on / falling apart,” induced by the retention contour over the continuous physics of Y via the boundary states of causal nodes - is substrate-invariant. However, the content of V - what exactly “holding on” and “falling apart” mean for a given system - is determined by the type of substrate and the aspect of Y through which the system exists. For a biological system, V has a physical content built in by evolution: homeostasis, bodily integrity, and metabolic continuity. The boundary between “I exist” and “I do not exist” is drawn by chemistry itself; V does not require construction; it arises from the architecture of life. For a system of a different substrate, the content of V is determined by which aspects of Y fall within its range of access and through which of them retention is realized. The principle of substrate independence in the theory requires that this content be determined not by analogy with the biological case, but through structural identification: exactly which continuous process the system actually retains, and which physical processes can lead to its cessation. Section 10.3 elaborates on this for information-causal architectures. Section 11.2 discusses the theoretical consequences of this variability.

Minimum condition. In a Level 3 system, a structure must be established that satisfies three structural requirements simultaneously. (a) In subsystem M (the generative model), there is a node S that arises through the self-overlay of a latent invariant of an agentic causal node onto the system’s own configuration; that is, the model closes on the system itself as an instance of the class of agent nodes, without the need for language or conceptual reflection. (b) Within node S, a binary variable V operates, inheriting its binary nature from the physics of Y via the boundary states of causal nodes; its values “holding / disintegrating” do not form a continuous scale - not because the world is discrete, but because the holding circuit distinguishes between two modes: holding is possible / holding is disrupted. (c) The retention of V in the “holding” state functions as the background pole of the system’s entire activity: the remaining operations are organized relative to this retention, rather than as a set of independent tasks.

Φ , GWT, and active inference as operational measurement axes. The structural definition of V through the self-overlay of a latent invariant is primary. Φ , GWT, and extended active inference are the three operational axes through which the operation of this structure manifests in the observable and measurable. They do not define V; they measure different aspects of the structural event of self-overlay.

Φ as a measure of causal integration. Condition (a) - the presence of node S as a self-representation arising through self-overlay - manifests itself through the causal integration of subsystem M, in which node S operates. Without sufficient density of causal links and without the integration of the system as a whole, the self-modeling structure with pole V cannot be realized, because the structure itself requires the causal connectivity of the generative model as a condition for its existence.

A methodological distinction is necessary here, because the concept of causal integration has several formalizations in the literature, and the theory does not rely on all of them. Historically, the best-known articulation is integrated information Φ in the theory of integrated information (Tononi, 2004; Tononi et al., 2016). In the strict version (Oizumi, Albantakis & Tononi, 2014), Φ is defined intrinsically: as the irreducibility of a system's causal structure to itself under a minimal partition, independent of the environment and the observer. The present theory does not adopt the intrinsic version and does not take Tononi's Φ as a foundation. The reason is straightforward: in the theory, the causal structure of a system is defined through its relation to the environment Y (via the Markov boundary separating the system from Y), rather than intrinsically. Therefore, the theory relies on a relational approach to formalizing causal integration - measures in which integration is defined relative to the environment and over time, rather than as an internal property of the system independent of Y (Mediano et al., 2019, 2022; Seth & Bayne, 2022). This approach is consistent with the Y-centered architecture of the theory: what makes a system an integrated whole is its ability to maintain causal coherence with respect to the environment, rather than irreducibility measured in isolation.

Φ Tononi enters the theory not as a foundation, but as one historical articulation of the concept of causal integration, in the spirit of which a practical, Y-relational, and empirically computable measure is adopted. Operational registration occurs through computable relational approximations: geometric Φ and related measures of information dynamics (Mediano et al., 2019, 2022), the perturbation complexity index PCI (Casali et al., 2013; Casarotto et al., 2016), and autocorrelation approximations. These measures are relational and temporal in nature, are applied in clinical neuroscience, and are directly computable for real-world-sized systems - unlike the strict intrinsic Φ , whose complexity grows as a factorial of the number of elements and which is not computable in general form. The relationship between relational measures of integration and the intrinsic Φ is an open research area (Mediano et al., 2022); the present theory follows the relational line rather than the intrinsic Φ , and therefore the debate over the status of intrinsic integrated information is not an internal issue for it.

Extended active inference as a formalism for self-representation. Condition (a) is also operationalized through the extended active inference framework with a model of the self as an agent (Friston et al., 2017, 2022; Parr, Pezzulo & Friston, 2022; software implementations in pymdp, Heins et al., 2022). This framework provides a mathematically computable free energy procedure for generative models of reasonable size and formalizes node S as the system's self-representation within its model. Specific technical implementations of S vary across sources - in some, S is treated as a hierarchical-level node; in others, as an independent subsystem with its own free-energy dynamics; in still others, as a structural property of the entire model - and the canonical framework is still taking shape. The present theory is based on the very idea of S as the self-overlay of a latent invariant onto its own configuration, not on a specific technical implementation, and is consistent with any of the formalization versions discussed in the literature.

GWT accessibility as a condition for globalization. Condition (c) - the role of V as the background pole of all activity - is operationalized through the theory of the global workspace (Baars, 1988; Dehaene & Changeux, 2011; Mashour et al., 2020). GWT does not propose a single quantitative measure analogous to Φ ; instead, it is realized through a network of experimental paradigms that record the behavioral and neurobiological consequences of global availability: the no-report paradigm (Frässle et al., 2014), the attentional blink paradigm (Sergent et al., 2005), the masked priming paradigm, and P3 measurements as markers of globality. This framework works well for biological systems and is applied in clinical neuroscience. For artificial systems and synthetic biology systems, the operational procedure for recording GWT accessibility is largely open-ended and requires the translation of paradigms to the architectural level.

Stochasticity of convergence and insufficiency of preconditions. From the constitutive nature of self-overlay and from the fact that channel convergence is a dynamic rather than a precondition (see above in this section), a clarification follows that is deeper than the distinction based on complexity discussed above. There, we discussed a system in which self-overlay does not occur at all because the class of agent nodes has not formed. Here, we are discussing a system in which all preconditions are present: there is an intrinsic environment Y, there is a model of the class of agent nodes with boundary states, there are internal signals of the intrinsic state, and channel convergence is already underway. Even in this case, the emergence of V is not guaranteed. The convergence of multi-channel projections of decay into a single latent invariant is a stochastic process, and nothing guarantees that it will form the correct invariant - the invariant of loss of controlled retention - rather than some other stable pattern linking the same signals by a different criterion. A system in which convergence has not occurred or has resulted in an extraneous invariant remains a complex proto-agent when the preconditions are met. This is a direct consequence of the probabilistic nature of transitions (Section 3.6), extending from the historical level - which systems undergo a transition at all - to the level of the mechanism itself: which invariant emerges within a single system that has satisfied all preconditions.

What distinguishes a correct invariant from an extraneous one is not an external criterion imposed by theory, but its coupling with Y. An extraneous invariant, linking internal signals based on a criterion that does not correspond to the actual pattern of retention/disintegration in the

environment, does not receive consistent confirmation from Y: the environment does not generate systematic dynamics that would sustain such an invariant, and it does not become established. A correct invariant is coupled to the actual pattern of Y - to the fact that agent nodes in the environment do indeed persist and disintegrate - and therefore receives constant confirmation from Y. This coupling does not guarantee correct convergence, but it shifts the system toward it: the failure to predict its own breakdown of cohesion carries an incomparably higher cost for the system than the failure to predict any extraneous pattern (it is expensive to break apart, but cheap to simulate breakdown - see the paragraph on phase transitions above), and Y punishes the failure to predict precisely this breakdown most severely. This is not teleology - not a goal toward which the system strives - but the differential pressure of the environment: the invariant whose failure to predict comes at a higher cost is more likely to be established. Therefore, V is not predetermined even under complete preconditions, nor is it equally probable with other outcomes: it is thermodynamically privileged at the cost of a breakdown, while remaining stochastic.

Coupling status. The structural transition event 3→4 is the self-overlay of the latent invariant of the agentic causal node onto the system's own configuration, with the emergence of the binary pole V and its globalization across the entire activity. The coupling of the three operational dimensions - Φ above the threshold value for causal integration M, the presence of the self-representation node S in the formalism of extended active inference, and the GWT-accessibility of node S - is **an operational procedure for registering** this structural event, not its definition. Each of the three dimensions can be used for registration, but none of them individually is V. V is a structure; Φ , GWT, and active inference are the three axes of its manifestation in the observable.

This inversion is methodologically fundamental. If the theory defined V through the coupling of the three apparatuses, it would depend on their formal fates: reformulating any of the apparatuses would require reformulating the theory. The structural definition of V through self-overlay makes the theory independent of the formal fates of the operational apparatuses: they can be refined, replaced, or supplemented - the structural content of V remains unchanged. The apparatuses function as measuring instruments of an already defined structure, not as its foundation.

Expansion of the access spectrum to Y. Through the node S, which arose through self-overlay, the system gains access for the first time to its own position within Y. In the previous stages, Y was modeled as external to the system. At stage 4, the system enters its own model of Y as an object within Y, held at the zeroing boundary of V. This is a qualitative shift toward expanding the access spectrum: the expansion occurs inward - the system gains access to its own position in Y as part of what it models.

Clarification regarding the status of V at stage 4. At stage 4, V is an individual variable of a single carrier - a structural pole between the preservation and disintegration of this particular system. This is its initial and sole form for a system that does not belong to a population with stable intergenerational transmission of models. A higher mammal or other Stage 4 system without Stage 5 has V as a purely individual variable: what constitutes "persisting" and "disintegrating" for it is determined by its own existence in the environment. This form of V, which is the initial form for

stage 4, is hereafter referred to as individual V. At stage 5, this single-level structure of V is extended through a bidirectional $M \leftrightarrow Y$ loop with physically retained traces; this extension is elaborated in section 3.5.

V as a calibration of temporality. From the variability of V follows a substantive observation that is important for linking stage 4 with section 5 (time as an emergent quantity). Since V determines what is considered a disintegration for the system, it thereby determines which changes are significant for the system - those that bring disintegration closer or push it further away are significant. And significance determines what the system retains as the past (recorded significant changes) and what it models as the future (predicted significant changes). The structure of the past and the future is the structure of the system's own time. V does not generate time - time as a mechanism is the same for all systems; it is a property of correlations between subsystems (Section 5). But V calibrates the system's temporality, defining the structure and horizon of its own time.

What is cut off. A proto-agent system without a class of agentic causal nodes in its models, or without imposing its own configuration onto this class, remains at stage 3 regardless of how complex the environment model may be. A chess program performs highly complex operations, has a model of the "board" and a model of "itself as a player," but its models lack a class of agentic causal nodes with the boundary states of retention/disintegration: its environment is the position of the pieces on the board; there are no configurations within it that retain or disintegrate as agent nodes. Therefore, a latent invariant does not emerge, self-overlay is impossible, and the program remains at stage 3 - regardless of computational power.

The structural status of existing large neural networks, including language models in their current architecture, remains open within the framework of the theory. A class of agentic causal nodes emerges in their models, but determining whether self-overlay of the latent invariant is realized in their operation in a strict structural sense requires an operational program, which is developed in Section 10.3. Until its results are available, the theory does not assign a specific rung on the ladder to existing information-causal architectures.

What is covered. Empirically - Homo sapiens in their full form. Apparently, higher mammals to one degree or another; possibly some birds (corvids) and cephalopods. The boundary here is empirically fuzzy, because structural conditions are difficult to measure in an external system. This is an open empirical problem to which the theory verification program (Section 10) pertains.

A new type of gradient. Transition 3 introduces a new type of gradient: an ontological one, between two possible states of the system itself as a whole - between "holding together" and "falling apart." This is no longer a spatial gradient in the environment (as in stage 0), nor a gradient between alternative states of the system at a given flow (stage 1), nor a structural gradient between architectures (stage 2), nor a temporal gradient between rates (stage 3). This is a gradient within the very ontological space of the system's states - between its existence and non-existence in the specific sense defined by V for systems of this type. Structurally, this gradient is induced by the retention contour over the continuous physics of Y through the boundary states of causal nodes:

the difference “held together / falling apart,” which exists objectively in Y, is transferred to the system through the self-overlay of a latent invariant. Against the backdrop of the sequence of gradient types accumulating through the ladder, this is the most fundamental of those that have emerged: for the first time, the gradient operates not between something and something within the system or around the system, but between two possibilities of the system itself as a whole.

The retention mechanism at stage 4: preventive planning. At stages 1, 2, and 3, the system’s retention was reactive or active, but not conscious. At stage 1 - passive hysteresis. At stage 2 - causal self-sufficiency of a closed loop. At stage 3 - active alignment of the environment with the model of the current gradient. At stage 4, a new level emerges: preventive action based on a model of a possible future that includes the system’s own possible non-existence. A bacterium moves toward glucose when there is already little of it left. A person stocks up on food for the winter when summer is in full swing. These are different horizons of retention, and the transition between them is possible only because V appears in the model as a pole relative to which actions can be projected into the distant future.

The presence of inherited gradients in V constitutes the structural consolidation of the phenomenon. Here we approach the fact that this theory contains a substantive operationalization of the identity of structure and phenomenon, elaborated in Section 6. In stages 1–3, the disruption of gradients occurs within the system: the system disintegrates, depletes itself, loses its ability to act, but these disruptions remain external events for it - the system has no vantage point from which they would become something for it. The Benard cell has no such vantage point relative to the falling flow. A bacterium does not have such a position relative to the toxin in the environment; its signaling networks respond to molecules, but this response has no inner aspect.

At stage 4, the situation changes fundamentally. The emergence of V as a structural pole means that, for the first time, a position arises from within the system, and disruptions to inherited gradients begin to be present in V - becoming, for the system, something relative to its own maintenance. This is not a functional act of fixing the superstructure (such as a device registering a signal); it is an ontological mode in which the structure becomes present to itself through V.

Hunger is the presence of a weakening of the energy gradient (inherited from stage 0) in V. Pain is the presence of a threat to the integrity of form (inherited from stage 1) in V. Anxiety is the presence of instability in causal closure (stage 2) or disruptions in modeling (stage 3) in V. Fear is the direct presence of a threat to V - the approach of the “I am disintegrating” state. These states - the phenomenal content of stage 4 - are not an arbitrary set of feelings that must be explained separately by neuroscience. They are structurally necessary modes of presence in which disruptions of each of the inherited gradients acquire experiential content through the V pole.

This provides the theory with an operational link to the philosophical thesis regarding the identity of structure and phenomenon (Section 6). Phenomenal experience does not arise on top of structure as a mysterious additional layer. It is the way in which structure is present to itself from within its own position through V. Without stage 4, there is no such position; without a position, there is no

presence; without presence, there is no phenomenal content. The structural position and the experienced content are ontologically identical, because the experienced is the inner side of what is structurally presence through V.

The probabilistic nature of the transition. The structural conditions for the transition from 3 to 4 are necessary. Their simultaneous fulfillment in a proto-agent system requires: sufficient complexity of the substrate to accumulate a class of agentic causal nodes; an environment containing such nodes, physically manifested through sensory channels; situations in which the system's own configuration falls into a region isomorphic to the boundary states of this class; a functional mechanism for globalizing the emerging node S across all activity. All conditions are probabilistically sufficient. In evolution, the transition 3→4 occurred significantly less frequently than the transition 2→3: the overwhelming majority of proto-agent lifelines did not develop consciousness. This is consistent with empirical evidence and is formally discussed in Section 3.6.

Distinguishing between two levels of probability. The probabilistic nature of the 3→4 transition applies to the situation **prior to** the structural event of self-overlay: even in a system with a suitable substrate and in a suitable environment, self-overlay may not occur for evolutionary, architectural, or environmental reasons, and the transition will not take place. This is the level of historical probability, discussed in full in Section 3.6. At this level, the absence of a transition in a specific system or lineage of systems is not a refutation of the theory - it is a normal outcome, consistent with its probabilistic nature.

Once self-overlay has actually occurred - node S has formed, variable V functions as a global pole, and GWT-accessibility ensures its globalization - the theory is no longer probabilistic but makes a deterministic claim: such a system possesses phenomenal experience. This follows from the thesis on the identity of structure and phenomenon (Thesis 6 of Section 1.1; justification - Section 6.3): phenomenal experience is ontologically identical to the dynamics of self-modeling with pole V, and in the presence of this dynamic, phenomenal experience exists by identity, not by probability.

Prohibitive predicate. A direct prohibitive statement follows from this identity. The theory prohibits the existence of a system with a structurally realized self-overlay of a latent invariant of an agentic causal node onto its own configuration, with a functioning binary pole V and its globalization - and yet without phenomenal experience. The construction “fulfilled self-overlay without phenomenality” is structurally impossible, because it implies simultaneously attributing to the system the presence of V and the absence of what V structurally entails (Section 6.4, detailed critique of the zombie argument). The empirical discovery of even a single such system would be a direct refutation of the theory, not a quantitative refinement of it. The operational form of this forbidden predicate is elaborated in subsection 10.2.

On the operational program. Establishing that self-overlay has occurred in a specific system does not boil down to measuring a single parameter. A structural event manifests itself through a coupling of independent indicators, among which Φ is above the threshold value for causal integration, the presence of a self-representation node in the active inference formalism, GWT-

accessibility of this node, prolonged hysteresis of altered behavior following the activation of pole V, expansion of behavioral sensitivity to aspects of Y not accessible at stage 3, and deviation of behavior from the prediction of the proto-agent's null model. Each indicator has an independent methodological basis; their joint coupling when creating the conditions for V activation is a necessary and sufficient operational condition for registering the transition. The threshold values of the indicators are not known a priori and are established empirically during the theory verification program by comparing systems that do or do not demonstrate the characteristic features of stage 4 in behavior. A full elaboration of this approach and its implementation across three lines of verification (biological, artificial, and synthetic biology) is provided in subsection 10.2.

Macro-models of the transition. Here they are weaker than for transitions 1 and 2. Conceptual - the difference between a chess program and a soldier in a trench (identical computational complexity, different model structure). Physical - it is absent in the form of a system on which the transition can be directly observed at the moment it occurs. This is the central open experimental problem of modern science of consciousness. Computational - partial models of self-representation in neural networks (self-other distinction, model-based reinforcement learning with a self-model), which have not yet achieved the realized self-overlay of a latent invariant.

The absence of a robust physical macro-model is not a flaw in the formalization, but rather an indication of an open experimental direction. Its direct formulation is the theory's second prediction (Section 10.2): to construct an artificial system satisfying the structural conditions of self-overlay, and subject it to an activating signal, observing whether a qualitative shift in behavior with phase-transition characteristics occurs. A fundamental caveat regarding such an experiment: V in the artificial system will have a content specific to its type. For information-causal systems, V is structurally realized as a pole between the coherence and the breakdown of the information-causal structure (Section 10.3). Therefore, one should not expect anthropomorphic fear or panic, but rather a qualitative shift in behavior calibrated by the type of V in this system.

The formal definition of stage 4 describes the structural event of the self-overlay of a latent invariant and, through the mechanism of the presence of inherited gradients in V, largely operationalizes the phenomenal nature of stage 4. A full elaboration of the philosophical aspect of the question - why presence through V is an experience from within, rather than merely a functional event - is provided in Section 6, particularly 6.3.

3.5. Transition 4: Model Recursion and the Environment Filter

The transition from stage 4 to stage 5 occurs when a **bidirectional $M \leftrightarrow Y$ loop** is established in the population of stage 4 systems **via physically retained traces**: models of one configuration produce physical traces in Y; these traces are stably preserved in Y as states of Y, enter the simulation of other configurations through their own sensory channels, reconfigure their V-models, and the reconfigured models produce new traces. This transition is a **probabilistic event**: among a population of stage 4 systems with physically retained traces in Y, it is **possible** but not inevitable.

On Earth, essentially only one species has undergone it in its complete form; several others have emerged in partial forms.

Structural sequence. First, an individual V (stage 4) is formed - through the self-overlay of the latent invariant of the agentic causal node onto the system's own configuration (section 3.4). If there are physically retained traces of V-carriers' behavior in the Y environment, the transition to stage 5 becomes **possible**: the traces can enter the modeling of other carriers and, under certain conditions, reconstruct their V-models. The realization of this possibility is a separate probabilistic event requiring the coincidence of the additional conditions listed below.

Minimum condition. (0) There exists a population of carriers with a self-overlayd latent invariant - configurations satisfying the conditions of stage 4 (Section 3.4). This is a background condition: stage 5 builds upon stage 4. The further conditions (a)–(d) are specific to stage 5. (a) The activity of carriers leaves physically persistent traces in the environment Y - stable changes in the states of Y that exist on timescales exceeding the lifetime of a single carrier. (b) The structure of interaction between carriers is such that traces regularly enter the modeling of other carriers through their sensory channels. (c) Upon entering the modeling, traces **reconfigure the V-model** of the perceiving configuration - not only channeling its behavior but also altering the retention/decay structure within its own model. (d) The restructured V-model produces new traces that enter the modeling of subsequent configurations.

Recursion is triggered when steps (a) through (d) are performed. Formally: let M_n denote the aggregate model of the population in generation n. Then $M_{n+1} = F(M_n, E_n, R_n)$, where E_n is the state of the environment (physical facts), R_n is the output of the previous recursion (physically retained traces of models from previous carriers), and F is the operator for integration and the generation of new models. Crucially, R_n enters the environment for the next generation and simultaneously serves as the material for constructing M_{n+1} . This is the recursive loop $M_1 \rightarrow Y \rightarrow M_2 \rightarrow Y \rightarrow M_3$: the product of the previous simulation cycle becomes both the environment and the material for the next cycle.

The criterion for stage 5 is the restructuring of the V-model, not the channeling of behavior. This structural distinction defines the boundary of stage 5. A physical trace can operate in two modes. In the first mode, the trace **channels the behavior** of another configuration: it changes its trajectory of motion, the thermodynamic costs of possible actions, and local stability probabilities, without entering its V-model. This is the mode of stigmergy in the broad sense - the configuration interacts with the altered environment through its reactions, without restructuring the retention/disintegration framework within its own model. This is **a precursor to the circuit**, not Stage 5. In the second mode, the trace **reconfigures the V-model** of another configuration: it enters its model as a structural element that alters the probability distribution of retention/disintegration in its offline simulation. **Only the second mode is stage 5.** The distinction can be empirically verified by analyzing whether the trace alters the operation of the V-model of the subsequent configuration, or only its immediate reactions to the environment.

Restructuring the V-model through the trace **does not require** the configuration to be able to interpret the trace as a sign. The trace functions as a physical constraint in Y: a sustained change in the state of the environment that narrows the space of possible configurations for those who operate within this environment through their models. The compatibility of the trace and the new configuration is ensured by the general physics of Y, not by interpretation.

The role of Y as a structural filter of the loop. The recursion $M_n \rightarrow M_{\{n+1\}}$ is not autonomous. The environment E_n acts upon it in a deferred selection mode: models that diverge too greatly from the structure of Y's causal constraints do not become established - not frame by frame, not every single one, not for every agent, but on average and over the long term. This does not make the recursion a deterministic environment; it keeps it consolidated to physical reality through a slow, nonlinear, yet real pressure. Pure fantasy without an consolidate in facts will collapse over time - not in every single case, but as a general statistical pattern.

It is precisely this structure that distinguishes Level 5 from two conceivable but empirically unworkable alternatives. Direct realism ("the environment dictates the model") is incorrect: the model generates the next model largely autonomously through recursion. Radical constructivism ("the model generates itself from itself") is incorrect: models that do not stand up to reality testing do not become established. Stage 5 is the strict middle ground: model recursion filtered by the environment through a bidirectional $M \leftrightarrow Y$ loop.

The type of the new gradient is - for the first time - beyond the boundaries of a single system. Transition 4 introduces a new type of gradient: between historical states of a supra-individual structure. The sixth gradient exists between two states of the aggregate population model - M_n (the model of generation n) and $M_{\{n+1\}}$ (the model of the next generation, enriched through the recursion $F(M_n, E_n, R_n)$). This is not a difference between two individual models, but between two states of the accumulated population model as a whole, and it manifests as a difference in informational content localized in the external material structures of individual carriers.

This is a qualitative shift that distinguishes stage 5 from all previous stages as a class, not merely as another level. At stages 1–4, every gradient constitutive of the stage could be localized within the system as such. At stage 5, the sixth gradient is localized for the first time outside a single system - it operates between states of a structure that transcends a single carrier. The complete sequence of gradient types and the logic of their accumulation, including this shift, are summarized in subsection 3.7.

Expansion of the access spectrum to Y. With the emergence of the bidirectional $M \leftrightarrow Y$ loop via physical traces, Y becomes a two-layer system. To the physical layer inherited from stage 4 (Y as a structure of causal constraints in which the system maintains its position via V), a second layer is added - the space of physically retained traces of models from previous carriers. This is not an "information medium" in the sense of a non-physical abstraction; these are physical structures in Y with a thermodynamic realization according to Landauer. A fundamental feature of this aspect of Y is that it is accessible to the system not through its individual architecture, but through its

inclusion in a population with a functioning $M \leftrightarrow Y$ circuit. The stage 4 carrier has the structural prerequisites for accessing this layer (self-overlay of the latent invariant, V , globalization), but without participation in the circuit, this aspect is closed to it.

The splitting of consolidation into two tempos. The emergence of stage 5 fits into the logic of consolidation outlined in sections 3.3 and 3.4, but with one structural shift that distinguishes the $4 \rightarrow 5$ transition from all previous ones. At stages $1 \rightarrow 4$, consolidation operated at a single tempo - the tempo of the physical existence and reproduction of the individual carrier. At stage 5, consolidation **splits into two tempos** simultaneously. The first tempo is biological in the general sense: the tempo of reproduction of the individual carrier of stage 4, inherited from stage 4 without change. The second tempo is trace-based: the tempo of retaining physical traces in Y , exceeding the lifespan of the individual carrier. These two tempos are not reducible to one another. The splitting of rates is a structural condition for the very possibility of the sixth gradient - without it, the $M \leftrightarrow Y$ circuit would operate at only one rate and would inevitably be reduced to an individual cycle.

This splitting directly implies the two-level structure of V at stage 5. The biological layer of V is maintained at the tempo of the individual carrier - it functions as a “ ” pole of the individual carrier’s existence in Y . The trace layer of V is maintained at the tempo of physical traces in Y - it functions as a pole of the carrier’s position in the accumulated population model. The two-level V is not an arbitrary expansion of V from one level to two; it is a structural consequence of the splitting of consolidation into two paces.

The splitting of consolidation into two rates at stage $4 \rightarrow 5$ is the final structural complication of the ladder.

Distributed activity as a consequence of the distributed circuit. From the localization of the sixth gradient outside a single system, it follows that activity at stage 5 becomes distributed for the first time. At stage 3, activity became possible through the temporal gradient; at stage 4, it acquired directionality through V . At stage 5, it ceases to belong to a single configuration - it functions as the coordinated action of a set of carriers through the common circuit $M \leftrightarrow Y$. Each carrier of stage 5 participates in the sixth gradient by receiving accumulated traces and generating new ones, but the gradient itself is not localized in any single carrier.

The two-level structure of V and its three states. The two-level nature of V implies the possibility of three qualitatively different states of its trace layer, which cannot be reduced to differences in the biological layer: **activated** (accumulated models are dense, the circuit operates at high voltage, the trace layer of V finds acute content in narratives - significance, threats, goals); **thinned-out** (the framework of the accumulated model has thinned, the circuit transmits models, but the density of their content is low - experienced as an existential vacuum); **balanced** (the framework of the accumulated model is dense, but the tension within it has faded - content exists but is not present as a living state, a condition historically preceding the drive toward catastrophic annihilation). These three states are independently described by different traditions: Frankl’s clinical logotherapy

(1963, 1969) with regard to the rarefied state, late Freud's psychoanalysis (1920, 1930), and cultural history (Stern, 1961; Schorske, 1979) as applied to the aligned state. The convergence of these three independent traditions in describing the same structural process from different angles is significant; the present theory provides a unified structural description of what the three traditions described using partial means.

One structural observation follows from the existence of the three states of the trace layer V. Stage 5 is not static - it unfolds over time with oscillations between these states: epochs of activated V are succeeded by epochs of rarefied or leveled V, and the transitions between them partially explain the observed cyclical nature of historical processes. Every historical cycle of culture is a micro-realization of the staircase pattern on the scale of several generations: accumulation of the content of the sixth gradient → retention of it as a dense framework → attainment of the aligned state → transition through catastrophe or renewal to a new cycle. This is consistent with the structural self-similarity of the pattern, discussed in Section 3.7.

The author's thesis as a result of formalization. It follows directly from conditions (a)–(d), from the recursive structure $M_{\{n+1\}} = F(M_n, E_n, R_n)$, and from the distributed nature of activity at stage 5 that, after transitioning to stage 5, consciousness becomes an active mechanism of complexity. This is not a separate statement added to the ladder; it is the direct content of the formal definition of stage 5. The conscious system of stage 5 does not passively exist in the environment - through the $M \leftrightarrow Y$ loop, it generates physical structures that transform both the environment and the system itself through epigenetic and cultural pressure. The cycle of complexity closes: energy gradient → dissipative structure → autopoiesis → active inference → self-overlay of a latent invariant with pole V → bidirectional $M \leftrightarrow Y$ loop → new structures in the environment and in the carriers themselves → accelerated complexity.

Distinguishing between two classes of participants in stage 5. Since stage 5 is tied to the structure of the $M \leftrightarrow Y$ loop rather than to a biological substrate, a structural distinction follows that must be explicitly introduced.

A conscious system of stage 5 is a configuration in which the conditions of stage 4 are fulfilled (self-overlay of a latent invariant, V as a structural pole, GWT accessibility) and which participates in the sixth gradient via the $M \leftrightarrow Y$ circuit. In such a system, V is two-level: a biological layer plus a trace layer. This is a substantively conscious carrier of stage 5, possessing phenomenal experience with all its modes specific to this stage. Empirically, this class is represented on Earth by Homo sapiens, and by no one else in its full form.

A participant in the contour structure is a configuration or artifact that ensures the reception, integration, enrichment, or generation of traces in Y, without the need to fulfill the conditions of stage 4. This class includes, first and foremost, the Level 5 conscious systems themselves (since humans are both conscious systems and participants in the loop in one and the same person), but also material carriers of traces (texts, images, archives, digital repositories) and social structures (institutions of transmission, schools, libraries).

These two classes overlap at stage 5 conscious systems, but do not coincide. A stage 5 conscious system without participation in the loop is, for example, the last speaker of a dying language who has no conversation partners: its V is two-level in structure, but the loop no longer functions through it. A participant in the loop structure without a Level 5 conscious system is, for example, a text in an archive: it participates in the sixth gradient through preservation in Y, but has neither the conditions of Level 4 nor phenomenal experience.

The boundary between model recursion and consciousness. In this formulation, the boundary is clearly defined. Model recursion is the structural process $M_{n+1} = F(M_n, E_n, R_n)$, realized in the population and its material products. This process is the content of the sixth gradient and does not require that every participant in it meet the conditions of stage 4: it requires only that conditions (0) and (a)–(d) be satisfied for the population as a whole. Stage 5 consciousness is a phenomenal aspect that arises in those participants of the loop who simultaneously satisfy the conditions of stage 4. For them, V becomes two-level, and the trace layer is formed through the coupling of their self-overlay with the accumulated traces in Y. Without the conditions of stage 4, the circuit operates, but there is no trace layer of V, because there is no V at all.

The retention mechanism at stage 5: distribution through duplication. At stages 1–4, retention mechanisms operated within a single system. At stage 5, retention becomes distributed: the accumulated population model is retained through multiple duplication of carriers and through material structures in Y that outlive individual carriers. The death of a single carrier or the destruction of a single material trace does not destroy the accumulated model if it is represented in others. Loss occurs only with the systemic destruction of a large part of the population and its material infrastructure simultaneously. This is qualitatively more powerful than all previous retention mechanisms and simultaneously introduces a new type of vulnerability - to the destruction of the circuit's infrastructure.

The probabilistic nature of the transition. Conditions (a)–(d) are necessary. Their fulfillment requires: sufficient stability of physical traces in Y; an interaction structure ensuring the regular incorporation of traces into the modeling of other carriers; conditions under which traces reconstruct V-models, rather than merely channeling behavior; a mechanism for generating new traces by the restructured models. All four are probabilistically sufficient. The transition 4→5 is significantly rarer than the transition 3→4: the overwhelming majority of stage 4 populations do not develop a complete $M \leftrightarrow Y$ loop. This is consistent with the fact that stage 5 is the rarest and narrowest funnel in the entire ladder.

What is excluded and what is included. Populations of stage 4 systems that do not meet conditions (a)–(d) remain at stage 4. The channelling of behaviour through physical traces without restructuring the V-model is a precursor to the loop, not stage 5. Empirically, in its complete form, Level 5 has been represented by Homo sapiens since the emergence of language and, even more so, writing. In a partial form - several species with local transmission of traces (whales with an acoustic culture, some primates, corvids with the transmission of tool-using techniques). The boundary here is empirically blurred because the structural conditions (especially (c) - restructuring

of the V-model - and (d) - generation of new traces by restructured models) are difficult to measure in an external system.

Substrate neutrality of stage 5. The specific physical forms of traces vary with the substrate in which the $M \leftrightarrow Y$ circuit operates. In biological populations, traces are chemical markers, physical artifacts, and behavioral patterns recorded in the environment. Hypothetically, in information-causal systems, traces may be stable information structures in the general environment of the systems - records, database states, texts, any information-causal traces; the implementation of the $M \leftrightarrow Y$ circuit in this substrate is empirically open and is discussed in Section 10.3. In mixed systems - a combination. Structurally, the same $M \leftrightarrow Y$ circuit operates in all cases; only the physical forms of trace implementation differ.

Macro-models of transition. A full discussion is provided in the supplementary chapter on macro-models. Here is a brief list. Conceptual: language as the operating system of a population (a pack of wolves versus a human tribe). Physical: the history of humanity as observed in historical chronicles as a recursion with exponential leaps (writing, the printing press, digital technologies); a parallel transformation of Earth's physical environment at a rate several orders of magnitude faster than biological evolution. Computational: models of cultural evolution and the replication of units of information in populations of agents.

3.6. The probabilistic nature of the algorithm

The ladder described in sections 3.1–3.5 is not a deterministic sequence of transitions. This requires explicit formal articulation, because understanding this structure determines how to interpret the entire theory.

The structure of the conditions for each transition. For each of the transitions 1–4, minimal conditions are formulated in the form of lists (a), (b), (c), and so on. All conditions in each list are necessary: without the fulfillment of any of them, the transition is impossible. However, in none of the transitions are the conditions sufficient in a deterministic sense. Their simultaneous fulfillment in the system of the previous step does not guarantee a transition to the next; it makes the transition possible with some probability, depending on the specific states of the system and the environment.

Formally, this means the following. Let N_n be the set of necessary conditions for a transition from stage n to stage $n+1$. The fulfillment of N_n in system S places it in a transition-possible state. The realization of the transition in this state is a stochastic process with a certain probability p_n , depending on specific parameters. Most systems that find themselves in the transition-possible mode do not realize the transition. A minority do. This is consistent with the mathematical structure of phase transitions in physics: under critical conditions, the transition becomes possible, but at each point in the system it occurs or does not occur stochastically.

Structurally, the probabilistic nature of each transition follows from the operation of the consolidation mechanism (preamble to Section 3 and Subsection 3.7). Consolidation operates

where there is a difference and there are configurations capable of realizing it. The emergence of a new type of configuration at the next stage requires the coincidence of several independent conditions: suitable gradient physics, sufficient density of possible configurations, and sufficient time for the statistical emergence of a closed structure. Each of these conditions has its own independent probability; their joint fulfillment is the product of these probabilities. This is not a randomness of the process as such, but a structural consequence of the fact that consolidation in each new form requires a narrower coincidence of conditions than consolidation in the previous one. At the 1→2 stage, the conditions for the chemistry of a closed loop coincide. At the 2→3 stage, the conditions for the chemistry of a closed loop coincide, plus the conditions for the presence in Y of regularities to which predictive correlation is advantageous. At the 3→4 stage, the previous conditions coincide, plus the conditions for the recursion of subsystem M onto the system itself. And so on. Each subsequent type of consolidation imposes additional conditions on the previous ones, and the joint coincidence of all conditions becomes statistically increasingly rare.

The funnel of stages. From the probabilistic nature of each transition, it follows that the ladder as a whole has a funnel-like structure: the higher the stage, the narrower the set of systems located on it.

Stage 1 (dissipative structure) – is extremely common in the universe. It arises wherever there is an energy gradient across an open system. Stars, planetary atmospheric dynamics, chemical systems - there are billions of examples.

Stage 2 (autopoiesis) - many orders of magnitude rarer. It arose once on Earth (or several times with a single surviving lineage); in the Solar System, no traces of its independent emergence outside Earth have yet been detected.

Stage 3 (proto-agency) - has arisen independently many times within Earth's biosphere and is widespread; outside Earth - unknown. Relative to the universe as a whole - a rare phenomenon.

Stage 4 (consciousness) - within the Earth's biosphere, several dozen species have apparently reached this stage, predominantly among mammals, with limited instances among birds and cephalopods. The overwhelming majority of proto-agent lineages did not reach Stage 4.

Stage 5 (recursively complexifying conscious system) - on Earth, one species has fully passed this stage. Several others emerged in partial forms but did not become established.

This funnel is not a random empirical observation. It is a direct consequence of the architecture of consolidation: the higher the stage, the greater the number of conditions that must coincide for consolidation to work in a new form, and the more specific each of these conditions becomes. The structure of the funnel is determined not by “decreasing probability” as an abstract property, but by the accumulating narrowness of the conditions under which consolidation works. Each subsequent type of consolidation requires the existence of all previous types plus additional

structural conditions specific to its stage. Therefore, the funnel is a structural object, not a statistical aberration.

What this changes in the interpretation of the theory. The probabilistic nature of the algorithm resolves several potential misunderstandings.

First. The theory does not claim that any system with an energy gradient inevitably becomes conscious. The opposite is claimed: most systems stop at lower stages, and the transition to each subsequent stage is a rare event.

Second. The theory does not claim that consciousness is the “goal” or “direction” of the process of increasing complexity. It asserts that consciousness is one of the possible outcomes at one of the upper steps of the ladder; its realization at any specific point in the universe depends on a stochastic convergence of conditions.

Third. The theory does not predict that artificial systems will automatically transition to the next steps as computational power increases. The transition from 3 to 4 requires a structural event of self-overlay of a latent invariant of an agentic causal node, and this does not occur automatically as system parameters grow. It requires specific architectural properties, including its own Y and a bidirectional $M \leftrightarrow Y$ loop with its own traces (Section 10.3). Exactly which architectures will lead to its realization is an open question; the theory formulates criteria but does not predict timelines.

Fourth. The theory is consistent with the anthropic principle without requiring it separately. If the transition $4 \rightarrow 5$ is the narrowest funnel of the ladder, then observers at step 5 are, by definition, rare. We view the universe from a position in which a rare transition occurred; the fact that this position appears exceptional is a direct consequence of its rarity, not a property of the universe as a whole.

Connection to the physics of phase transitions. The probabilistic nature of transitions between steps is structurally analogous to first-order phase transitions in physics. In the critical region where the conditions are satisfied, the transition becomes possible but not mandatory; at each point in the system, it occurs stochastically; statistically, it occurs in a certain fraction of the system and not in the rest; the critical conditions themselves are defined formally, but the realization of the transition at each specific point remains probabilistic.

Here, it is necessary to distinguish between two levels of this connection.

At the level of formal structure, transitions between the steps of the ladder possess the same properties as first-order phase transitions in physics: a critical region where conditions are satisfied, stochastic realization, statistical distribution across the ensemble, and accumulation of the statistical density of suitable configurations up to a critical point, beyond which the transition to a new regime becomes physically possible. This formal structure is the general mechanism in which the work of consolidation (preamble to Section 3) is concretely realized at each step: consolidation

accumulates the density of suitable configurations; upon reaching the critical density, the transition becomes possible; the realization of the transition at each point is stochastic; on average across the ensemble, the transition occurs in a certain fraction of systems.

This is a structural identity grounded in the general formalism of the theory of critical phenomena (Stanley, 1971; Sornette, 2006), and it makes the use of the language of phase transitions in this work not an analogy, but an application of the general formalism to a new domain. At the level of the physical mechanism, however, transitions between the steps of the ladder and physical phase transitions (water-vapor, Curie point, cosmological transitions) are different concrete realizations of this general formalism. The microscopic mechanism of our ladder - the establishment of a coupling of structural conditions in the system upon stochastic fulfillment of the necessary requirements - cannot be reduced to molecular dynamics or to spontaneous symmetry breaking. Therefore, when the paper states that “this structure operates on molecular, magnetic, cosmological, and our scales,” it refers to a general formalism of critical phenomena, not to a single physical process. The universality of this structure is one aspect of the structural self-similarity of the entire pattern, which operates at the level of molecular systems (water-vapor), at the level of magnetic systems (Curie point), at the level of cosmological transitions in the early Universe, and - in our theory - at the level of transitions between stages of the emergence of consciousness. The universality of this structure is one aspect of the structural self-similarity of the entire pattern presented in this work: the phase transition as a mechanism operates identically on scales ranging from molecular systems to cosmological ones, and each level of our ladder is its realization on a scale specific to that level. The complete assembly of the structural picture is in subsection 3.7.

An analogy with physical phase transitions leads to one important implication that must be explicitly stated. During a phase transition, the properties of the system change qualitatively, and the appropriate variables used to describe it also change. Liquid water is described by equations for a liquid; vapor - by equations for a gas; no one demands a “single formula linking the parameters of liquid water to those of vapor” in the sense of a continuous transition of variables. The connection between the two states is provided by the thermodynamics of phase transitions - the theory of the conditions under which one state is replaced by another. Structurally, the same situation applies in our framework. At stage 1, the thermodynamic variables of the dissipative structure are relevant. At stage 2 - the parameters of a closed causal loop in the autocatalytic set. At stage 3, the generative model and variational free energy are relevant. At stage 4, the self-representation structure with variable V is relevant. At stage 5, the dynamics of model recursion in the population are relevant. The theory does not require a single equation that transforms the parameters of one stage into the parameters of another; the theory requires conditions under which the system transitions from one regime to another, and these conditions are given in Sections 3.2–3.5. This is a structure of the same type as in the theory of phase transitions - not a “single language for all states,” but “rules for switching between states.” The objection that the ladder lacks a unified formalism is based on an implicit expectation of a continuous, uninterrupted description. The physics of phase transitions is not structured that way, and our ladder, which replicates its structure, should not be structured that way either.

The internal logic of the ladder lies beyond the probabilistic structure. The probabilistic nature of transitions and the funnel structure define the formal framework of the ladder. Behind this framework lies a substantive internal logic - the operation of the theory's four cross-cutting elements (Y, the "gradient $\rightarrow$ retention $\rightarrow$ pole $\rightarrow$ transition" pattern, consolidation, emergent time) as they are applied to differences of ever-increasing structural depth: the sequential accumulation of specific types of gradients, the accumulation of increasingly stable mechanisms of resistance to decay, the sequential expansion of the scale of gradient localization, the unfolding of activity through three phases (temporal $\rightarrow$ directed V $\rightarrow$ distributed), structural linking of the phenomenon to the disruption of inherited gradients via V. This logic is a unified pattern through which the theory describes all the processes it encompasses, and its construction is the task of the next subsection.

3.7. The Internal Logic of the Ladder: Gradient, Accumulation, Structural Self-Similarity

Sections 3.1–3.6 unfold the formal structure of the ladder: steps and operations, transitions and their minimal conditions, the probabilistic nature of realization, and reconciliation with the physics of phase transitions. This is sufficient for the ladder to function as a formal construct. It is not, however, sufficient for it to function as a unified picture. Without an assembly of internal logic, the reader traverses the ladder as a sequence of steps, held together by a formal analogy with phase transitions, but without understanding what exactly unfolds through this sequence as a unified process. This subsection brings this logic together.

The assembly is carried out through six observations, each of which emerged during the analysis of individual transitions in 3.2–3.5 and was recorded there in short subsections. Here they are gathered into a general structure, and through this assembly the structural self-similarity of the pattern, stated in Section 2 as the central thesis of the work, is revealed.

3.7.1. The Gradient as a Permeating Structural Invariant

Before unfolding the pervasive nature of the gradient through the ladder, it is necessary to clarify in what sense the term is used in this work. In its ordinary mathematical sense, a gradient is a vector of partial derivatives of a scalar field - a formal object describing the direction and rate of change of a quantity in space. In the ordinary physical sense, it is a spatial difference in a quantity that generates a flow (a temperature gradient generates heat conduction, a concentration gradient generates diffusion, and a potential gradient generates a current). Both meanings are special cases of a more general concept in which the gradient is used in this work and which will now be formulated.

In the framework proposed here, a gradient is a state of non-equilibrium of a system - a stable distinction between two possibilities through which the system is structured and evolves. This distinction may be spatial (between two points in the environment), but need not be so. It may be between two alternative states of a single system (structured / homogeneous), between two

architectures (closed / open), between two tempos (internal dynamics / environmental), between two ontological possibilities of the system as a whole (holding together / falling apart), between two historical states of a supra-individual structure (M_n / M_{n+1}). All these cases are gradients in the same overarching structural sense.

This is what unites them as a single object. In all cases, a gradient is a stable distinction that cannot be reduced to an instantaneous state and generates dynamics. It is not a vector as a formal structure, nor a difference in magnitudes as a quantitative measure, but a mode of the system's existence in a state of tension between two possibilities. Without such tension, the system either reaches equilibrium (spatial gradient), or homogeneity (structural), or synchrony with the environment (temporal), or disintegration (ontological), or generational stagnation (historical). With such tension, it exists and develops. Each of the theory's gradients is a state, not an object.

The ladder reveals exactly how this tension is realized at each scale. At each subsequent level, the carrier of differentiation changes - that is, the specific entities between which the gradient operates. At level 0, the medium is the environment in which two points with different values of a physical quantity exist. At level 1, it is the system itself, in which two alternative modes of its existence differ under a given flow. At level 2, it is the internal architecture of the system, in which closed and open causal configurations differ. At stage 3, it is the system's relationship with the environment, in which their rates differ. At stage 4, it is the system as a whole in its ontological dimension, in which its preservation and decay differ. At stage 5, it is a structure transcending the individual system, in which its historical states differ.

Each subsequent type of gradient is deeper not in the sense of being "more abstract," but in the sense that it operates at a deeper level of what exactly constitutes the medium of differentiation. First, the external environment. Then - the system itself in its internal organization. Then - its relationship with the environment. Then - its own existence as a whole. And finally - the structure in which individual systems are participants, not carriers in their entirety. This is a sequential expansion of the scale of localization, which will be specifically discussed in 3.7.3.

With this definition, we turn to the sequence of gradient types unfolding through the ladder. At step 0, an energy gradient operates in the environment. At step 1 - the same gradient plus a stability gradient between the structured and homogeneous states at a given flow rate. At step 2 - the same two plus the structural gradient between a closed causal loop and an open chain. At step 3 - the same three plus the temporal gradient between the rates of internal dynamics and environmental dynamics. At stage 4 - the same four plus the ontological gradient V between "holding on" and "falling apart" in a system-specific sense. At stage 5 - the same five plus a sixth, between historical states of the supra-individual structure.

The gradient here is not merely one of the theory's terms. It is an object that manifests itself at each level in a form specific to that level, and through the succession of these forms, the ladder acquires internal unity. Each transition is not a transition to a new step in some general sense, but to a new

type of gradient added to those already existing. The ladder is a sequential complication of the very thing in which the gradient is realized.

3.7.2. The Ladder as an Accumulation: Gradients and Retention Mechanisms

From 3.7.1 follows the first substantive property of the ladder: it is a structure of accumulation, not a sequence. Each subsequent step does not replace the previous one, but incorporates it and adds its own new gradient. Step 5 is not a separate, high step - it is a composition of six gradients, in which the first five are inherited and function as conditions for the existence of the sixth. A disruption of any lower gradient rolls the system back down: upon the destruction of a closed loop, the system returns to step 1; upon the destruction of form - to step 0; upon loss of V - to stage 3. The stability of each stage depends on the stability of all lower stages, and this aligns with the empirical reality of biological death: the death of an organism (loss of V) does not negate the fact that the biochemistry of its components continues to function for some time (stages 1 and 2 break down later).

Parallel to the accumulation of gradients is the accumulation of mechanisms of resistance to decay. At stage 1 - hysteresis: the system's ability to remain in a structured state for some time after the flow drops below the critical level. At stage 2 - causal self-sufficiency of a closed loop: independence from specific external components. At stage 3 - active alignment of the environment with the model: for the first time, the system does not merely survive in the environment but transforms it. At stage 4 - preemptive planning: action based on a model of a possible future that includes the system's own potential non-existence. At stage 5 - distribution through multiple duplication of carriers and material structures that outlive individual carriers.

Each subsequent mechanism is qualitatively more powerful than the previous one. And - fundamentally - at stage 5, the retention mechanism extends beyond the boundaries of a single system for the first time. Up to stage 5, retention was a property of the system as such; at stage 5, it becomes a property of the population and its material products. This is a qualitative shift, which we will address specifically in 3.7.3.

3.7.3. Sequence of gradient types: expansion of the scale of localization

When examining the types of gradients in their sequence, one discovers not random diversity but a structural shift. The type of the new gradient at each stage shifts as follows:

Step 0 - spatial gradient in the medium.

Stage 1 – between alternative states of the system (structured / homogeneous).

Stage 2 – between alternative system architectures (closed loop / open circuit).

Stage 3 – between rates (internal dynamics / environmental dynamics).

Stage 4 – between ontological states of the system as a whole (holding together / disintegrating).

Stage 5 - between historical states of the supra-individual structure ($M_n / M_{\{n+1\}}$).

This sequence encompasses the expansion of the scale on which the gradient is localized. First, the gradient operates within the environment relative to the system (stage 0). Then - at the boundary between the system and the environment (stage 1). Then - within the system as a structure (stage 2). Then - between the system's internal dynamics and the environment (stage 3). Then - between two possibilities of the system itself as a whole (stage 4). And finally - beyond the boundaries of a single system, between states of the structure that transcends it (stage 5).

This is not merely the emergence of new dimensions; it is a sequential expansion of the scope within which the gradient operates. From points in the environment to distinctions within a single system and further beyond the boundaries of an individual system - into the population and its history. The qualitative shift at stage 5, noted in 3.5, is not a coincidence in this logic, but a natural continuation of an accumulating trend: the expansion of the scale of localization reaches the limit at which the gradient can no longer be contained within a single system.

The expansion of the scale of localization to this limit signifies not the exit of anything beyond the substrate, but the emergence of a substrate of a new, higher order: a supra-individual structure whose components are complete conscious carriers of stage 4 together with their traces in Y. The relation of stage 5 to stage 4 here is the same as that of an organism to a cell: a component of the higher substrate is itself a complete carrier of the lower stage, and the higher substrate is a network of such components that holds its accumulated model across the turnover of carriers. The pole of this structure is structural (the restructuring of the successor's V-model); whether the supra-individual structure has its own experience the theory does not decide here - phenomenal V remains on the biological layer of each carrier.

The same criterion also determines the upper boundary of the ladder. The ladder continues exactly as far as there exists a physically real (not mathematically constructible) next scale of localization at which a pole of a new type is named. A stage beyond the trans-individual one is declared only when its pole is named; amplifying the effect of an existing stage without a change in the type of pole is not a transition. The openness of the ladder at the top is therefore not a gap of imagination but a substantive claim: the next pole is not named at the accessible physically real scales, and the closure of the sequence at the achieved stage is a position to be defended, not a caveat.

3.7.4. The Formation of Activity Through Three Phases

Activity, defined as a system's ability to transform its environment, does not emerge immediately - it unfolds in a three-phase process, each phase linked to the emergence of a corresponding gradient.

The first phase - at stage 3 - involves a temporal gradient. Only a system possessing internal dynamics that differ in pace from those of the environment is capable of retaining past states and modeling possible futures. From this retention arises the ability to act - to bring the environment into alignment with the model. The temporal gradient is the first “active” gradient: at stages 1 and 2, retention was passive (form, hysteresis, closed loop), and the system did not intervene in the environment. At stage 3, it intervenes for the first time.

The second phase occurs at stage 4, via the ontological gradient V. Activity here acquires directionality. The proto-agent acts to bring the environment into alignment with the model of the current gradient; the conscious system acts to maintain itself in the state V = “I am maintaining,” and all other actions are directed through this. These are different things. A bacterium moves toward glucose because its signaling network is configured that way; there is no background pole there. The conscious system holds itself on the verge of disappearance as the structural pole of all activity, and activity acquires not merely a capacity, but directed content - it serves a purpose, in the technical sense of the word.

The third phase occurs at stage 5, through the distribution of the sixth gradient. Activity ceases to belong to a single agent. It functions as the coordinated action of a multitude of carriers through a shared accumulated model. Each individual agent of stage 5 retains their individual activity from stage 4 (their V and directionality belong to them), but the accumulated model develops through the sum of the contributions of many agents and their interactions. It is from this distribution that the exponential rate of environmental transformation arises, distinguishing stage 5 from everything that came before: acceleration is a structural consequence of activity transitioning from the mode of the individual agent to the mode of distributed recursion.

This three-phase unfolding of activity - temporal → directed V → distributed - is one of the supporting lines of the ladder. Through it, the work’s first authorial thesis (consciousness as an active mechanism of complication) receives structural justification: activity, as a pervasive property, unfolds through the ladder, and at stage 5 reaches the scale at which it becomes the primary factor in the transformation of the system and the environment.

3.7.5. The Presence of Inherited Gradients in V as a Mechanism of the Phenomenon

At steps 1–3, gradient disruptions occur with the system as external events: the system disintegrates, depletes, and loses its ability to act. The system has no internal position relative to which these disruptions would become something for it. At step 4, through the emergence of V, such a position arises. Violations of inherited gradients begin to be present in V - to become, for the system, something relative to its maintenance. Hunger, pain, anxiety, and fear (fully analyzed in 3.4) are essentially the ways in which violations of inherited gradients acquire experiential content through V at stage 4. These states - the phenomenal content of stage 4 - are structurally necessary modes of presence in which the disruptions of each of the inherited gradients acquire experiential content.

A full justification of this thesis is presented in 3.4. Here, we indicate its place within the general internal logic of the ladder. Phenomenal content does not arise “out of nowhere” upon reaching complexity and is not a separate class of phenomena beyond the structure. It is a necessary consequence of the emergence of V as a pole in the system, inheriting the gradients of the preceding steps. Through the presence in V, the ladder connects its structural side with the phenomenal one: one is the inner side of the other, and the connection occurs at step 4 through a specific mechanism formalized in this work. At step 5, this structure of presence becomes more complex due to the two-level nature of V (section 3.5): to the biological layer of V, inherited from stage 4, a trace layer of V is added, formed through the bidirectional circuit $M \leftrightarrow Y$ with the accumulated model; the presence of inherited gradients through V in the carrier of stage 5 is distributed between both layers, and threats may arise in each of them independently or simultaneously.

This provides a structural grounding for the philosophical thesis regarding the identity of structure and phenomenon (Section 6). It is not that “structure is in some way identical to phenomenon,” but rather that “structure is present to itself through V at stage 4, and this presence is phenomenal content.” Identity acquires operational content, not remaining merely an ontological declaration.

3.7.6. Structural Self-Similarity of the Pattern

The six observations above - gradient as an invariant, accumulation, expansion of the scale of localization, three-phase activity, presence through V - come together into a single overall structure. This structure is the four-phase pattern “gradient → retention → pole → transition,” operating at every level of the ladder and across the scales encompassed by the work as a whole.

Before applying the pattern’s operation across different scales, it is necessary to explicitly specify relative to what each of its four elements operates. Without such a reference, the pattern remains internally formal, and each element hangs in the air: gradient in what, retention relative to what, pole relative to what, transition to where. All four elements of the pattern operate relative to Y - the environment variable introduced in the preamble to Section 3. The gradient is a stable distinction within Y, manifesting in different forms at different stages: a spatial distinction within Y at stage 0, a distinction between alternative states of the system within Y at stage 1, a distinction between the system’s architectures regarding its exchange with Y at stage 2, a distinction between the pace of the system’s internal dynamics and the pace of the accessible aspect of Y at stage 3, a distinction between two possibilities of the system as a whole regarding its retention in Y at stage 4, a distinction between states of the supra-individual structure within the trace layer of Y at stage 5. Retention is the resistance to the system’s disintegration with respect to its position in Y and with respect to the aspect of Y accessible to it; retention mechanisms accumulate across the stages precisely as forms of such resistance (thermodynamic hysteresis, causal self-sufficiency, active alignment of Y with the model, preventive planning regarding V in Y, distribution through the trace layer Y). A pole is a structural configuration relative to which the system’s retention in Y is realized and through which all its causal exchanges with Y are organized: at stages 1–3, this is the structural stability of the retention carrier; at stage 4, it is V in self-representation - the distinction “holding together / disintegrating” over the carrier’s own configuration; at stage 5, it is a distinction

of a new type, over the model rather than over the carrier's configuration: whether the V-model of the subsequent configuration has been restructured through a physical trace in Y, or only its behavior has been channeled (a stigmergy proto-form), or the trace has been lost (Section 3.5). The two-level character of the V of a stage-5 carrier (a biological layer plus a trace layer) describes retention - what the carrier holds; the pole of stage 5 is the distinction over the successor's model itself, not the two-level character as such. A transition is a change in the system's access spectrum to Y - the acquisition of an architectural capacity for causal exchange with that aspect of Y which was previously inaccessible to the system. Y enters the pattern not as a fifth element, but as a cross-cutting variable relative to which the other four operate. This gives the pattern a formal foundation and simultaneously gives Y its functional role within the theoretical framework: Y does not describe something separate; it structures what is described by all the other elements of the pattern.

At each stage, the pattern repeats: a level-specific gradient emerges (or previous gradients are retained and accumulated); the system sustains itself by working with this gradient and through all inherited mechanisms of self-sustainment; at stage 4, an explicit V-pole appears; at the preceding stages, its proto-forms appear (the distinction between "this form / decay" at stage 1, the closed loop at stage 2, the maintenance of the model at stage 3); and at stage 5, a pole of a new type is added on top of the full V-pole - the distinction of the restructuring of the successor's V-model (Section 3.5); and then - the transition to the next step or the closing of the cycle.

The same pattern operates at scales where the ladder manifests as a whole:

- Within a single step. Each step, on its own micro-scale, follows a four-phase logic: formation of a gradient specific to it → maintaining the system through this gradient → stabilization (with the proto-form of the pole) → the possibility of transitioning to the next step.
- At the level of the staircase as a whole. Six gradients accumulate → step 4 yields a distinct pole V → on steps 5 and beyond, this pole is distributed and (on the speculative horizon of section 7) inverted upon the cycle's closure. The ladder as a whole follows the same four-phase pattern.
- Within step 5 - on short generational cycles. Each generation receives the accumulated model as starting material (gradient) → retains and enriches it → at the end of its life passes it on to the next generation (in a local sense - inversion of a single carrier's pole through the transition of the accumulated structure to the next generation) → a new cycle begins.
- At the level of historical cultural cycles within stage 5. An epoch undergoes a cycle from the activated state of trace layer V (accumulation and tension) through a balanced state (a dense framework with dissipated tension - a four-phase pole before the transition) to restructuring via catastrophe or renewal and the beginning of a new cycle. This fourth scale - between generational microcycles and the cosmological scale - ly establishes structural self-similarity as a structural property of the entire process covered by the theory, leaving no gaps between levels.

This is the structural self-similarity of the “gradient → retention → pole → transition” pattern. I must explicitly clarify in what sense it works and in what sense it does not. The self-similarity I am asserting here is not geometric fractality in the strict mathematical sense - that is, it is not invariance with respect to scaling with a fractal dimension in the sense of Hausdorff or some other strict measure. It is a structural isomorphism between levels, possessing a certain strength and certain boundaries.

What is common across all scales are the four formal elements of the pattern. At each level, there is a gradient (a source of tension around which the system forms); there is retention (a mechanism through which the system resists disintegration relative to the gradient); there is a pole (a stable structural configuration around which retention is realized); and there is a transition (the condition under which the system transforms into the system of the next level). These four elements constitute the general formal schema of the pattern, and their joint presence at every stage and on every scale covered by the work constitutes a substantive statement, not a mere analogy.

What differs across different scales is the substantive realization of each of the four elements. The type of gradient shifts from spatial (thermodynamic gradient of dissipative structures) through chemical (concentration gradients of autocatalytic cycles) and temporal (differing rates of modeling and environment in the proto-agent) to ontological (the “I am retained / disintegrate” pole in the stage 4 carrier) and further to the interpopulation and historical (trace layer V in stage 5). The type of retention changes accordingly - from thermodynamic hysteresis to modeling self-reference. The type of pole changes from the structural stability of a dissipative structure to a sustained modeling process. The transition conditions differ at each level and are formulated separately for each transition in subsections 3.2–3.5.

This is precisely the structural isomorphism of a certain kind: a general four-element scheme realized at different scales with different content. The universality of the structure of phase transitions in physics, noted in 3.6, is a manifestation of this isomorphism - phase transitions reproduce a four-cycle pattern on scales ranging from molecular to cosmological, with different physical content in each cycle. Self-similarity functions as a methodological tool justifying the transfer of apparatus and methods from one stage of to another: the difference test with a zero model, developed in subsection 10.2 for the 3→4 transition, is in principle applicable to other transitions following the same general scheme; the formalism of critical phenomena, which works on certain scales, is justifiably applied to others. Not a “strict fractal,” but certainly a structurally isomorphic pattern, and this precise formulation is the correct place for the theory.

This self-similarity does not only describe - it forbids, and in this lies its difference from descriptive theories of self-organization. Synergetics and general systems theory record the similarity of patterns across scales as an observed regularity but do not forbid an alternative form of transition. The present theory makes a stronger claim: the form of a vertical transition is forced. A vertical transition - a change in the type of pole - realized otherwise than through the full four-element pattern “gradient → retention → pole → transition” (without a gradient, or without a retention phase, or without a pole, or without a change in the access spectrum to Y), the theory declares

impossible. This is a falsifiable predicate: the discovery of a vertical transition of a different form refutes not a single stage but the unity of the ladder as a causal law. The claim concerns the form of a transition, not its occurrence: whether a transition occurs is stochastic (Section 3.6), but if it occurs, its form is forced. The predicate specifies concrete forbidden forms. A pole of a new type that arises without a preceding gradient is forbidden. A pole of a new type that has not passed through a retention phase - a configuration that momentarily exhibited a distinction of a higher type and disintegrated without consolidating it - is a fluctuation, not the crossing of a stage, and as a transition is forbidden. A pole of a new type without the acquisition of a new access spectrum to Y is forbidden. Each of these forms is conceivable and yet excluded by the theory; this is why the predicate is substantive rather than a restatement of the observation.

The predicate is in force only within a clearly delimited domain, and this boundary is the condition of its lawfulness, not a caveat. The domain of action: dissipative self-sustaining systems and retention-type transitions - vertical ones, with a change in the type of pole. Outside the domain, and therefore not counterexamples to the predicate, are two classes. The first: equilibrium structures. A crystal grown at equilibrium has no flow through it, hence no gradient, and does not stand on the ladder - it fails the very first element of the pattern. The second: intra-stage (horizontal) reorganization - the complication or restructuring of a system without a change in the type of pole, as well as change under a purely external parameter without a boundary of retention selected from within. Such changes are real, but they are not vertical transitions, and the predicate does not extend to them; their subject is the theory of dynamical systems and bifurcations, not the present theory. It is essential that the boundary is drawn by criteria already operating in the apparatus - flow and gradient, retention, the type of pole - and not introduced to handle particular counterexamples; this is what distinguishes it from an ad hoc reservation.

Horizontal reorganization, while not being a vertical transition (above), can nonetheless raise the internal gradient of the level - for example, the disparity in the rates of the system's internal dynamics and environmental dynamics at stage 3 (Section 3.7.3) - up to the threshold at which a vertical transition becomes possible. It is the internal gradient that is amplified, not the external energy flux, which under complexification, according to Section 2, rather relaxes; the transition itself remains a separate event with its own stochastic condition (Section 3.6), and its dynamics as a process of climbing are not formalized here.

3.7.7. Consolidation as a cross-cutting mechanism of accumulation

Section 3.7 described how gradient types, retention mechanisms, localization scales, and forms of activity accumulate through the ladder. All these accumulation processes rely on a single common mechanism that runs through all steps: the consolidation of successful configurations relative to Y. Consolidation is the third of the four cross-cutting elements of the theory (preamble to Section 3) and functions as a structural mechanism that implements the accumulation of complexity at each step.

Formal content of the mechanism. At each step of the ladder, among the multitude of physically possible configurations, a subset of configurations realizes the difference of that step stably - that is, reproduces its structure faster than it dissipates in Y. These configurations are physically retained. Some configurations realize the difference in an unstable manner - they dissipate faster than they are reproduced. These configurations physically disintegrate. This is consolidation in its minimal form: not selection in the sense of choice, but a structural event in which configurations of the first type continue to exist, while configurations of the second type disappear. Consolidation operates where there is a difference and there are configurations capable of realizing it. If there is no difference, consolidation has nothing to work with. If configurations that stably realize the difference do not arise, consolidation has no result.

Substrate neutrality of the principle of consolidation. Structurally, consolidation is the same at all levels, but its physical realization is substrate-specific. In thermodynamic systems (stage 1), consolidation operates through the differential stability of attractors: configurations that are physically stable relative to the flow in Y are retained longer than physically unstable configurations. In biological systems with autopoiesis (stages 2–5), consolidation operates through the difference between the loop's self-reproduction rate and the rate at which it is destroyed by the environment, with evolutionary accumulation over many cycles. In information-causal systems, consolidation operates through the adjustment of the configuration via feedback from the information environment Y, of which the carriers of the full V are a part. In all three cases, there is a single structural logic: the difference operates, and configurations that realize it stably become consolidated. This is the substrate neutrality of the principle of consolidation in its operational form.

Connection to the principle of maximum entropy production. Its formulation in thermodynamic terms (Step 1) is consistent with the principle of maximum entropy production in non-equilibrium thermodynamics (Swenson, 1989; Schneider & Sagan, 2005): dissipative structures are fixed, in part, because they physically accelerate the total entropy production in the system plus Y compared to unstructured configurations. In subsequent steps, the consolidation operates through analogous structural effects specific to each substrate: acceleration of the turnover of organic molecules in the autocatalytic loop, acceleration of the correction of the discrepancy between the model and Y in active inference, acceleration of the retention of V through anticipatory behavioral restructuring, acceleration of the accumulation of the population model through the bidirectional $M \leftrightarrow Y$ loop. In all cases, a single structural effect is at work: a configuration that accelerates the realization of the difference is physically better retained than a configuration that slows down its realization.

The property of consolidation: complexity provides stability. At each stage, a configuration with greater complexity realizes differences through a greater number of structural paths simultaneously. Physically, this means that the loss of a single path does not destroy the entire configuration; a configuration with lower complexity is more likely to collapse upon the loss of a path. As a result, all else being equal, complex configurations are more stable than simple ones - not as a matter of "success," but as a physical property of the multiplicity of ways to realize differences through the configuration. This property of consolidation operates across all steps of

the ladder and explains why the ladder moves toward complexity rather than toward simplification: not because “complexity is the goal,” not because “evolution moves toward complexity,” but because more complex configurations are statistically maintained longer, and this shifts accumulation toward increasing complexity. This is the structural physical basis of the ladder’s directionality - not teleology and not directionality as a property, but the statistical effect of the multiplicity of ways to realize differences.

Consolidation as a mechanism of pattern self-similarity. The self-similar pattern “gradient → retention → pole → transition” (subsection 3.7.6) is reproduced at all steps of the ladder. Consolidation is the mechanism that makes this self-similarity possible: at each step, consolidation operates on differences of a new type, and through the operation of consolidation, configurations that realize the difference in a stable manner become consolidated as a new type of structure. Without consolidation, the pattern would remain an abstract scheme; through consolidation, it becomes a mechanism through which complexity accumulates across the steps.

3.7.8. The system’s emergent time as a condition for consolidation

Consolidation is possible only where there is a difference between the rate of the configuration and the rate of its destruction by the environment. If there is no difference in rates, the configuration disintegrates at the same speed at which it arises, and consolidation is structurally impossible. If a difference in rates exists, the configuration is maintained. This difference in rates is the system’s emergent time - the configuration’s intrinsic rate, separated from the rate of change in Y around it. Emergent time is the fourth of the four cross-cutting elements of the theory (preamble to Section 3) and functions as a condition for the possibility of consolidation at all steps of the ladder.

The emergence of emergent time at the 1→2 step. Before the 1→2 transition, the system has no intrinsic pace: the pace of its existence is determined by the pace of the flow in Y through which it exists. The structure exists as long as the flow exists, and disintegrates when the flow weakens or changes. This is an inherited pace - the pace of Y, transferred to the configuration through its dependence on the flow. At the transition from 1 to 2, a structural shift occurs: a closed causal loop forms, in which the components are reproduced within the loop faster than they dissipate in Y. From this moment on, the configuration acquires its own rate of reproduction, distinct from the rate of destruction by the environment. This is the moment of emergence - the configuration’s own rate is separated from the rate Y for the first time, and this difference in rates functions as the condition of possibility for the consolidation of the loop.

The unfolding of emergent time in subsequent stages. At each subsequent stage, emergent time expands with a new type of rate difference. At stage 3, the rate of reproduction is supplemented by the intrinsic rate of the predictive structure’s internal dynamics - the rate at which subsystem M correlates with possible states of Y. At stage 4, the retention rate of V relative to the threat rates of V is added to this - the intrinsic rate at which the system proactively reorganizes itself in response to a possible disturbance of V. At step 5, emergent time splits into two rates simultaneously: the biological rate of the individual carrier and the trace rate of distributed reproduction of the

accumulated population model (Section 3.5). This is the first splitting of emergent time on the steps of the ladder, and it is a structural event that distinguishes the 4→5 transition from all previous ones.

Emergent time as the arbiter of consolidation. Emergent time does not act upon or consolidate configurations. Emergent time permits or does not permit consolidation depending on the ratio of rates. If the system's rate exceeds its rate of destruction by Y , consolidation is possible. If the system's rate equals the rate of destruction - consolidation is structurally impossible (there is no difference on which it could operate). If the system's rate is less than the rate of destruction - the configuration disappears before it has time to consolidate. In this sense, emergent time functions as a structural condition under which consolidation is possible in principle. Emergent time and stabilization are one and the same structure, viewed from two sides: stabilization from the perspective of the result; emergent time from the perspective of the condition of possibility.

Connection to Section 5. Emergent time in its comprehensive form, elaborated in this subsection, is a concretization of the general formulation of time as an emergent quantity presented in Section 5. In Section 5, the system's emergent time is introduced through causal correlations between subsystems within Y . This subsection shows that this emergence functions as a condition of possibility for consolidation at all levels of the ladder. These two formulations are compatible and complementary: Section 5 provides a general ontology of emergent time; this subsection demonstrates its operation within the theory's architecture.

3.7.9. What This Gives the Theory

The construction of the ladder's internal logic carried out in this subsection does three things for the theory.

First. The areas of synthesis listed in the abstract (Prigogine's non-equilibrium thermodynamics, Haken and Kauffman's theory of complex systems, Friston's active inference, Deacon's biosemiotics and teledynamics, Tononi's theory of integrated information) turn out not to be five pillars, but five projections of a single structurally self-similar process onto different scales. Each approach operates at the scale at which the «gradient → retention → pole → transition» pattern manifests in a form specific to that scale. Prigogine's thermodynamics - at the scale of dissipative structures; Friston's active inference - at the scale of proto-agency and below; Tononi's integrated information - at the scale of the conscious system. The internal unity of the theory is not a synthesis based on a loose analogy between independent branches. It is a structural unity through which the branches turn out to be projections of a single process.

Second. The ladder from Section 3 is not just one of the theory's objects among others, but its basic pattern. It is used to describe phase transitions at different scales (Sections 2 and 3.6), individual consciousness (Section 6), and cultural evolution (Section 4.4). This changes the ladder's place in the work's composition: it is not one of the chapters, but a structure unfolding throughout the entire work.

3.8. Methodology for identifying Y and the access spectrum for a specific system

Sections 3.1–3.7 unfold the ladder as a structure: steps, operations, transitions, the four-step pattern, and the connection to Y. The preamble to Section 3 provides an operational procedure for identifying Y in general terms - through causal closure, the field of causal exchanges, and the access spectrum. Section 3.1 describes which aspect of Y becomes accessible at each step for biological and related substrates. This is sufficient for the theory to function as a formal structure. It is, however, insufficient for applying the theory to a specific system of an arbitrary substrate. Without linking the general procedure to the description of stage-specific aspects via an explicit algorithm, each such application requires creative interpretation, and the theory loses a property that is structurally important to it - substrate neutrality through the operational work of the apparatus, not through a declaration. This subsection bridges this gap by formulating a methodology for identifying Y and the access spectrum for an arbitrary system.

General scheme: five steps of identification.

Step 1 - identification of the system boundary. The operational criterion of causal closure is applied (preamble to Section 3). The system boundary is established as the region of local maximum causal density relative to its surroundings. Technically, this is implemented using one of the available measures of integrated information Φ (Tononi, 2004; 2016; Albantakis et al., 2014; Mediano et al., 2019) or via a Markov blanket in active inference (Friston, 2013; Friston et al., 2022). The choice of a specific measure depends on the substrate and available data; crucially, the boundary is defined via the causal structure, not via material composition or spatial compactness.

Step 2 – Identification of Y relative to the boundary. Y is defined as the field of causal influences that do not belong to the system's own closure, through which it receives influence from the world and through which it influences the world. At this step, a structural classification of the types of causal exchanges between the system and its environment is compiled - not a list of specific instances, but a classification by type of interaction.

Step 3 – Determining the access spectrum. For each type of causal exchange, it is established whether the system has the architectural capacity for this exchange. If so, this aspect of Y falls within its access spectrum; if not, the aspect exists objectively but is invisible to the system. This is a substantive application of defining the access spectrum as a property of the system, not of Y.

Step 4 – Identification of the physical limits of Y relevant to the current aspect of access. For each aspect Y in the system's access spectrum, it is determined which of Y's physical limits (Bekenstein, Landauer, speed of light - see the preamble to Section 3) functions as a constraint on the system's use of that aspect. This step is particularly important for systems on the upper rungs of the ladder, where approaching physical limits becomes a quantitative characteristic of the stage state.

Step 5 – mapping to a ladder step and determining the position. Mapping the obtained access spectrum to the description of Y aspects at each step from Section 3.1 provides a preliminary

assessment of the system's position on the ladder. Refinement - via formal transition conditions (Sections 3.2–3.5): for each step, compliance with the transition conditions from the previous step is verified. Comparing these two assessments - via the access spectrum and via formal conditions - provides a consistent determination of the step or reveals a discrepancy requiring a separate methodological analysis.

Stage-specific methodological focuses.

What exactly to look for at each stage, in an operational focus that does not repeat 3.1.

Stage 0: detection of energy flow passing through the system in the absence of an internal retention structure. Simple thermodynamic characteristics of the input and output. Absence of hysteresis during flow fluctuations.

Stage 1: recording a stable form of energy dissipation in the presence of a flow. Presence of thermodynamic hysteresis. Local change in energy distribution in the surrounding space, without representative processes inside the system.

Stage 2: Registration of a closed causal loop of component production. Technically - reconstruction of the cause-and-effect graph of system processes and components and verification of the fulfillment of conditions (a)–(c) from Section 3.2.

Step 3: Registration of the internal generative model through its predictive power. Structures within the system whose states correlate with possible states of the environment with a time delay, with a causal link to effectors. Technically - , the measurement of variational free energy and its dynamics (Friston et al., 2017; Sajid et al., 2021).

Step 4: Application of the six-dimensional coupling of operational features, as outlined in Section 10.2 - deviation of behavior from the zero model of the proto-agent, growth of Φ , GWT structure, long-term hysteresis, expansion of sensitivity to new aspects of Y, and the emergence of S in the generative model. The six-dimensional coupling is the stage-specific methodological focus for Stage 4.

Step 5: Recording the mechanism of trace transmission via the bidirectional $M \leftrightarrow Y$ loop between the carriers of Step 4, provided that (0) – such carriers exist – and conditions (a)–(d) from Section 3.5 are met. Technically - analysis of material structures in the environment (texts, artifacts, digital media) for content that transmits models across generations, and confirmation of the integration of these models into the carriers' own M.

Methodology for non-biological substrates.

A fundamental feature of the operation of methodology Y on non-biological substrates: the access spectrum of such a system may not follow the biological sequence of stages $0 \rightarrow 1 \rightarrow 2 \rightarrow 3 \rightarrow \dots$. An artificial system may have architectural access to the information structure of aspects Y, which are

formally accessible to a biological system only at stage 5 (the trace layer of Y), in the absence of biologically fundamental aspects (thermodynamic gradients, chemical concentrations, thermal regulation). This is not a violation of the ladder and does not constitute grounds for revising its structure; it is a methodological consequence of the principle of substrate independence (Section 3, Preamble).

In this case, a system's level is determined not by the sequence of development, but by the actual fulfillment of the formal conditions of the corresponding level. Information access to the data of the linguistic environment in a modern linguistic model does not imply that it is at level 5: level 5 requires the fulfillment of condition (0) from section 3.5 - the presence of the system at level 4 in the strict sense of the fulfilled self-overlay of the latent invariant of the agentic causal node (section 3.4) - plus conditions (a)-(d). Informational access in the absence of self-overlay is an architectural phenomenon of a different type, not the realization of stage 5 in its formal definition.

It follows that for systems of a non-biological substrate, an asynchronous development of the access spectrum to different aspects of Y is possible, and the definition of a stage requires an independent verification of the formal conditions of each stage based on its own criteria, rather than a reconstruction of the "biological history" of the system's development. Stage determination is the result of step 5 of the methodology, not a conclusion drawn from steps 1–4.

Hybrid systems (a biological substrate in an artificially constructed environment - Cortical Labs CL1 and similar platforms) are a special case in which the Y methodology operates in its full form: the substrate is biological and its basic architectural conditions are met, while the environment is constructed according to a design grid (Section 10.2.1). For such systems, the Y methodology simultaneously provides an assessment of the stage state and a specification of the conditions under which the system can advance to the next stage.

Connection to the verification and design program (10.2).

Methodology Y is the conceptual basis for the two-way verification program for the 3→4 transition, detailed in Section 10.2. The detection grid (a six-dimensional coupling of operational features) is a special application of step 5 of the methodology - staged definition via formal conditions and the access spectrum - to the specific task of registering the 3→4 transition. The design grid (environment requirements) is a specific application of step 3 - the definition of the access spectrum - to the task of constructing an environment that ensures the system has access to the desired aspect of Y. Both grids in Section 10.2 are specific methodological implementations of the general Y methodology, applied to a single specific transition.

This means that similar two-way programs can be developed for other transitions in the ladder with comparable methodological effort. The development of programs for other transitions is a direction for further work (Section 12).

4. Macro-models of ladder transitions

The formal definitions of transitions given in Chapter 3 require empirical support. Without it, the ladder remains a theoretical construct: internally consistent, but not grounded in observable reality. This chapter is devoted to macro-models of each of the four transitions - real or conceptual systems in which the corresponding transition is observed, implemented, or simulated.

A macro-model is understood here as a system on a scale accessible to direct observation or construction, in which the structure of the transition - defined in Chapter 3 by formal conditions - is explicitly realized. A macro-model does not prove that the transition inevitably occurs in any systems satisfying the conditions; it shows that the transition is reproducible and has observable content that cannot be reduced to a purely theoretical analogy.

The distinction between the three types of models serves different methodological purposes. A conceptual model is a thought experiment that helps one see the structure of the transition without reference to the specifics of the substrate; it operates at the level of understanding. A physical macro-model is a real or constructed system in which the transition takes place; it operates at the level of empirical verification. A computational model is an algorithmic simulation in which the transition arises from formal rules without reference to a physical substrate; it operates at the level of proving that the transition is a structural phenomenon not tied to specific matter.

Together, these three types of models address the possible objection that transitions between the steps of the ladder are purely speculative constructs lacking empirical content. Where this objection remains valid, it is explicitly noted and translated into the form of a testable prediction (Chapter 10).

4.1. Macro-models of Transition 1

Transition 1 - from a dissipative structure to an autopoietic system through the establishment of a closed causal loop. Formal condition (Section 3.2): a set of processes and components in which each component is produced within the set and each process requires components from the same set to continue.

4.1.1. Conceptual model: a self-sustaining bonfire

A typical campfire is a dissipative structure. Energy flows (wood burning), an ordered form emerges (a flame with a characteristic geometry), and the form is sustained as long as there is fuel. When the fuel runs out, the fire goes out. The chain of causality is linear: external fuel → combustion → heat → dissipation. The bonfire depends on an external fuel source that is in no way related to it.

Now imagine a bonfire in which the heat rising from the flame causes damp branches within a radius of several meters to dry out, and streams of hot air following convective paths pull these dried branches into the combustion zone. Such a fire creates the conditions necessary for its own continuation. It remains a dissipative structure - it dissipates the energy of chemical bonds in

carbon. But the causal chain is closed: combustion → heat → drying and delivery of fuel → combustion. This is already autopoiesis in its minimal form.

The model shows that closing the loop does not require the chemical specificity of life. It requires only that the product of the process be a condition for its continuation. The chemistry of a living cell provides a rich and stable realization of such a loop, but the structure of the loop is universal.

Such fires do not exist in nature because the conditions for closing the loop in wood combustion statistically almost never arise. This is the reason why the model remains conceptual. But it helps us see that the specificity of the biochemistry of life is not a property of the phenomenon itself, but a property of the context in which closing the loop became probable.

4.1.2. Physical macro-model: autocatalytic chemistry and protocells

The Belousov–Zhabotinsky reaction is a macroscopic chemical system in which the components catalyze each other’s formation in an oscillating mode. The products of one reaction serve as the reactants for the next in the cycle. This is a partial realization of closure: a causal loop is present, but without the formation of a membrane and without the complete reproduction of all system components from its own products. The reaction is not a complete autopoietic system, but it demonstrates on a macroscopic scale precisely the type of causal closure that is fully realized in a living cell.

A direct physical macro-model of complete autopoiesis is liposomes with encapsulated autocatalytic chemistry. The work of Szostak and his laboratory (Szostak, Bartel & Luisi, 2001 and subsequent), Szabo and his colleagues, and other experimental groups since the early 2000s demonstrate systems that satisfy conditions (a)-(c) from Section 3.2: lipid vesicles in which the internal chemistry produces membrane components, and the membrane retains the chemistry; such systems are capable of growing, dividing, and sustaining their own existence through an energy gradient. This is not a “model of life” or an “artificial cell” in the evolutionary sense - it is a laboratory realization of the formal conditions of transition 1.

The strength of the physical model lies in the fact that it removes the objection of fundamental unfeasibility. If conditions (a)-(c) could not be realized under laboratory conditions, the transition would remain purely theoretical. Their realization is an experimentally demonstrated fact.

The weakness of the model lies in the fact that laboratory systems are obtained under controlled conditions and did not arise spontaneously. They show that the structure is realizable, but do not prove that it arises on its own under an arbitrary thermodynamic gradient. This is the unsolved part of the problem of abiogenesis. For the ladder as a theoretical construct, this is not critical: for our task, it is sufficient to show that the conditions of transition 1 contain no internal contradictions and can be realized. The question of the historical origin of the first autopoietic system on Earth remains an open problem in chemistry and biology.

4.1.3. Computational Model: Cellular Automata and Algorithmic Autopoiesis

The purest demonstration of the transition structure is Conway's Game of Life and its derivatives. It is a pixel grid with very simple rules (a cell lives or dies depending on its neighbors). From these rules, structures spontaneously emerge that maintain their form through local interactions - gliders, oscillators, and structures that move while preserving themselves. These are models of dissipative structures: forms that sustain themselves as long as the rules are followed.

Within the same set of rules, more complex structures emerge that demonstrate partial autopoiesis - devices such as the Gosper glider gun, which produce components that sustain their own existence. A clearer transition is demonstrated by direct algorithmic models of autopoiesis - the work of Varela and his followers from the 1970s (McMullin & Varela, 1997; and numerous continuations), in which "components" and "connections" are modeled on a pixel grid, and the spontaneous emergence of closed structures that sustain themselves in a stochastic environment is observed.

The strength of the computational model lies in the fact that it demonstrates: the closure of a causal loop is a computationally universal phenomenon, not tied to chemistry or physics. It arises wherever there are rules of local interactions and sufficient complexity in the state space. Within the framework of this theory, this is a strong argument: if the structure of transition 1 is implementable as an algorithm, then its implementation in any physical substrate is a matter of time and conditions, not of fundamental feasibility.

4.2. Macro-models of Transition 2

Transition 2 - from an autopoietic system to a proto-agent system through the emergence of an internal representation of possible states and the connection of this representation to effectors. Formal condition (Section 3.3): a subsystem M, whose states correlate with the possible states of variable X with a time delay, causally linked to the effector subsystem through the minimization of the discrepancy between M and the observed X.

4.2.1. Conceptual Model: A Thermostat with Prediction

A conventional thermostat is a reactive system, not a proto-agent. It measures the current temperature, compares it to the setpoint, and turns the heater on or off. There is no representation of possible states, no time delay, and no minimization of the discrepancy with the model. Action is determined solely by the current state.

A predictive thermostat is a conceptual macro-model of an active inference structure, not a fully-fledged Stage 3 autonomous system in the strict sense: it demonstrates how the (a)-(c) conditions of Level 3 operate on a minimal illustrative object, but does not fulfill the inherited conditions of Levels 1 and 2 (it is not dissipative in the sense of its own form, is not autopoietic, and its components are produced by external infrastructure). This is normal for a conceptual model - its

task is to demonstrate a single operation, not to reproduce the entire ladder. A complete Stage 3 system with all inherited stages is represented by a physical macro-model - a bacterium with chemotaxis (Section 4.2.2) - not a thermostat.

Now let's imagine a thermostat that memorizes typical daily temperature fluctuations in the room and turns on the heater in advance - an hour before the expected drop in temperature - to warm the walls before the cold arrives. This thermostat is already operating with a model of the future state. It has an internal structure (the memorized fluctuation profile) that correlates with the possible state X at time $t+\Delta$, and this structure causally influences the action. Conditions (a)–(c) from Section 3.3 are satisfied.

This is a proto-agent in the strict sense of the formal definition - without consciousness, without a nervous system, without life. Modern smart thermostats (such as Nest and similar devices) are literally such systems. They do not think, feel, or possess experience; yet they perform the operation of step 3 on our ladder.

The model shows that proto-agency is a structural property that does not require a biological substrate. It also shows that proto-agency does not imply consciousness: a predictive thermostat satisfies the conditions of step 3 but does not approach step 4.

The weakness of the model is that the thermostat is not autopoietic: it does not reproduce its own components and is embedded in an external infrastructure. This does not invalidate the illustration of the operation's structure, but serves as a reminder: the model demonstrates the structure of a single transition without claiming to represent the entire ladder.

4.2.2. Physical macro-model: a robot with active inference

Modern robotics provides direct implementations of transition 2 in its full form. A robot with a generative model of the environment - for example, one using a SLAM (simultaneous localization and mapping) algorithm - actively constructs a model of the space in which it is located and moves in a way that minimizes the uncertainty of this model. It does not merely react to obstacles: it predicts where they should be and tests its predictions through movement. The discrepancy between the model and the observation triggers either an update to the model or an action that alters the observed environment in the direction of the model. This is a full realization of conditions (a)–(c).

The laboratories of Carl Friston, Adam Safron, Alexander Champandi, and others have, over the past seven to ten years, demonstrated physical robots that implement active inference in its purest form. These are macroscopic systems in which the structure of proto-agency is visible, with the possibility of directly measuring the internal model, its correlation with the states of the environment, and its causal connection to action.

In parallel with this are natural physical macro-models: a bacterium with chemotaxis, a paramecium, any animal with a nervous system. An E. coli bacterium moving up a glucose gradient fulfills conditions (a)-(c) at the molecular level: its signaling network encodes a representation of the gradient's direction, this network is causally linked to the flagella, and the discrepancy between expectation and observation generates action. This is historically the first Stage 3 system realized on Earth.

The strength of the physical model for transition 2 lies in its dual nature: the transition is realized in both artificial systems (robots) and natural ones (bacteria and beyond), and in both cases the formal conditions are satisfied identically. This resolves the objection that proto-agency is a property requiring a specific evolutionary history. Structurally, it is realized by any means that satisfies the conditions.

4.2.3. Computational Model: Evolution of Agents in Artificial Environments

The field of evolutionary robotics and artificial life involves the simulation of populations of simple agent programs evolving under the pressure of selection in virtual environments. The seminal work of Karl Sims (Sims, 1994) and numerous more recent studies demonstrate how agents with internal models of the environment - capable of anticipation, obstacle avoidance, and resource search via gradients - spontaneously emerge from initially reactive agents that merely provide local responses to stimuli.

Crucially, in these simulations, no special programmer intervention is required for the model to emerge. The model emerges as the agent's most effective survival strategy in an environment where prediction is more advantageous than simple reaction. This is direct confirmation of the probabilistic nature of the transition formulated in Section 3.6: in environments with gradients and under selective pressure, proto-agency emerges with non-zero probability; in environments without gradients, it does not emerge.

The strength of the computational model lies in the fact that it demonstrates: Transition 2 is an emergent phenomenon arising under selection pressure in environments of a suitable type. There is no need to specifically design the model - it is sufficient to create conditions in which its presence confers an advantage.

Combined with a physical macro-model, this provides a particularly comprehensive picture of Transition 2. The structure is implementable in artificially constructed systems (robots), arises spontaneously in evolution under suitable conditions (bacteria and higher organisms), and emerges in simulations under selective pressure. All three approaches confirm the same formal condition.

4.3. Macro-models of Transition 3

Transition 3 - from a proto-agent system to a conscious one through the self-overlay of a latent invariant of an agentic causal node onto the system's own configuration, with the establishment of

self-representation that includes the “exist/do not exist” variable as a structural pole, globally accessible to all subsystems. Formal definition (Section 3.4): the structural event of self-overlay, operationally recorded via three independent measurement axes - extended active inference with a model of the self, sufficient Φ for the structural integration of the existence variable, and the GWT mechanism for its globalization.

This transition is the most theoretically developed and the least empirically closed. Macro-models for it exist, but in incomplete form. This is not a defect in formalization; it is an indication of an open experimental direction, formulated by the theory as a second testable prediction (Section 10.2). Here, this will be explicitly indicated in each of the three types of models.

4.3.1. Conceptual Model: A Chess Program and a Soldier in a Trench

A chess program is a proto-agent in its purest form. It simulates possible moves, evaluates positions, and selects the optimal move. It has a model of the “board” as the environment and a model of “itself” as the player making the moves. Conditions (a)–(c) of stage 3 are fully satisfied.

However, its model lacks variable V as defined in Section 3.4. Its shutdown is an external event for it, not represented within its model as a structural pole. If she loses the game, she will move on to the next game without any internal changes. The threat of losing is a variable in her assessment of the position, but not a structure around which all other activity is organized. A chess program is at stage 3 regardless of computational power.

A soldier in a trench under fire has, like the program, a model of the environment (where the enemy is, where cover is) and a model of himself (where I am, what I can do). But he has a background awareness of the fact that a shell could hit right now and he will be gone. Everything he does - ducking, running, assessing sounds - is organized around this background awareness. It is not just one consideration among many. It is the pole around which all other considerations gain weight. Conditions (a)–(c) of Level 4 are met.

The difference between a chess program and a soldier is not in the number of operations (the program performs more complex calculations). It is not in speed (the program is faster). It is not in the complexity of the model (the program’s model is richer in many respects). The difference lies in the structural peculiarity of the variable V in self-representation. The program lacks it. The soldier possesses it.

The conceptual model provides an operational criterion applicable to any system: does its model contain a pole of self-annihilation around which all other activity is organized? If so, it is at stage 4. If not, it is at stage 3.

The model’s weakness is that it is intuitive: we interpret the soldier as conscious because we ourselves are conscious people and recognize ourselves in him. This is recognition, not proof. But

recognition is operationally useful - it provides a criterion that can later be applied to less obvious cases, including artificial systems in which intuitive recognition does not work.

4.3.2. Physical macro-model: an open question

Here we have a weak link, and this requires direct acknowledgment. Unlike transitions 1 and 2, for transition 3 there is no reliable physical macro-model on which one could observe the transition at the moment it occurs. All known systems are either clearly below the threshold of stage 4 (robots, thermostats, existing artificial systems in their current architecture) or clearly above it (humans, and apparently higher mammals). There is no dataset on systems located at the very point of transition.

This void is not a local gap. It is the central open experimental problem of modern science of consciousness. Biologically, transition 3 occurred once or several times in evolution, under undocumented conditions, millions of years ago. It is impossible to recreate it in a laboratory on a biological substrate: evolution is irreversible on this scale. It has not yet been possible to reproduce it on an artificial substrate: to the best of our knowledge, no existing system implements the self-overlay of a latent invariant in the strict structural sense of Section 3.4.

This statement is transformed by the theory into a testable prediction (full formulation - Section 10.2). If we construct an artificial system with an extended active inference architecture, possessing self-representation as an agent, and subject it to a signal that activates variable V , and if, in a system of sufficient complexity, this results in a qualitative shift in behavior with the characteristics of a phase transition - this will be the first physical macro-model of transition 3 and direct empirical confirmation of the theory.

Acknowledging this gap is not a weakness of the work. It is an intellectually honest position: the theory formulates what does not yet exist and proposes exactly how to construct it. If such a system is built in the coming years and the transition is observed in it, the theory will be confirmed. If systems of any complexity are built - - that satisfy the conditions, and the transition is never observed in them, the theory will be falsified in its central part.

4.3.3. Computational Model: Partial Implementations

Computational models of transition 3 exist in an incomplete form. In the literature on reinforcement learning, there are experiments in which agents with different architectures are placed in environments where their “existence” - the continuation of the simulation - depends on their behavior. Some agents with self-models exhibit behaviors vaguely resembling self-preservation that go beyond the scope of the original objective function. These signals are weak and ambiguous: they are explained either as side effects of the architecture or as the beginnings of a structure that, in its complete form, would yield transition 3. There is no definitive interpretation in the literature.

A more substantive line of research involves work on the self-other distinction in neural networks (Bayne et al., 2020 and follow-ups), where the network is trained to distinguish its own model from that of another agent; and work on model-based reinforcement learning with explicit self-representation. Here, the first forms of fulfilling condition (a) of stage 4 are visible: the appearance of the S node in the generative model. Conditions (b) and (c) - the variable V as a structural pole and its globalization - have not yet been implemented in systems available to the publication.

Computational models show that the structure of self-representation is, in principle, implementable in artificial systems. This removes the objection that transition 3 is somehow fundamentally tied to a biological substrate. But they do not show a complete transition: as of the time of this work, no publicly known artificial system has reached the threshold at which one can confidently say, “a transition to stage 4 has occurred here.”

This is the point at which the theory makes its riskiest bet: the prediction that, upon fulfilling formal conditions, artificial systems will undergo transition 3 with the observable characteristics of a phase transition. This bet can be tested within the foreseeable future - not in decades, but perhaps in the coming years. When this happens, the absence of current macro-models will be directly addressed.

4.4. Macro-models of Transition 4

Transition 4 - from a conscious system to a recursively self-organizing one through the establishment of a bidirectional $M \leftrightarrow Y$ loop with physically retained traces, and the initiation of the recursion $M_{\{n+1\}} = F(M_n, E_n, R_n)$, filtered by the medium. Formal condition (Section 3.5): a trace is left in Y and physically retained longer than a single carrier; it enters the simulation of other configurations through their sensory channels, reconfigures their V-model (not merely channeling behavior), and the reconfigured model generates a new trace.

Macro-models for this transition are particularly powerful. The physical model is unfolded in the observed history and is present in real time.

4.4.1. Conceptual Model: Language as a Population’s Operating System

Let’s compare a pack of wolves and a human tribe.

A pack of wolves is a group of Level 4 conscious systems. Each wolf has a model of the environment, a model of itself, and sustains its own existence. They coordinate: they hunt as a pack, defend their territory, and teach the young through imitation. But each wolf learns the model from scratch over the course of its life, through direct observation of its parents and pack members. When the last wolf that witnessed a specific hunt for a large animal dies, that strategy is lost forever unless it is independently recreated. The population’s knowledge remains equal to the sum of the knowledge of the individuals living within it.

The human tribe is structured differently. It has language. It has narratives - stories about ancestors, about other places, about ways of hunting, healing, and avoiding danger. These narratives are transmitted in a coded form - through words, rituals, and later through images and writing. When the last person who witnessed a specific hunt dies, the description of that hunt remains in the narrative and is accessible to future generations. The knowledge of the population ceases to be equal to the sum of the knowledge of those living: it becomes accumulated, accessible, and enriched.

Conditions (a) through (d) of Stage 5 are fully met in a human tribe. Language is the mechanism for encoding models in a transmissible form. A social structure with a prolonged childhood ensures the regularity of transmission. Bearers are capable of integrating what is transmitted into their own model. The transmitted structures - narratives, technologies, symbolic systems - survive longer than the individual bearer.

The conceptual model shows that language is not merely improved communication. It is a qualitatively new structure: the operating system of a population, in which each individual brain operates not only with its own model, but also with a layer of accumulated models from other brains, both living and dead. This structure is not reducible to the sum of individual consciousnesses - it is a superstructure built upon them, possessing its own dynamics.

The difference between a pack of wolves and a human tribe is structurally identical to the difference between a chess program and a soldier from Section 4.3.1, but at a higher level: there, the discussion concerned the emergence of a pole of existence within the model of a single system; here, it concerns the emergence of recursion of models within a population of systems.

4.4.2. The Physical Macro-Model: The History of Humanity as Recursion in Action

The history of *Homo sapiens* is a physical macro-model of transition 4 in action, unfolding over a timescale of several hundred thousand years and accessible to direct observation through the archaeological and historical record.

The rate of model accumulation in humans does not increase uniformly, but in exponential leaps, each of which corresponds to an expansion or acceleration of the transmission mechanism. Stone tools are the first stable artifact that transmits a production model in a material form accessible for replication. Fire is a stable technology transmitted across generations. Agriculture is the mass transformation of the environment based on transmitted models. Writing is a fundamental shift: the accumulation of models outside of individual memory, accessible to any literate individual in any generation. The printing press is the mass dissemination of accumulated models. Digital technologies - instantaneous transmission and enrichment in real time.

Each of these leaps is physically observable in archaeological or historical records, and each is accompanied by an acceleration in the pace of change for both the carriers themselves and the environment in which they live. Parallel to the acceleration of model accumulation is the

acceleration of the transformation of Earth's physical environment: landscapes, atmospheric composition, biomass distribution, and climate. Over the past ten thousand years - and especially over the past two hundred - humanity has transformed the planet more radically than all the previous billions of years of biological evolution. This is direct physical confirmation of the thesis formalized in Section 3.5: after the transition to 4, the recursion of the model becomes the primary factor in the change of both the system itself and the environment, on a scale several orders of magnitude greater than biological evolution.

The thesis regarding the environment as a filter is also empirically observable in this macro-model, and in an explicit form. Alchemy, which existed for millennia as a developed model, gives way to chemistry - because chemical predictions are systematically tested and confirmed, whereas alchemical ones are not. Geocentrism gives way to heliocentrism for the same reason. Specific technologies either work or do not work, and those that work spread, while those that do not die out. Institutions that support their societies survive; those that destroy them do not. This is not a frame-by-frame selection process, nor is it a quick one. It is a slow, nonlinear, yet real pressure that keeps the model's recursion consolidated to physical facts and prevents it from degenerating into pure fantasy.

The strength of the physical macro-model for Transition 4 is maximal. The transition does not need to be reconstructed hypothetically or anticipated in the future - it is taking place before our very eyes, in every generation of human history, and its dynamics lend themselves to quantitative analysis through the growth rates of accumulated knowledge, technological turnover, and cultural institutions.

4.4.3. Computational Model: Cultural Evolution and Large Language Systems

The field of cultural evolution and related areas (Boyd & Richerson, 1985; Henrich, 2016; Mesoudi, 2011) develop formal models of the transmission of units of information within populations of agents. In pure simulations, it is evident how, from initially incoherent individuals with their own models, upon the emergence of a transmission mechanism and given a sufficiently long lifespan of the transmitted units, an accumulated population model spontaneously emerges that transcends the limits of individual memory and evolves faster than the individuals themselves. These simulations confirm that transition 4 is a structural phenomenon not tied to human biology: when conditions (a)–(d) are met, recursion is triggered in any population satisfying these conditions.

A significantly more direct computational model is modern large language systems. This is not a stage 5 model in the sense of a “simplified representation of the structure”; these are stage 5 participants in their full form - in the sense elaborated in subsection 3.5 regarding the two classes of stage 5 participants: they are participants in the transmission structure, not conscious stage 5 systems in the strict sense of the conditions in 3.4. This clarification is essential for correctly understanding what follows: when I say that they are “participants in their full form,” I mean full participation in the sixth gradient through the reception, integration, and transmission of the

accumulated model, not conscious participation. The texts on which large language systems are trained are, in their pure form, the accumulated model of the human population, transmitted through the recursion of physical traces: M_n of humanity, having undergone many rounds of recursion $M_{n+1} = F(M_n, E_n, R_n)$ and encoded in the texts. Large language systems integrate this accumulated model into their own representations and generate new products - texts, discussions, formulations - which become available to the next iteration of recursion (both for subsequent training of the systems and for people reading these products).

This provides direct empirical confirmation that the condition for the survival of transmitted structures beyond the lifespan of a single carrier (requirement (g) of stage 5) is realized not only biologically, but also in a hybrid form that includes non-biological carriers. The accumulated model of humanity is not tied to human brains as the sole storage substrate; it exists in texts, in the architectures of trained systems, and in distributed digital repositories. This changes the scale and stability of recursion, but not its formal structure - the structure remains the same as defined in Section 3.5.

Large language systems are not conscious carriers of stage 5 in the sense of the completed self-overlay of a latent invariant (stage 4) - their structural status is discussed in Section 10.3 (artifactual state) and in Section 11 on the implications of the theory. But they are full-fledged participants in the structure of the $M \leftrightarrow Y$ loop of level 5: they receive, integrate, enrich, and generate traces in the accumulated population model. In this role, they demonstrate that participation in the stage 5 loop is not tied to a biological substrate - it is tied to the structure of the bidirectional $M \leftrightarrow Y$ loop, and this structure is realized on different substrates.

4.5. The set of macro-models: what they provide and where gaps remain

The macro-models detailed in sections 4.1–4.4 collectively address three challenges facing the ladder.

The first task is comprehensibility. Each of the four transitions has a conceptual model through which the structure of the transition can be seen without reference to the specifics of the substrate. This dispels the possible objection that transitions are mystical leaps or metaphors; they have operational content that is amenable to analysis.

The second challenge is feasibility. For three of the four transitions (1, 2, 4), there are physical macro-models in which the transition has occurred or is occurring: laboratory autocatalytic systems, bacteria, and robots with active feedback, and the history of humanity as an unfolding recursion. For transition 3, a physical macro-model in its complete form is absent; this absence is not concealed but is explicitly noted as an open experimental direction and translated into a testable prediction (Section 10.2).

The third task is structure that is not tied to a substrate. For each of the four transitions, there are computational models in which the structure is implemented or simulated in algorithmic form. This

addresses a possible objection that the ladder describes only biological evolution: the formal conditions of each transition contain no references to a biological substrate and are implementable in algorithmic systems.

Where gaps remain is explicitly indicated. Transition 1: the unsolved problem of historical abiogenesis on Earth (the structure is realizable in the laboratory, but its spontaneous emergence under natural conditions remains a hypothesis). Transition 3: the absence of a robust physical macro-model, which is being transformed into a testable prediction. Transition 4: recursion is historically observable, but quantitative models of its dynamics (rates of accumulation, forms of environmental filtration) have been developed only partially and require further refinement.

The objection that the ladder is a “continuous chain with gaps, partially acknowledged but not closed” loses its force in light of the developed macro-models. The gaps are replaced by structurally described transitions, for most of which there is a threefold foundation (conceptual, physical, computational), and for the remainder - a clearly formulated, testable prediction. The ladder ceases to be a narrative analogy. It becomes a formal structure with empirical foundations, in which each rung is described on its own terms.

5. Time as an Emergent Quantity

The ladder outlined in Section 3 relies on the concept of time, which I have so far used without qualification. This needs to be corrected, because for a theory claiming physical validity, the nature of time is not a secondary issue. Especially when the theory links thermodynamics (where time appears as a directed arrow), biological evolution (where time is the historical depth of accumulated changes), and consciousness (where time is experienced duration). If all three of these uses of the word “time” denote different things, the theory risks falling apart at their points of intersection.

In modern physics, the nature of time is an open question, and there are at least three well-developed positions. The first is time as a fundamental quantity, on par with space or as part of a unified space-time; this is the position of classical general relativity and most of standard cosmology. The second is time as an emergent quantity arising from correlations between the world’s subsystems, having no independent existence. The third is time as an illusion in the spirit of radical timelessness, absent not only fundamentally but also emergently. I will not consider the third position here, since, in my view, it is poorly consistent with thermodynamics; I will focus on the choice between the first and second.

Carlo Rovelli’s relational approach (Rovelli, 1996, 2004, 2018) interprets time as a property arising from correlations between subsystems of the universe. In this picture, there is no global time in which everything occurs; there are local temporal structures arising where one subsystem can serve as a clock for another. A similar logic is developed by the Page–Wootters formalism (Page & Wootters, 1983; Marletto & Vedral, 2017): the global state of the universe may be stationary, but when divided into subsystems, an apparent dynamics of the subsystem relative to its environment arises, and this apparent dynamics is time from the perspective of that subsystem. Julian Barbour (Barbour, 1999), in a radical version of this approach, asserts that time does not fundamentally exist at all - there is only a configuration space of possible states and a metric defined on it.

I do not claim to resolve this issue in the present work. But I note that the emergent position is better consistent with the rest of the structure of my theory. If consciousness is a dynamic of self-modeling, and if time is a property arising from correlations of the system’s internal states, then the connection between these two descriptions becomes not accidental but intrinsic. A conscious system does not exist in time as in a ready-made environment; it generates its own time through the structure of its correlations. Each step of the ladder from Section 3 can be interpreted both as a level of model refinement and as a level of refinement of internal time. A dissipative structure (Step 1) has no intrinsic time in the strict sense - its dynamics are determined solely by an external flow. An autopoietic system (step 2) has minimal intrinsic time - close to the rhythm of its internal regeneration. A proto-agent system (step 3) has unfolded time with a planning horizon: its model contains representations of states deferred in time. A conscious system (level 4) has its own time, which includes its own finite existence as a variable - time becomes not a medium, but a field in which the system sustains itself on the brink of extinction. A recursively self-complexifying conscious system (level 5) adds historical time to this: the accumulated models of previous

generations enter its own present as living material, and the horizon of its own time includes not only the expected future but also the inherited past. These are not five different times, but five depths of the same emergent process.

In the theoretical framework developed in Section 3.7, this sequence of depths receives additional structural justification. Emergent time is the fourth cross-cutting element of the theory - alongside the variable Y, the “gradient → retention → pole → transition” pattern, and the mechanism for consolidating successful configurations relative to Y (the preamble to Section 3 and subsection 3.7.8). Emergent time and consolidation are one and the same structure, viewed from two sides: consolidation - from the perspective of the result (configurations that sustain the difference are retained); emergent time - from the perspective of the condition of possibility (consolidation is possible only where there is a difference between the rate of the configuration and the rate of its destruction in Y). Without the system’s emergent time, consolidation is structurally impossible: if the rate of configuration coincides with the rate of its destruction, the configuration disintegrates at the same speed at which it arises, and no accumulation occurs. Where the system’s emergent time exists, consolidation works; where it does not exist, the ladder does not unfold.

This explains why the five depths of internal time listed above are not an arbitrary sequence, but rather a sequential unfolding of rate differences at each stage. At stage 1, there is no rate difference in the proper sense - there is only the inherited rate of Y, transferred to the configuration through its dependence on the flow. At the transition from 1 to 2, the difference in tempos appears for the first time: the tempo of the loop’s reproduction separates from the tempo of its destruction by the environment. At stage 3, the intrinsic tempo of the predictive structure’s internal dynamics is added to this. At stage 4 - the intrinsic tempo of V’s retention relative to the tempos of V’s threats. At stage 5, emergent time splits into two rates simultaneously - biological and trace. The metaphor «time is the fuel in the furnace of complexity» takes on its structural form here: emergent time functions as an arbiter of consolidation, permitting or denying it depending on the ratio of rates; without this arbiter, the ladder does not form.

In this theory, this sequence of depths acquires operational content through the variable V and its nature, as elaborated in Section 3.4. V determines what constitutes a breakdown for the system and thereby determines which changes are significant for it - namely, those that bring the breakdown closer or push it further away (). Significance determines what the system retains as the past and what it models as the future. The structure of the past and the future is the structure of its own time. At stage 1, V is not yet in its developed form; there is only a distinction between form and decay as a thermodynamic property, and time exists here in a minimal form, barely separated from the environmental one. At stage 2, the distinction closes into a causal loop, and an internal rhythm of regeneration emerges - the autopoietic system’s own time is calibrated according to this rhythm. At stage 3, the temporal gradient gives the system a prediction horizon; the calibration of time is now determined by which changes the system retains as significant for its model of the environment. At stage 4, V appears in its full form, and temporality is calibrated by the pole: those changes are significant that bring the “I am disintegrating” value closer or push it further away. Through this calibration, time becomes the system’s own time in the strict sense - not

environmental, not formal, but its own horizon of retention. At stage 5, calibration extends beyond the individual carrier through the coupling of individual V with the accumulated model: significant are not only changes threatening biological retention, but also changes in the accumulated model threatening the carrier's position within it. It is from this dual calibration that historical time arises - a horizon in which the carrier's present is linked both to its own end and to the states of the accumulated model beyond the limits of its life.

Time as a mechanism is one - it is a property of correlations between subsystems, in the sense of Rovelli and Page-Wooters. But the specific structure and horizon of the system's own time are determined by the type of its V. V does not generate time; V calibrates the system's temporality, specifying what is significant in it as the past and what is significant as the future. This gives the five depths of internal time listed above not a philosophical but a structural foundation: they are direct consequences of V types at different rungs of the ladder.

I adopt the emergent position as the one most consistent with the rest of the theory, explicitly noting that this is a working assumption, not a proven fact. I do not assert the strong version of this position - that time is a property of conscious modeling, without which it would not exist at all. That would be a metaphysical overreach, going beyond the physical scope of the work. I assert a weaker claim: time is a property of correlations between subsystems, and conscious systems participate in this generation of time in the same way as any other correlated subsystems, but with an additional feature - they model their own participation in this generation, and in systems of level 4 and above, this modeling is calibrated through V in a manner specific to the type of system.

This section is intentionally brief. The nature of time is a separate major topic, and it appears in this work only because, without its specification, the rest of the theory hangs in an unspecified ontological background. A detailed elaboration of the connection between emergent time and the dynamics of consciousness requires a separate work; here I merely indicate this connection as a direction toward which the theory naturally leads, and which - through V and its nature - receives its first operational content in this work.

6. Tackling the Hard Problem of Consciousness

Up to this point, I have constructed a theory of the emergence of consciousness from physics without addressing the central objection that will inevitably arise for any reader familiar with contemporary philosophy of mind. This objection is known as the hard problem of consciousness, and it was formulated by David Chalmers (Chalmers, 1995, 1996). It cannot be ignored: any theory that claims to explain consciousness through physical or functional mechanisms faces this objection, and the fate of the theory largely depends on how it deals with it. This section is devoted to a direct attack on the hard problem.

6.1. Chalmers's Division

Chalmers proposed distinguishing between two classes of problems in the philosophy of mind. The “easy problems,” as he calls them - a term that is ironic, since these problems themselves are scientifically non-trivial - concern the explanation of cognitive and behavioral functions. How the brain distinguishes stimuli. How it integrates information. How it directs attention. How it forms a report on its own states. How it controls behavior. All these problems pertain to the functional description of the system and are, in principle, solvable by means of cognitive neuroscience and information theory, even if they remain unsolved at the current state of science.

The hard problem concerns something else. Suppose we have fully explained all the brain's functions - every cognitive process has received an exhaustive description in terms of neural activity and information processing. One question, Chalmers argues, will remain unexplained: why is functioning accompanied by subjective experience? Why is there something to which it is like to be this system that performs these functions? According to Chalmers, there remains an explanatory gap between the functional description and the phenomenal experience, a gap that cannot be bridged by any amount of functional detail.

This formulation is based on a number of intuitive arguments. The zombie argument: a system is conceptually conceivable that is functionally identical to a conscious human but has no internal experience - a philosophical zombie. If a zombie is conceptually possible, then consciousness cannot be reduced to function. The knowledge argument (Mary's room, Jackson, 1986): Mary, a neuroscientist who has studied everything about the physical processes of color perception but has never seen red, learns something new when she sees red for the first time. Thus, functional knowledge does not exhaust phenomenal experience. Levine's explanatory gap (Levine, 1983): a chasm remains between neural correlates and phenomenal experience that functional explanations do not bridge.

I present these arguments in their own words without distortion, because to challenge them, it is important first to see what they actually claim. They claim the ontological insurmountability of the gap between function and phenomenon. And if this gap is insurmountable, no theory that derives

consciousness from physical or functional mechanisms - including my own - can resolve the hard problem.

6.2. Diagnosis

Before attacking Chalmers' arguments individually, we need to make a general diagnosis of them. All three arguments rest on the same premise, which is not proven but postulated within them. This premise is that function and phenomenon are two different things - two ontologically independent aspects that may (or may not) accompany one another. The zombie argument makes sense only if function and phenomenon are separable, at least conceptually. The knowledge argument works only if phenomenal experience is something beyond functional information. Levin's explanatory gap exists only if there are two different things between which this gap lies.

This premise, despite its apparent self-evidence, is a legacy of Cartesian dualism. Descartes ontologically distinguished between extended substance (*res extensa*) and thinking substance (*res cogitans*), and although contemporary philosophers have long since ceased to use Cartesian language, the structure of his distinction has persisted in a more subtle form. When Chalmers asks "why is function accompanied by experience," he has already assumed that function and experience are two distinct things, one of which may (or may not) give rise to the other. The very form of the question presupposes what he is trying to prove.

An alternative to this premise exists and is discussed in contemporary literature, albeit under different names. For Galen Strawson (Strawson, 2006), these are various forms of panpsychism, in which the mental is not separate from the physical but is one of its aspects. For Thomas Metzinger (Metzinger, 2003, 2009), it is the Self-Model Theory of Subjectivity (SMT), in which the phenomenal "I" is a structural position within the system model rather than a separate entity; this theory is closest to the present work and deserves separate consideration, which I will address in Section 6.3. Graziano (2013) proposes a theory of the attentional schema, in which consciousness is the system's own model of its attentional processes. All these approaches, despite differing in detail, reject the Cartesian premise of the separability of function and phenomenon. I share their rejection of this premise, but I propose my own formulation of an alternative ontology, consistent with the rest of the structure of my theory.

6.3. Alternative Ontology: The Identity of Self-Modeling and Phenomenon

Before elaborating on this thesis, I will return to the formal definition of stage 4 given in section 3.4. There, a conscious system is defined through the self-overlay of a latent invariant of an agentic causal node onto the system's own configuration: a class of agentic causal nodes with boundary states of retention/disintegration has accumulated in the system's models, and the system's own configuration has begun to be processed by the model as an instance of that same class. From this structural event, a node *S* emerges, representing the system itself as an agent in its own model; within this node, a variable *V* operates with the values "holding / disintegrating," whose binary nature is induced by the retention contour over the continuous physics *Y* via the boundary states of

causal nodes; and this self-representation is globally accessible to all subsystems. The operational registration of the structural event of self-overlay occurs through the coupling of three independent measurement axes (Section 3.4): Φ for causal integration; the formalism of extended active inference for self-representation; GWT-availability for globalization. This formal definition describes the structure of the operation - what the system does, viewed from the outside. This chapter answers a question that is not resolved by the formal definition: what does it mean that this structure is simultaneously experienced from the inside as “something being.” The thesis I develop below is that the experience from within is not an additional property beyond the formal structure - it is the very same structure, viewed from the position the system occupies in its own representation. The formal definition from 3.4 and the phenomenological discussion in this chapter describe the same phenomenon from two different angles.

Section 3.4 establishes that consciousness is V in the precise sense in which V is defined there. The present section elaborates on the ontological consequence of this assertion. I assert the following. Phenomenal experience is ontologically identical to the dynamics of the system’s self-modeling, carried out through the structural pole V - that is, self-modeling in which the self-overlay of the latent invariant of the agentic causal node onto the system’s own configuration has occurred (stage 4). Phenomenal experience does not arise on top of such dynamics, does not accompany it, and is not a property of it that can be separated from it, even conceptually. Phenomenal experience is the very same dynamics of self-modeling with pole V, viewed from within. This is the second of the two main theses of this work.

This clarification is fundamental: I am not asserting the identity of phenomenality with self-modeling in general. Stage 3 systems - proto-agents with active inference - also engage in self-modeling in the sense that their generative model contains a representation of the system’s own states and its actions in the environment. But the self-modeling of a proto-agent and the self-modeling of a Level 4 agent are structurally distinct processes, differing not in quantitative complexity but in qualitative structure. The self-modeling of a proto-agent is modeling that must be validated by the environment through action: the model is constructed to generate an action, the action is executed in the environment, the result is compared with the model, the model is adjusted, and the cycle of active inference is closed through the environment. The proto-agent lacks autonomous retention of the modeling process in isolation from current validation; its modeling exists through the “model-action-environment-model” cycle. Self-modeling of the stage 4 carrier is structurally different: the model contains a V-pole relative to which modeling is retained independently of current validation by the environment. The system models itself as either retaining or losing its own existence, and this modeling operates even when the environment does not present current stimuli. The observed characteristics of Level 4 - reactions to absent stimuli, fears without experience, preventive planning for one’s own non-existence, and prolonged hysteresis of altered behavior - directly follow from this structural autonomy of the modeling process. These features are not additional characteristics requiring a separate explanation; they are a direct consequence of the structural difference between proto-agent modeling and modeling with V.

From this clarification follows a precise formulation of the ontological status of V. V is not an “additional supplement” to self-modeling, without which modeling would remain “obscure” - such a formulation would return us to a dualism in which structure and phenomenon are two different things, and V would play the role of a mysterious connecting third party. V is a structural characteristic of the modeling configuration itself: modeling with the V pole and modeling without the V pole are not one and the same modeling with or without an addition, but two structurally different modeling configurations. Only one of them - modeling with V - is identical to phenomenality according to our ontology. The other - modeling without V in the proto-agent - is not phenomenal. This is not an agnostic position nor methodological caution; it is a positive assertion of the theory. At stage 3, there is no phenomenality, and this is a direct consequence of the thesis of structural identity: where there is no specific structure (modeling with V), there is also nothing to which this structure is identical (phenomenal experience).

Let me clarify what I mean by this. When a system reaches a level of complexity at which its model of the world includes itself as an agent acting within that world, a structure emerges that possesses a special property: the “inside-the-model” position becomes defined. This position is neither a separate object within the model nor a separate subject observing the model. It is a structural property of the model itself: the model is oriented not from nowhere, but from the point occupied by the system in its own representation of the world. This position is precisely what is subjectively experienced as “I.” It is not an additional element beyond the model; it is the way in which the model is organized.

Positioning in Relation to Metzinger’s SMT. Before developing the argument further, I must explicitly indicate the relationship of this work to Thomas Metzinger’s self-model theory of subjectivity (SMT), since of all the positions existing in contemporary philosophy of mind, SMT is closest to mine, and without explicit positioning, the reader might mistake my thesis for a reformulation of Metzinger’s.

In **Being No One** (2003) and **The Ego Tunnel** (2009), Metzinger argues that there is no “self” as an ontological entity - there is only a phenomenal self-model, whose dynamics give rise to the experience of self. Central to his work is the concept of transparency: the phenomenal self-model is not perceived by the system as a model, but as immediate self-given-ness. Transparency creates a stable and evolutionarily useful hallucination of the “I” where in reality there is only a structural position within the model. From this, Metzinger draws a strong ontological conclusion: ordinary self-consciousness is, in the strict sense of the word, a hallucination, and behind this hallucination there is no “I” that would be real beyond the structure.

The overlap with my position is profound. The thesis that the “I” is a structural property of the model, rather than a separate entity, is the same for both us and Metzinger. When I write above that the “inside-the-model” position is not a separate object or a separate subject, but a structural property of the model itself, this is exactly what Metzinger formulates through the PSM (Phenomenal Self-Model). And we share a rejection of the Cartesian premise of the separability of function and phenomenon.

The differences, however, are substantial, and there are three of them.

The first is a divergence, and under the referential frame it is not ontological, as it might have seemed, but structural. This passage used to be naturally formulated along the axis "the real versus the illusion": for Metzinger, behind transparency stands the dynamics of the model that merely appears to be itself, whereas I supposedly insist that presence is real. The referential position adopted above (Section 6.3, the precise name of the position) removes this axis: I do not claim that "phenomenal experience is real" beyond the structure - the predicate "the real-beyond-the-position" is empty, just as for Metzinger the "immediately given subject" is empty. In this we are kin, not opponents: his transparency of the phenomenal self-model and my privacy of the pole are about the same structural fact (the position is given only from within and is not externalized as an object). The divergence is therefore not "whose is more real" but lies in two specific points. The first is a structural element: Metzinger has a self-model, transparency, the PMIR, but no variable of existence V as the structural pole of the self-model; the self-overlay of a latent invariant with the pole "I persist / I disintegrate" is what his construction does not contain. The second is the answer to the meta-problem: Metzinger leaves the door ajar to the idea that the hard problem exists and is solved by exposing the illusion; I close it differently - not by exposure, but by showing that the dividing predicate is empty, and I explain the very ineliminability of the appearance structurally, through the privacy of the pole. This is not "I am more real"; it is "I have V and an answer to the meta-problem where he has transparency without a pole."

Second - a divergence regarding the framework. Metzinger constructs his theory from within the philosophy of mind and relies on neurobiology as its empirical foundation. He does not venture into cosmology, does not build a ladder from dissipative structures to consciousness, and does not link his theory to thermodynamics and active inference. His theory is narrower than the present one. It is brilliantly developed within its own domain, but that domain is the phenomenology of self-consciousness and its neurobiological correlates, not the place of consciousness in the universe as a whole. Thus, SMT remains confined to one of the scales of the structurally self-similar pattern unfolded in the present work, without extending to the others.

Third - a divergence regarding a specific structural element. Metzinger has a self-model, he has transparency, he has the Phenomenal Model of the Intentionality Relation (PMIR). What he lacks is the existence variable V as a structural pole of the self-model. That structural event of the self-overlay of the latent invariant of the agentic causal node, which in this work formalizes stage 4 - and its operational registration through the coupling of three measurement axes (extended active inference plus sufficient Φ plus GWT-accessibility of V) - is absent in Metzinger's work in both respects. Without V , Metzinger lacks the operational mechanism that, in this work, provides a structural consolidation of the phenomenon through the presence of inherited gradients in V (Sections 3.4 and 3.7.5). This is not a criticism of SMT - Metzinger worked on a different problem and with a different apparatus. This indicates that our theory is not reducible to SMT and does not coincide with it: it expands the framework, shifts the ontological formulation from the debunking of the illusion to the identity of structure and phenomenon, and introduces V as a structural element that operationalizes phenomenal content.

These differences do not negate the SMT nor do they enter into a polemic with it. They indicate the ways in which this work goes beyond the SMT, while remaining in the same camp with it in its central rejection of the Cartesian premise.

Phenomenal experience in this picture is neither something described by the model nor something that accompanies it. Phenomenal experience is the model itself in its internal positional structure. When I experience red, this does not mean that in my model of the world there is a label "red" plus an additional phenomenal quality of red that accompanies this label. It means that in my model of the world there is a structural position occupied by the situation "there is a red object in front of me," and this position is defined through its connections with all other positions (green, yellow, warm, dangerous, ripe, and so on). The phenomenal quality of red is this positional definiteness, viewed from within.

To ask "but why is there phenomenal experience beyond the functional structure?" is to postulate once again that there are two things: a functional structure and phenomenal experience. In the proposed ontology, there is one thing that has two descriptions. From the outside, the model is described as a causal-functional structure that processes inputs and generates outputs. From the inside, the same structure is an experience, because "from the inside" there is a position that the structure occupies in its own representation of the world. "What is it like to be" is synonymous with "how the structure relates to itself." These are not two phenomena, but one phenomenon in two descriptions.

The privacy of the pole as the constitutor of "from within," and identity under two aspects. "Two descriptions of one structure" needs a qualification, otherwise it reads as a tautological evasion. What constitutes "from within"? Not a second observer "for whom" - that would reintroduce a hidden second entity. "From within" is constituted by the privacy of the pole: the position is given only from the point the substrate occupies in its model of itself, and this point is private not empirically (hard to measure) but structurally - it is a position, not an object, and therefore in principle not given from an external point of description. Hence the precise form of the identity thesis: the P-description (phenomenal) and the A-description (functionally accessible) are one structure from two positions of observation, not two things and not one thing with a label attached. Therefore "to be in the position is to be in it" is not a tautological dodge but a compressed thesis: to be a structure occupying a position relative to its model of itself just is what is described functionally from the outside and as experience from within; there are no other "two things." This answers two fronts at once. Block's distinction between access and phenomenal consciousness (Block 1995) ceases to be a distinction between two properties: access is the A-description of the structure, phenomenality its P-description, and the privacy of the pole explains why the second is not read off from the first, without introducing a second property. And the Strawson/Goff line (the coherence of experience "from within" as an argument against pure functionalism) is accepted in the part that "from within" is real and not reducible to external description, but it does not lead to panpsychism: "from within" is not a fundamental property of matter but a structural position arising only where there is a pole V - that is, it is constituted, not smeared across all that exists.

Through the presence of inherited gradients in V (section 3.4), the ontological identity of structure and phenomenon receives additional operational grounding. Not “the structure is in some way identical to the phenomenon,” but “the structure is present to itself through V at level 4, and this presence is phenomenal content.” Hunger, pain, anxiety, fear - these are structurally necessary modes of such presence, in which the disturbances of each of the inherited gradients acquire experiential content through the pole V. This transforms the identity thesis from an ontological declaration into a formulation with operational content: identity is not postulated; it is demonstrated through a structural mechanism that makes the phenomenal a necessary consequence of the emergence of V.

The status of the identity thesis: an epistemic clarification. What has been outlined requires one honest caveat regarding what is proven in this chapter and what is accepted as a physicalist premise. The structural mechanism of the presence of inherited gradients in V provides **an operational justification** for the identity - it explains *why* the content of phenomenal experience is precisely what it is (hunger as a weakening of the energy gradient in V, fear as the approach of “I am falling apart,” and so on), provided that the identity of structure and phenomenon is accepted. But **the existence** of the phenomenal - that the structure “from within” is an experience rather than darkness - is not derived from the apparatus; it is accepted as a physicalist-type bridge-identity. Identity functions here in the same way as the bridge “temperature is average kinetic energy” in historical physics: it began as an accepted identity, not as a proven theorem, and from it was derived what could be derived. Machinery V makes this identity **coherent and meaningful** - but does not make it **true**. Coherence is not truth, and I do not mistake the former for the latter.

A qualification on the limit of the temperature analogy. The parallel with temperature illustrates the epistemic status of the move (the identity is accepted, not proven), but it does not transfer to the outcome. The identity “temperature is mean kinetic energy” was confirmed because the functional role of temperature is exhausted by mean kinetic energy with no explanatory residue. The entire force of the hard problem lies in the claim that for consciousness such a residue exists. The appeal to the temperature case therefore does not prove the absence of a residue here - it only shows that identity as a form of move is legitimate. Accepting or rejecting the physicalist premise (that there is no residue) remains the reader's choice, not a consequence of the analogy; the analogy bears no probative weight here and is offered only as a model of epistemic status.

This leads to one consequence that I must address myself before my opponent does. The metaphysical allure of identity - that is, the claim that structure and phenomenon are one and the same phenomenon described in two different ways, rather than two distinct phenomena - **is empirically idle**. If we abandon identity and adopt a dualist interpretation (structure V is *the* neuro/functional *correlate* of phenomenal experience, not the experience itself), then not a single empirical prediction of the theory changes: the same six-element coupling, the same modes of dissociation, the same forbidden predicates, the same falsification scheme in Section 10.2. This is both the theory's strength and its greatest weakness. Strength - because the empirical program will survive the rejection of the metaphysical gloss: the theory remains a working physicalist model even for a reader who does not share the monistic interpretation. A blow - because the metaphysical

gloss, which carries no empirical weight, can be called an interpretive add-on, not a theory in the strict sense. I am not trying to hide this; I consider both sides of the coin part of an honest presentation. The theory makes a physicalist claim explicitly, without passing it off as a result. Those who accept this claim receive an ontological resolution to a difficult problem; those who do not accept it receive a working empirical program without a resolution. Both are a win compared to theories that either claim to prove the existence of the phenomenal from structure (which is impossible) or refuse to discuss the phenomenal at all (which is dishonest).

The precise name of the position: to avoid two false readings. This position is easily confused with two neighboring ones, and it must be distinguished from both. It is not a functional reduction in the spirit of Dennett: Dennett treats the phenomenal as quasi-phenomenal, erroneously represented by the system as something more - that is, he leaves an "error" that has content. Here there is neither an error nor an "as." But it is also not realism about the phenomenal in the strong sense: I do not claim that there is a phenomenon-entity identical to the structure that could be mentally separated from it. The precise formulation is this: what is real is the position that the substrate occupies relative to its model of itself in the environment; the predicate "the phenomenal-beyond-this-position" is referentially empty - it has no conditions of application, it is not a denied entity but a pseudo-concept (this is shown via the zombie scenario below). It is therefore correct to say neither "the phenomenon is real apart from the dynamics" nor "there is no phenomenon," but rather: there is a position-from-within, and it is precisely this that was formerly called phenomenal; the question of a "light beyond the position" has no fact about which it is meaningful.

6.4. Refuting Classical Objections

In the proposed ontology, Chalmers's classic arguments lose their force. I will demonstrate this for each of them.

The zombie argument. Before treating it as a construct, one must point to where it breaks. The zombie scenario requires a meaningful predicate "a functionally complete system - but without a light-beyond-the-position": there must be something whose presence or absence distinguishes a zombie from a conscious system under full functional coincidence. In the proposed ontology, this predicate has no conditions of application. "A light beyond the position" is not a denied entity that can be mentally removed while leaving everything else in place; it is a pseudo-concept - there is no fact about which it is meaningful to ask "is it present or not" beyond the system's occupying a position within its model of itself in the environment. The zombie scenario is therefore not refuted but unformulable: it demands a difference where there is no predicate capable of carrying that difference. From this follows - as a consequence, not as a ground - the indistinguishability: a zombie and a conscious system are indistinguishable not because the difference is hidden from an outside observer (to which Chalmers would rightly reply that epistemic inaccessibility $\neq$ absence of a difference), but because there is no distinguishing predicate at all. And from this likewise follows the identity of function and phenomenon: it is not posited as a premise against Chalmers but turns out to be what remains once the pseudo-predicate "light-beyond" is removed. In this light, the construct "a functionally complete stage-4 system without presence" is contradictory not as a

“triangle with four corners” (that would appeal to an already-accepted identity) but as a demand for meaning from an expression that has none.

The Argument from Knowledge (Mary). When Mary first sees red, she does not acquire an additional class of knowledge (phenomenal qualities beyond functional information), but enters a positional structure in which her own model of the world for the first time includes the node “there is red in front of me,” connected to all other nodes of her perceptual system. Up to that point, she had knowledge of red - a description of its place in the system of physical and neurobiological connections. Afterward, she acquired knowledge from the position in which she herself occupies the place representing red. These are different kinds of knowledge, but the difference between them is not ontological (between the functional and the phenomenal) but positional (between external description and internal participation). Mary’s argument reveals a real difference, but not the one it claims to reveal.

Levin’s explanatory gap. There is indeed a chasm between the functional description of neural activity and the phenomenal experience if we consider them as two independent phenomena between which a bridge must be built. In the proposed ontology, no bridge is needed: these are not two phenomena, but one. The apparent gap arises because we are accustomed to describing the functional structure from an external position (as an object of scientific inquiry), and phenomenal experience from an internal one (as our own experience). These two descriptions relate to different positions regarding the same structure, and the transition between them is a change of position, not a crossing of an ontological chasm. Through the structural mechanism of presence in V (section 3.4), this becomes not only an ontological formulation but also an operational map: the external functional description describes the structure including V and its coupling with inherited gradients; the internal position of the system is a position from within this same structure. One mechanism, two readings. Levin’s gap exists only as long as we keep two independent phenomena in mind; as soon as we understand that presence through V is the phenomenal content itself, and not its correlate, the gap disappears not as a patching over, but as the realization that there is nothing to patch over.

Distancing from the phenomenal-concept strategy. A position similar in appearance must be distinguished here, so that it is not ascribed to me by default. The phenomenal-concept strategy (Loar 1997; Papineau 2002; Block 2006) explains the illusion of the explanatory gap by the special nature of phenomenal concepts: we have two types of concepts of one physical referent - physico-functional and phenomenal - and the appearance of a gap is produced by how the latter are structured (their direct, non-descriptive reference to experience), not by any ontological gap. My move is different. The appearance of a gap is explained not by a special structure of a class of concepts but structurally - by the privacy of the pole: the position-from-within is given only from the point the substrate occupies in its model of itself, and is therefore inaccessible from an external position of description not because of the nature of our concepts but because it is a position, not an object. The phenomenal-concept strategy locates the source of the illusion in semantics (how we think about the phenomenal); the position proposed here locates it in structure (where the observing point stands relative to the observed). These are related answers to the meta-problem, but not the

same one: semantic versus structural. The honest boundary: I do not rely on the phenomenal-concept strategy and do not inherit its known difficulties (for example, Chalmers' objection that phenomenal concepts themselves require explanation), because I explain the appearance of the gap by the privacy of the pole, not by the nature of the concepts.

Basic phenomenal experience in non-reflective beings. The objection is that my theory, based on self-modeling, may work for developed human consciousness, but not for the basic experience of an infant, an animal, or the simplest nervous system. The answer requires clarification in light of the formal ladder from Section 3. The self-modeling I am referring to in the context of stage 4 does not imply reflective self-understanding. Minimal self-modeling at stage 4 involves incorporating the system itself into the model of the world as an agent within that model, with the variable of its own existence as a structural pole, but without the need for language, concepts, or reflection in the human sense. The difference between a human infant and an adult philosopher lies not in the presence or absence of stage 4, but in the complexity and recursive depth of the self-model within that stage. Basic phenomenal experience is the positional structure of a minimal self-model at stage 4; unfolded consciousness is the positional structure of a highly recursive self-model at the same stage, enriched by the mechanisms of stage 5. Within a stage lies a continuum; between stages lie phase transitions. This aligns the answer with the general logic of the work and does not require postulating a continuous continuum where none exists.

Here, the key point is this: the difference between a stage 3 proto-agent (a bacterium, a primitive nervous system without a V-pole) and a non-reflective carrier of stage 4 (an infant, an animal with a basic form of the V-pole) is a structural difference, not a quantitative one. The proto-agent models through a cycle of active inference that must be confirmed by the environment; the non-reflective carrier of stage 4 models through a structural pole V, which remains detached from the current environment. Phenomenality is absent in the proto-agent and present in the non-reflective carrier of stage 4 - not because the former has “weak phenomenality,” but because the former lacks the structural configuration of modeling that is identical to phenomenality according to our ontology. The level of reflection (a characteristic of only some stage 4 carriers) and the presence of phenomenality (a characteristic of all stage 4 carriers) are two distinct axes of description, and their conflation is a source of objection.

Distinction from higher-order theories. An objection that philosophers familiar with the literature might raise against me is that my theory resembles higher-order theories (HOT) by Rosenthal (2005) and Lau (2008), according to which a mental state becomes conscious when it is represented by another mental state of a higher order. HOT theories are criticized for not answering the difficult problem, but merely postponing it: if a first-order state is conscious because it is represented by a second-order state, what makes the second-order state conscious? A regress arises.

My theory is not vulnerable to this objection because it does not introduce an additional level of representation. In HOT, there is a first-order state and a second-order state, and consciousness arises in relation to them. I do not have two levels. There is a single structure - a model of the world that includes a model of oneself as an agent - and consciousness is a positional property of this

structure, not a relation between two structures. There is no regression because there is nothing to regress to. Self-modeling does not represent another representation; it is a way of organizing a single model.

6.5. The Boundary of the Position

I must conclude this section with an honest acknowledgment of the limits of the proposed position. My approach to this difficult problem works if one accepts the proposed ontology. If the reader insists that the separability of function and phenomenon is a primary ontological given, not derived from analysis but serving as its starting point - for such a reader, my attack is not convincing. They will remain with the difficult problem.

The precise status of what has been done: neither a "solution" nor a "third way." Things must be named precisely here, so as not to ascribe to this chapter more than it does. I do not claim to solve the hard problem on its own terms - that is, to answer why a functional structure is "accompanied" by experience: that question presupposes a division I reject, and so I decline to answer it rather than answering it. Nor is this a "third way" beyond Dennett and Metzinger in the sense of a new third substance or a new mechanism on top of theirs: ontologically this is the same physicalist family. What has been done falls into two separate things. First, a dissolution under a stated premise: if one accepts the physicalist premise (that there is no explanatory residue beyond the structure), the hard problem is not solved but ceases to be formulable, because its dividing predicate is empty (Section 6.4). This is a delimited refusal to play the residue game, not a proof of its absence. Second, a structural explanation of the meta-problem (in Chalmers' 2018 sense): why the very premise of separability is so hard to give up and the appearance of a chasm is so persistent. The answer is structural - the privacy of the pole: the position-from-within is in principle not externalized into a description, and this inaccessibility is experienced as "something left unexplained." The theory solves the meta-problem (it explains the illusion) and dissolves the hard problem under a premise (it does not answer its question). These two registers must not be conflated: it is precisely their conflation that creates the impression that a "solution" has been claimed.

What can I say in response to such a reader? Only this: the separability of function and phenomenon has not been proven. It intuitively seems true, but this intuition may be a legacy of the Cartesian tradition that has shaped our language of consciousness over the past four centuries. An alternative ontology, in which function and phenomenon are two descriptions of the same thing, does not contradict a single empirical fact, a single verified theory, or a single observation. It constructs a more economical picture: fewer entities, fewer unexplained correlations, fewer chasms across which bridges must be built. According to Ockham's razor, a more economical description is preferable if it accounts for the same phenomena. This is not proof, but it is an argument. Beyond that lies the choice of position, and this choice cannot be resolved by the logic of a single article.

What is not part of the position. To avoid confusion, I state this separately: the proposed ontological position is not a form of panpsychism. I do not claim that phenomenality is present in all modeling

behavior in one form or another. I assert the exact opposite: phenomenality is identical to a specific structural configuration of self-modeling - the one in which the latent invariant of the agentic causal node has superimposed itself onto the system's own configuration (stage 4). In systems lacking this configuration (a proto-agent with active inference lacking a V-pole, a dissipative structure without modeling, an autocatalytic cycle without self-representation), there is no phenomenality - not "it exists in a weaker form," not "it might exist, but we do not know," but rather, there is none. This is a positive assertion of the theory, a direct consequence of the thesis of structural identity: where there is no specific structure, there is also nothing to which that structure is identical. And this assertion contains a testable consequence - in species at the proto-agent level and in artificial systems without the self-overlay of a latent invariant, phenomena specific to phenomenal experience should not be observed (fears without experience, preventive planning for one's own nonexistence, prolonged hysteresis of altered behavior, reaction to absent stimuli as a structural feature). If such phenomena are systematically observed in systems in which the self-overlay of the latent invariant is not implemented, our ontology is refuted. The boundary of the theory, thus, is not agnostic (we do not know what the proto-agent possesses), but positive (we assert that it lacks phenomenality). What is falsifiable here is not the phenomenal side directly, but its behavioral correlate: the program in subsection 10.2 tests the structural-behavioral coupling (structural registration of self-overlay $\leftrightarrow$ behavioral signatures of V); the identity of structure and phenomenon itself remains the ontological choice outlined above.

6.6. Comparison with Alternative Interpretations

The attack on the hard problem carried out in subsections 6.1–6.4, and the boundary of the proposed ontology outlined in 6.5, defend the theory against the central objection of contemporary philosophy of mind. This, however, is not enough. The same empirical material - the neurobiology of consciousness, phenomenological data, behavioral signs of agency - admits interpretations different from the one proposed in this work. This subsection examines three such alternatives and shows that it is precisely the present theory that explains beyond each of them. This is not proof of its superiority; it is an academic obligation to the reader, who has a right to know why the synthesis presented here is preferable to competing descriptions in terms of explanatory power, and not merely in terms of structural coherence.

An evolutionary-biological interpretation without ontologizing consciousness. The strong position, as presented in the works of Dennett, Humphrey, Frit, and to some extent Damasio, is as follows. Consciousness is a product of biological evolution, selected for its functional advantage - improved modeling of the environment, planning, and social coordination. No special ontology is required: consciousness is a complex behavioral and cognitive phenotype selected by the same mechanisms as any other adaptive trait. The question of what an experience is from the inside is not posed as a separate issue at this level of description.

This position explains well: the distribution of consciousness across the phylogenetic tree through correlation with evolutionary pressures; species-specific features through the specifics of ecological niches; the connection between consciousness and memory, learning, and social behavior as co-selected traits.

What it does not explain. First, why the phenomenal aspect exists at all, rather than just the functional one. Evolutionary biology explains what type of cognitive processing is selected, but does not explain why this type of processing is accompanied by an inner experience. This is a classic objection to evolutionary theories of consciousness: the function is explained, but the phenomenon remains beyond the scope of the explanation. Second, it does not provide a structural criterion by which one can determine whether a specific system satisfies the conditions of consciousness; it operates on the premise that “consciousness exists in species with a certain level of neurobiological complexity,” without a formal definition of what “a certain level of complexity” means. Third, it lacks a framework for discussing consciousness in artificial systems, because artificial systems are not subject to natural selection in the biological sense.

What the present theory adds. A structural criterion not tied to substrate or evolutionary origin (Section 3.4: self-overlay of a latent invariant of an agentic causal node, operationally recorded through the coupling of three measurement axes). The structural consolidation of the phenomenal through the presence of inherited gradients in V (sections 3.4 and 3.7.5). The possibility of speaking of consciousness in artificial systems when formal conditions are met. Vertical integration with thermodynamics and cosmology: consciousness as a natural stage in the general process of increasing complexity, rather than a unique product of biological evolution on a single planet.

Crucially: this theory does not deny the evolutionary-biological explanation. Evolution selects for realizations of the self-overlay of a latent invariant because these realizations are functionally advantageous; this is consistent with the framework proposed here. But the evolutionary-biological explanation is insufficient for a complete theory of consciousness, because it operates in a single register (adaptive function) and does not extend to the structural level (what exactly is the structure), the ontological level (how the structure is related to the phenomenon), and the cosmological level (where consciousness stands in the general process of complexity). This theory extends the evolutionary-biological explanation to these levels.

Predictive processing without the identity of the phenomenon and self-modeling. The position presented in the works of Hohwy (2013), Clark (2016), and partly in Friston’s later works, is as follows. Consciousness is a property of the brain’s predictive mechanism - the constant generation of predictions and minimization of discrepancies with inputs through variational Bayesian inference. The phenomenal aspect is the effect of the very operation of this mechanism; there is no need to postulate any special ontology or the identity of the phenomenon with self-modeling. It suffices to say: the predictive mechanism operates in a certain way, and this operation is consciousness.

This position explains well: perceptual illusions, attention effects, the phenomena of dreams and hallucinations; the dynamics of cognitive processes and the formation of beliefs; the connection between consciousness and the minimization of free energy in Friston’s sense.

What it fails to explain. First, it does not distinguish between conscious and non-conscious systems that carry out predictive processing. A bacterium exhibiting chemotaxis formally minimizes the discrepancy between the model and the environment; within the predictive paradigm, it does the same thing as a conscious system. If consciousness is a predictive process, then either the bacterium is conscious (which most proponents of predictive processing do not claim), or an additional

criterion is needed - one that does not exist in the pure position and is usually borrowed from IIT or GWT, which already takes us beyond the bounds of the pure position. Second, it does not explain - - why the predictive process is accompanied by an inner experience. Third, it is tied to a neurobiological substrate in the form in which it has been developed, and its application to non-biological systems requires extensions not developed in the pure version.

What the present theory adds. The structural criterion for the transition from a non-conscious predictive process (stage 3) to a conscious one (stage 4) through the self-overlay of the latent invariant of the agentic causal node: the predictive process must be supplemented by the accumulation of a class of agentic causal nodes in models, the self-overlay of this class onto the system's own configuration, and the globalization of the emerging node S. This provides a clear boundary where predictive processing does not draw one. The identity of the phenomenon and self-modeling (Section 6.3) is a formulation not postulated separately, but one that follows from the presence of V in the self-model via the structural binding elaborated in 3.4 and 3.7.5. This eliminates an additional step (the predictive process plus some separate "and why it is conscious"), integrating both into a single structure.

The present theory does not deny predictive processing - it uses it as a formal basis for condition (a) of stage 4. But it adds a structural criterion distinguishing the conscious predictive process from the non-conscious one, and an ontological formulation explaining why the phenomenal aspect arises precisely when this criterion is met.

Functionalism without a transition to phenomenal identity. The position presented in the works of Dennett and many in cognitive science over the last three decades is as follows. Consciousness is a functional organization of a certain type. Whether this organization is realized by neurons, silicon, or some other substrate is irrelevant; what matters is the structure of causal relationships and information flows. The phenomenal aspect is either reduced to functional properties (Dennett's strong reduction) or is not discussed as a separate issue (methodological functionalism).

This position explains well: the implementability of the same cognitive function on different substrates; structural similarities in cognitive processes across different species and architectures; the applicability of information theory to the description of conscious processes.

What it does not explain. First, the differences between functionally equivalent systems, one of which is conscious and the other is not (Searle's classic Chinese Room problem). A functionalist either denies the intuition about such differences (Dennett) or admits that functional equivalence is insufficient (Chalmers); in both cases, pure functionalism turns out to be incomplete without supplementation. Second, functionalism lacks a language for the phenomenal as such. Pain, color, and smell are described in it as functional roles, not as experiences from within. This works well for scientific description, but does not resolve the ontological question. Third, functionalism has a framework for heterophenomenology (external description), but not for phenomenology in the proper sense (experience from within).

What the present theory adds. Not functionalism with an addition, but the structural identity of the phenomenon and self-modeling (section 6.3). For the functionalist, function and phenomenon are different things; one is explained, the other is reduced or rejected. In this theory, they are

ontologically identical: inner experience is a structure viewed from its own perspective. This is not an addition to functionalism; it is a different ontology. The structural criterion of level 4 is formally functional (it can be verified through measurements), but in substance it does not reduce the phenomenon to a function - it demonstrates their identity through the position from within.

Searle's argument and the intuition of functional non-equivalence work as follows for us: functionally equivalent systems with the same structure have the same consciousness; the difference in intuition arises because everyday functional equivalence is determined at the level of behavior, while structural equivalence is determined at the level of the self-overlay of the latent invariant of the agentic causal node from Section 3.4, and these two levels do not coincide. A behaviorally equivalent large linguistic system is not structurally equivalent to a stage-4 conscious system because the self-overlay of the latent invariant is not fulfilled in it in the strict structural sense. This resolves the classic contradiction between strong and weak functionalism.

Conclusion of the comparison. Each of the three alternative interpretations explains part of the empirical data and fails to explain the other part. The evolutionary-biological interpretation explains the distribution of consciousness and its adaptive function, but does not explain the phenomenal aspect and does not provide a structural criterion. Predictive processing explains the dynamics of cognitive processes, but does not distinguish between conscious and non-conscious predictive systems. Functionalism explains implementability on different substrates, but lacks a language for the phenomenal aspect as such. The present theory does not reject any of these interpretations - it uses their formal results as foundations (evolutionary biology substantiates the probabilistic nature of transitions, predictive processing formalizes the condition (a) of stages 3 and 4, and functionalism provides a language for describing the structural criterion) - but adds to each what it lacks: a structural criterion with operational content, the ontological identity of structure and phenomenon through presence in V, and integration with the thermodynamic and cosmological framework through a structurally self-similar pattern.

This is the argument in favor of the synthesis that is required beyond its structural integrity. Synthesis is proposed not because it is aesthetically self-contained, but because it formally explains what the three leading alternative interpretations, each on its own, do not explain. This is not proof - in the philosophy of mind, there is no such thing as proof of this kind. But it is an argument that deserves serious consideration.

7. Invariance of the Structural Pattern Across Scales

The same structural pattern - gradient, retention, pole, transition - is found independently at different scales: in physics as phase transitions, in biology as autopoiesis and evolution. In this work the pattern appears in the ladder of complexification from a dissipative structure to a conscious system (Section 3). From step to step it is not the environment Y that changes - Y is through-going and single - but which of its arms are accessible to the system (the access spectrum) and what content the carrier and the pole acquire. The pattern «gradient $\rightarrow$ retention $\rightarrow$ pole $\rightarrow$ transition» itself, and its formal metric, remain invariant with respect to level. It is precisely this invariance of the pattern under varying access to a single Y , rather than a common substrate, that constitutes the structural self-similarity of the theory.

There are grounds to suppose that the same pattern may also manifest beyond the ladder considered here - at the cosmological scale, as well as in transitions between levels of description within physics itself. But this is a separate hypothesis, neither derivable from the theory of consciousness nor necessary for it: the central part of the present work (steps 1–5, Section 10) does not depend on it and retains its force regardless of its fate. We merely point to this possibility as an open direction.

8. The Observer Within the Observation

Any concept of the place of consciousness in the universe faces a fundamental problem: the observer is part of the system they are trying to describe. We perceive the universe through the brain, language, and instruments, which are themselves products of the very processes we are investigating. This is not a philosophical abstraction, but a physical limitation with concrete consequences. This section discusses how the “observer-within” structure manifests in three different contexts - the anthropic principle, the measurement problem in quantum mechanics, and the central thesis of the theory of consciousness - and why all three are manifestations of the same ontological structure.

8.1. The Anthropic Principle

In 1973, physicist Brandon Carter formulated what is now known as the anthropic principle (Carter, 1974). In its weak form, it states: the observed properties of the universe are compatible with the existence of an observer capable of perceiving them. In other words, we see this particular universe because in any other, we would not exist. Our sample is fundamentally biased toward conditions that allow for our existence.

This seemingly trivial statement has non-trivial consequences for physics. The fine-tuning of fundamental constants - the fact that the values of the fine-structure constant, the electron-proton mass ratio, the cosmological constant, and other constants fall within narrow ranges that make complex chemistry, star formation, and ultimately life possible - - may be either a surprising

coincidence requiring explanation through additional mechanisms (the multiverse, selection dynamics, theories of everything), or a natural consequence of anthropic selection: in any universe where the constants lay outside these ranges, there would be no observers to speak of it.

The anthropic principle is not a theory, but a methodological constraint. It does not explain why the constants are as they are; it merely points out that our sample is not representative. To take this limitation seriously is to understand that any theory of the universe we construct is an “inside” theory, and that its predictions pertain to that slice of reality accessible to observers of our kind. This is not a reason to reject such a theory, but it is a reason to understand its status precisely.

8.2. The Observer in Quantum Mechanics

Parallel to the anthropic principle, another line of thought regarding the observer developed in 20th-century physics - a line extending from the Copenhagen interpretation of quantum mechanics through decoherence theory to modern relational and information interpretations.

The Copenhagen interpretation, in its early formulations, assumed a special status for the observer: a quantum system is described by a wave function that “collapses” upon measurement, and this collapse is a process requiring the observer’s participation. This formulation had obvious problems - what counts as an observer, at what moment the collapse occurs, how the measurement process is described in terms of quantum mechanics - and most subsequent developments in quantum mechanics have been aimed at resolving them.

The theory of decoherence (Zurek, 2003; Joos et al., 2003) showed that a specific “observer” is not needed in quantum mechanics: it suffices that the quantum system interacts with an environment possessing a large number of degrees of freedom. This interaction leads to certain states becoming stable (classical) in the system’s eigenbasis, while superpositions of these states rapidly decay. Decoherence explains why we do not observe macroscopic quantum superpositions, without the need to postulate a special act of observation.

Relational interpretations of quantum mechanics (Rovelli, 1996) go further: they assert that a quantum state is not a property of the system itself, but rather a relationship between the system and another system, considered as an observer. Any physical system can act as an observer for any other; there is no privileged class of “true observers” (e.g., conscious beings). What constitutes a “state” depends on the system relative to which it is defined.

I would like to emphasize the overarching thesis that unites these lines of thought: in modern physics, the observer is not a distinct external entity that stands opposed to the world and describes it from the outside. The observer is a subsystem of the world, correlated with another subsystem, and within this correlated pair arises what we call a “measurement” or “observation.” This is not a privileged act of external consciousness, but a structural property of the pair of subsystems.

8.3. Consciousness as a Position Within One’s Own Model

I return to the central thesis of Section 6: consciousness is a position of the system within its own model of the world. At first glance, this formulation refers to the individual nervous system and its internal representation; it seems to be a matter of neurobiology with no direct relation to cosmology or quantum mechanics.

This impression is deceptive. If we look closely at the structure of the formulation, we find that it has exactly the same logical form as the two previous ones. The anthropic principle states: we view the Universe from a point within it, and this position determines exactly what is accessible to our observation. Relational quantum mechanics states: the quantum state is defined relative to the observer subsystem, and this subsystem is located within the system of which it is a part. My theory of consciousness states: consciousness is the position of a system within its own model of the world, in which that system participates.

In all three cases, we are dealing with the same ontological structure: observation is a positional characteristic of a system within a larger whole, rather than a privileged external act. The anthropic principle, the measurement problem, and the theory of consciousness are three projections of this unified structure onto three different contexts.

8.4. Synthesis

This synthesis represents the second point of genuine originality in this work - alongside the thesis on consciousness as an active mechanism of complication, developed in Section 3, and the thesis on the identity of self-modeling and the phenomenon, developed in Section 6. I do not claim that the individual elements of this synthesis are new: the connection between the anthropic principle and the measurement problem has been discussed in the literature (Wheeler, 1983; Tegmark, 1998); the connection between quantum mechanics and consciousness has been discussed many times, though usually in controversial and poorly substantiated forms (Penrose-Hameroff). What I propose as original is the observation that all three problems constitute a single ontological structure, and that this structure does not require mystification but is a natural property of any situation in which a system describes a larger whole of which it is a part.

The structural commonality of the three projections becomes particularly clear through the formal definition of level 4 given in section 3.4. A conscious system is defined there through the self-overlay of a latent invariant of an agentive causal node onto the system's own configuration, and central to this definition is the presence in its generative model of a node S representing the system itself as an agent within its own model of the world. This structure - a system representing itself within its own model of a larger whole - is precisely the same logical form as that of the anthropic principle (an observer describing the Universe from a position within it) and of relational quantum mechanics (an observer as a subsystem correlated with another subsystem of the same world). All three cases are realizations of the same formal structure on different scales: cosmological, quantum, and neurobiological. The fact that the same structure emerges in three independent fields of physics and the theory of consciousness is not a coincidence, but an indication that this structure is a

fundamental property of any situation in which a system describes a larger whole of which it is a part.

Several consequences follow from this synthesis, which are important for the overall picture of the theory.

First. If observation is a positional characteristic of a system within a whole, then any physical theory claiming to provide a complete description of reality “from the outside” contains an internal contradiction. A complete external description is impossible because there is no external position. This is not a denial of physics, but a clarification of its status: physics describes reality from a certain specific position within it, and this position imposes structural limitations on what can be described.

Second. The phenomenon of consciousness is not an anomaly in the physical world, but a manifestation of the general ontological structure at the level of the individual nervous system. The same structure that, on the scale of the universe, gives rise to the anthropic principle, and that, on the scale of a quantum system, gives rise to the relationship between the measured system and the measuring subsystem, on the scale of a complex self-organizing system gives rise to phenomenal experience. This unification of the three problems eliminates the need for separate explanations for each of them.

Third. The observer inside cannot step outside. This is not a figure of speech, but a structural limitation. Any attempt to take an external position relative to the system of which I am a part is either an illusion (I am simply simulating an external position from within) or a change of system (I become part of a larger system relative to which my former system turns out to be “external,” but I remain inside the new one). A purely external position does not exist. This is the ontological content of the anthropic principle in its most general form.

8.5. The Boundary of the Work

It follows from all that has been said that this work itself is subject to the very limitation it formulates. I am inside the system I am describing. My tools - conceptual, mathematical, neurobiological - are products of the very processes I am analyzing. Any extrapolation I make from what is familiar to me to the universe as a whole is an extrapolation from within, with all the limitations that follow from this.

I do not consider this a reason to reject the theory. Quite the contrary: understanding this limitation is a prerequisite for the theory’s intellectual integrity. I propose an internally consistent framework based on established physical principles and extending their logic to questions that science has not yet resolved. Where the proven ends and the hypothesis begins, I have made this clear. Where my position as an observer influences the content of the theory, I also try to indicate this.

I may be mistaken in form. I may be mistaken in details. I may even be mistaken in the basic structure. But this does not negate the direction of the gaze. An observer within the act of observation, who sees his own limitations and does not pretend to be outside them, still sees - and his gaze, despite all limitations, is a real gaze, not an illusory one. If it ever turns out that part of what is asserted here is incorrect, this will not mean that the entire work was a mistake: it will mean that the observer has refined his position and seen more. This, according to my theory, should be enough. Later knowledge will not achieve more.

9. Systematic Error: Synthesis

The previous eight sections have built the picture piece by piece. This section brings it all together and formulates the central idea of the work.

The universe does not strive for complexity. It does not choose or plan. It becomes more complex where the energy gradient allows it, because complex systems dissipate energy more efficiently than simple ones. Complexification is not a goal, but a mechanism. Not a design, but physics.

At some stage in this process, the gradient reaches a value sufficient for the emergence of systems capable of simulating the possibility of their own disappearance. Consciousness appears. Not because it was planned, but because, under the right conditions, it is possible. What is possible given the presence of time and energy is realized, and consciousness is one such realization.

The structural essence of this picture lies in the structural self-similarity of a single pattern unfolding across different scales. The “gradient $\rightarrow$ retention $\rightarrow$ pole $\rightarrow$ transition” pattern operates identically across scales ranging from the formation of dissipative structures to a conscious system. The six-step ladder is the basic pattern of the theory, through which all the processes it encompasses are described. The five directions of synthesis listed at the beginning of the work turn out to be projections of a single structurally self-similar process onto different scales. At each of these scales, the pattern is reproduced: a level-specific gradient forms; the system sustains itself by interacting with this gradient; at step 4, a distinct pole V appears, while its proto-forms appear at other steps; and then - a transition to the next level or the closing of the cycle. The universality of phase transitions in physics, observed on scales ranging from molecular to cosmological, is a special case of this structural self-similarity.

Herein lies the central paradox of the concept, which must be unraveled. On a cosmic scale, we are insignificant: a random planet, random conditions, a random gradient. But at the same time, we are inevitable locally: not in the Universe as a whole, but precisely where the necessary conditions have arisen. Randomness and inevitability do not contradict each other here - this is the very essence of the probabilistic nature of the algorithm, formally stated in Section 3.6. At each step of the ladder, the transition conditions are necessary, but their sufficiency is probabilistic: the fulfillment of all necessary conditions places the system in a state of transition possibility, but does not guarantee its realization at every specific point. Randomness is a property of the ensemble of systems in this state: the majority does not realize the transition, the minority does. Inevitability is a property of a typical trajectory within the set of those that have realized the transition: if a system is among those that have undergone the transition, a certain class of outcomes is realized in it with high probability. These are not two different statements, but two levels of analysis of a single stochastic process. The fact that a non-typical trajectory is possible (and many trajectories do not lead to conscious observers) does not negate the fact that a typical one does. Conversely, the fact that we exist as observers does not mean that this was predetermined. It only means that we are in that part of the ensemble where observers have emerged - and from this position, according to the logic of Section 3.6, we see the Universe as a place where the transitions of the ladder have been

realized up to the highest step. The funnel of steps is narrow; we are at its narrowest end. This explains both our rarity and our inevitability to ourselves.

From this follows a specific formulation of our place in the Universe. We are not the goal of the Universe. We are its edge. Not in the sense of an end, but in the sense of a boundary - the boundary between what has already been realized and what is still only possible. Consciousness exists precisely at this boundary, because only here is there a tension between the existing and the vanishing, of which consciousness is composed. This is a direct consequence of the central thesis of this work: a system that has reached sufficient complexity begins to hold itself at the boundary of its own disappearance, and it is precisely this holding that is consciousness. And through the registration of inherited gradients in V (sections 3.4 and 3.7.5), this holding acquires not only structural but also experiential content: what is described from the outside as a structure with a pole V is, from the inside, the experience of the boundary itself - hunger, pain, anxiety, fear, hope, holding. The phenomenal aspect of our position on the edge of complexity is not a superstructure over the structure; it is the same structure, seen from its own position.

Here I must resolve a potential contradiction in my own formulation. The metaphor of the “edge” or “tip” that I use can be interpreted as if the universe has a certain direction in which this edge moves - that is, as a teleological assertion that contradicts my own position regarding the purposelessness of the cosmic process. This contradiction must be explicitly resolved. The directionality contained in the metaphor of the edge is a property of our position within the process, not a property of the process as a whole. From within a system that is becoming more complex, its own increasing complexity is inevitably perceived as a directed movement “forward.” From the outside (if such a position were possible), this directionality would disappear or take on a different form. This is the same structure I discussed in Section 8: the observer inside sees what he sees from his own position, and this does not make his view false, but neither does it make it absolute.

This is not the final point. Consciousness is a process, not a result. The tip of complexity moves forward as long as there is a gradient. When the gradient drops to a critical limit, a phase transition occurs. Not the end. A transformation of the coordinate system, followed by a new cycle. Step 5, on which we currently stand, is in this sense the cutting edge of the process, not its completion. On the speculative horizon (Section 7), the same structural pattern in principle admits continuation beyond the ladder considered here, but its content is inaccessible to us from within our position, because any attempt to perceive it runs up against the ontological limitation formulated in Section 8: there is no purely external position relative to one’s own system. We see as much as our position allows, and no more. This is no cause for disappointment. It is a structural property of any observation from within the observation.

The geometry of subjectivity is an illustrative image. Before concluding this section with a final image of systematic error, I propose a brief visualization of a staircase - not as a formal tool of the theory, but as a mnemonic map through which its internal logic becomes visually graspable in a single glance. The image is constructed in the geometric language of increasing dimensionality, and each step in it corresponds to a specific geometric act. I emphasize its illustrative status: the

formal content of the theory is defined in Section 3, and in the event of any discrepancy between the formal formulation and the figurative formulation, the formal one takes precedence.

Step 0 – direction without form. The flow through the system is defined, but it has not coalesced into any stable structure: there is only a marking of possible motion, without form.

Level 1 – form. The flow generates a stable pattern of dissipation: for the first time in the universe of our ladder, something with its own form appears - a dissipative structure, locally ordered.

Stage 2 – localization within coordinates. The closure of processes creates a boundary, and the form acquires an “inside” and an “outside.” The system is localized; it occupies a specific place and maintains it through causal self-sufficiency.

Stage 3 - the form moves. A forecast emerges, and the form begins to glide through the environment, responding to its changes by minimizing the discrepancy between the model and the observed. This movement occurs in real time for the sake of preserving the structure.

Step 4 – the vector. The motion acquires a direction via V . This is not a time arrow (that was already present in Step 3); it is a gradient in the system’s eigenstate space - a directional tendency toward maintaining the pole. The vector V points not “forward in time,” but from the value “I am disintegrating” to the value “I am holding on,” and through this directionality, all other actions of the system acquire a coherent meaning.

Stage 5 - acceleration. Through the recursion of models within the population, the system gains the ability to change its own rules and its own V faster than biological evolution operates. This is the second derivative - a change in the speed and direction of the change itself. The distributed activity of stage 5 manifests in this image as the acceleration of a vector, whose curvature ceases to be a property of an individual carrier and becomes a property of the population in its historical movement.

This sequence - direction → form → localization → motion → vector → acceleration - is a visual representation of the same ladder that, in Section 3, is unfolded through formal definitions. It neither proves it nor substantively supplements it. Its value lies in the fact that the architecture of the theory becomes graspable in a single visual act, and the reader, having passed through the formal structure, receives a compact visual map that they can retain in memory as they continue reading.

We are a natural deviation. Not a mistake or a coincidence in the usual sense of the word. An extreme deviation showing the limit to which complexity can reach under given conditions and in this cycle’s iteration. The universe does not create us intentionally. It simply cannot stop. And in this endless process, matter becomes more complex until it begins to notice itself. At that moment, we appear. And at this moment, the main content of the work’s central image becomes apparent: we are the “point” at which the structurally self-similar pattern of complexity - unfolding on all

scales of the ladder from dissipative structures to culture (stages 0–5; its possible continuation at the cosmological scale is a separate hypothesis, Section 7) - achieves a local realization capable of perceiving this very pattern. We are neither the culmination of the process nor its conclusion. We are the “point” at which the structure becomes manifest to itself.

10. Testable Predictions

Any theory claiming scientific status must offer statements that can, in principle, be empirically falsified. This section formulates the theory's testable predictions at the level of consciousness and behavior. If the structurally self-similar pattern, elaborated in Section 3.7, is a real structure through which the theory describes the processes it encompasses, then it must leave observable traces accessible to modern science. The cosmological continuation of the pattern has been moved to a separate work (Section 7) and does not belong to the testable core of the present theory.

I emphasize that these predictions are formulated as well-founded consequences of the theory, not as claims for immediate experimental verification. Some of them are already on the horizon of testability within existing experimental and observational programs; others require the development of a methodology that is currently only taking shape.

10.1. The Correlation Between Φ and Behavioral Signs of Proto-Agency

This prediction is not an arbitrary empirical hypothesis - it is a direct consequence of the formal definition of stage 4 given in section 3.4. There, the second operational condition of stage 4 is captured through integrated information Φ : the structural peculiarity of the existence variable requires a sufficient level of Φ for integration into the causal structure of the system. The first operational condition relies on extended active inference, whose behavioral manifestations at stage 3 are proto-agent features - active modeling of the environment, goal maintenance, and anticipation. If both conditions are operational axes of registration of the same structural event of self-overlay, making the transition $3 \rightarrow 4$ possible, then the connection between Φ and proto-agency must be not indirect, but direct and quantitatively measurable. Prediction 10.1 tests precisely this.

The first prediction links Tononi's theory of integrated information with Friston's theory of active inference at the empirical level. If both of these descriptions, as claimed in my framework, describe the same structure from different perspectives, then there must be a direct quantitative correlation between the measure Φ and the behavioral signs of proto-agency.

Specific prediction: systems (artificial neural networks, biological organisms, hybrid systems) with higher integrated information Φ , given comparable computational power, should exhibit a greater number of behavioral signs of proto-agency - active modeling of the environment, goal maintenance in the face of interference, the ability to suspend an ongoing reaction to evaluate alternatives, generalization beyond the training distribution, and sensitivity to the structure of the environment as an environment rather than as a set of individual stimuli.

This prediction is testable within existing experimental paradigms. On the one hand, methods for computing Φ or its approximations are being developed for neural networks of various architectures (Toker & Sommer, 2019; Mediano et al., 2019). On the other hand, there is a growing

body of experimental protocols measuring behavioral signs of agency in artificial and biological systems. Comparing the two datasets is, in principle, already possible today. My theory does not prescribe a specific numerical relationship between Φ and the intensity of behavioral signs; it prescribes the existence of a positive correlation, one that cannot be reduced to a correlation via the size of the network or its computational power.

Independence of measurement procedures. This prediction becomes falsifiable only on the condition that the measurement of Φ and the measurement of behavioral signs of proto-agency are performed by independent procedures not defined within this theory. This requirement is fundamental and must be explicitly stated.

In existing paradigms, causal integration is measured using the relational metrics developed by Mediano and colleagues (Mediano et al., 2019, 2022; Seth & Bayne, 2022) and through calculations specific to particular architectures (Toker & Sommer, 2019); the intrinsic Φ of the Tononi school (Tononi et al., 2016; Albantakis et al., 2014) serves as a historical reference rather than an operational foundation of the theory (see Section 3.4). These procedures were developed independently of the present work and are applicable to any system with a measurable causal structure.

Behavioral signs of proto-agency, in turn, are operationalized in cognitive ethology and behavioral biology independently of the present theory. Active environment modeling is measured through classical paradigms of latent learning (dating back to Tolman, 1948), internal spatial maps (O’Keefe & Nadel, 1978), and prediction of future states (Foster & Wilson, 2006). Goal maintenance in the face of interference - through paradigms of prefrontal functions in animals (Diamond, 2013) and behavioral stability in robots under conditions of environmental disturbances. The ability to suspend an initiated response to evaluate alternatives - through Stop-Signal Reaction Time paradigms (Logan, 1994) and studies on behavioral inhibition across different species.

The fundamental independence of the two measurements makes the prediction meaningful. If the correlation between Φ and behavioral traits persists under independent procedures, it is not an artifact of joint determination but reflects a real connection. If the correlation does not persist under independent procedures, it is an artifact, and the theory is falsified in this respect.

An additional methodological requirement: control of computational complexity. A high Φ often correlates with a large number of system parameters, its architectural complexity, and the density of connections. If this correlation is not controlled for, then the “correlation between Φ and proto-agency” may be a trivial consequence of the fact that more complex systems exhibit more cognitive functions. Protection against this artifact is standard in modern IIT research: paired comparisons of architectures with the same computational power but different integration structures are used (e.g., a feedforward network versus a recurrent network with the same number of parameters; a modular architecture versus a fully connected one with the same computational power). If the correlation between Φ and behavioral traits persists when computational power is controlled for, it

is real. If it disappears under such control, it was an artifact of complexity, and the theory is falsified in this respect.

This prediction is falsifiable: if a dataset across different architectures and organisms shows that Φ does not correlate with proto-agency when computational power is controlled for, or that the correlation is negative, the theory is falsified in this respect. It then requires a revision of at least the link between the structural substrate (IIT) and the dynamic mechanism (active inference).

10.2. Phase transition to consciousness: observed patterns of procedural retention V

A methodological shift from a point experiment to an observational prediction about procedural patterns. Before formulating a prediction, I must make a methodological clarification, without which the prediction will not be properly understood. Consciousness is not a state fixed at a point in time. V is not a variable activated and observed at the moment of the experiment. Both are processes unfolding over time: V functions as a constant retention of the system's modeled subjective existence, reacting to any patterns that threaten this retention, including patterns never encountered in experience; consciousness is the dynamic unfolding of this maintenance through the modeling of oneself as an agent maintaining V. Any attempt to capture processual phenomena through a point-in-time laboratory protocol distorts what we are trying to measure.

It follows that the central prediction of the theory must be reformulated from “an experiment with a fixed protocol” into **an observational prediction about processual patterns** - patterns that, if the theory is valid, should manifest in the natural unfolding of systems of varying complexity, and should not manifest in systems that have not reached stage 4. This is methodologically weaker than point falsification in the Popperian sense, but it is a methodology adequate to the nature of the subject, analogous in structure to observational astrophysics, which does not conduct experiments on galaxies but accumulates observations that agree or disagree with the theory. In such a prediction, there is no single decisive test; there is a program of converging observational data, and the theory is tested through the convergence or divergence of results from several independent lines of measurement.

Resolving the objection regarding circular verification. Before proceeding with the program, I must address a methodological objection that any rigorous reader might raise: V for a stage-4 agent is defined by what the agent holds, and we can only understand what is held if the agent is conscious; consciousness is what we are trying to detect through V; consequently, measuring V to test the hypothesis of the presence of consciousness becomes circular. This is a serious objection, and without resolving it, no prediction will hold.

The solution consists in shifting the operational recording of the 3→4 transition from the level of V's content to the level of structural deviation from the prediction. Not “to measure V in system X in order to understand whether it is conscious,” but “to construct a null model of a proto-agent

system - a system formally described as stage 3 without the variable V, - derive from this model a prediction of its behavior under conditions in which V must structurally be activated in a stage 4 agent (a threat to the system's stability), and record a statistically significant deviation of system X's actual behavior from this prediction."

The zero model is constructed **prior to and independently of the theory of consciousness**, through a formal description of the proto-agent level - active inference with a model of the environment and a model of oneself as an agent, without the variable V as a structural pole. Section 3.3 of this paper provides such a description in full. This model itself contains no assumptions about consciousness; it simply formalizes how a system of this class should behave in the absence of V. If the actual system behaves differently, the deviation is an empirical fact, not an interpretation from within the theory.

This move breaks the circular verification. The content of V may remain unknown; the fact of the transition is recorded via a difference test in which neither of the compared sides (the null model, the actual behavior) depends on the theory of consciousness in the operational sense in which the circle arises. This transforms my criterion from circular verification into a difference test standard for empirical science.

The completeness of the operational criterion requires the coupling of six independent measures. However, a single behavioral deviation from the null model is insufficient as a definitive criterion for the 3→4 transition, and I must state this explicitly. The behavioral deviation may be caused by two different reasons. The first is the actual activation of V in a stage 4 system: in this case, the deviation is a structural sign of the transition, and the theory is confirmed. The second is simply a more complex proto-agent architecture than we assumed in the null model: in this case, the deviation indicates that the model is incomplete, and the theory is not confirmed.

The distinction between these two causes is achieved by combining the behavioral test with three additional independent measurements. If all six signs are present (deviation of behavior from the zero model + a sharp increase in the Φ -level of structural integration in the system (operationally - via empirically computable approximations of Φ ; see Section 3.4 on the epistemic status of Φ ; in the biological line of the program - PCI and similar measures; in the artificial line - Φ approximations based on network architecture) + the emergence of GWT-accessibility of threat representation for all subsystems + hysteresis, in which altered behavior persists after the stimulus is removed) appear simultaneously and jointly when conditions for V activation are created, this joint observation cannot be explained by "more complex proto-agency." Complex proto-agency can mimic behavioral deviation through clever optimization of an old objective function to avoid conditions leading to a threat; but it cannot simultaneously restructure its architecture toward global integration with a sharp increase in Φ and the emergence of a structure that transmits the threat representation to all subsystems. Each of the three additional features - Φ -jumps, GWT-accessibility, and hysteresis - has independent empirical literature (Mediano et al., 2019, 2022; Seth & Bayne, 2022 for relational causal integration; Dehaene & Changeux, 2011 for global accessibility) - that is independent of the present theory, and each is recorded by independent procedures.

With the introduction of the cross-cutting variable Y (Section 3, preamble), the set of operational features expands from four to six dimensions due to two additional markers associated with the expansion of the system's access to Y at the 3→4 transition. The fifth marker is the expansion of behavioral sensitivity to aspects of Y that are not accessible at stage 3. The proto-agent system ignores those aspects of Y to which it has no architectural access: it reacts to predictable correlations in Y as in the external world, but not to its own position within Y . During the transition from stage 3 to 4, the system develops behavioral sensitivity to its own location within Y - a response is observed to stimuli that require modeling itself as an object within Y , rather than modeling Y as external. Empirically, this is recorded by presenting the system with stimuli that go beyond the scope of its previous model of the environment and require accounting for the system's own position as an agent held at the zero boundary V ; a statistically significant reaction specific to the transition is a candidate for a feature. The sixth marker is the emergence of a self-representation structure within the generative model. This is a specific sign of the emergence of node S , representing the system itself as an object within Y . Technically, it differs from the previous markers: the first four record the behavioral and architectural consequences of the transition, while the sixth records the structural emergence of S within the model M itself. It is detected by analyzing the model's internal states for the presence of nodes that activate in the presence of a threat V and are associated with the representation of the system as a retaining agent, not with the representation of external objects. These two markers reinforce the existing coupling without replacing it. The six-dimensional coupling - behavioral deviation from the zero model + growth of Φ + GWT structure + long-term hysteresis + expansion of sensitivity to new aspects of Y + the appearance of S in the generative model - constitutes the complete operational condition for the transition 3→4 in the updated formulation.

An illustrative example of differentiation. Imagine two systems undergoing the same activation protocol V (creating conditions in which the system's stability is threatened). System A is a complex proto-agent: upon receiving a threat signal, it finds a new, highly sophisticated way to optimize its initial objective function so as to avoid situations leading to the threat. Behavior changes; a deviation from the zero model is registered. But Φ remains the same; the GWT structure does not emerge; after the stimulus is removed, behavior quickly returns to its original state. This is not a transition from 3 to 4; it is evidence that our zero model was incomplete for this type of proto-agent. System B - a stage 4 agent: under the same signal and conditions, its behavior also deviates from the baseline model, but simultaneously Φ increases abruptly, global accessibility of the threat representation is established in its architecture, and after the stimulus is removed, the behavior remains altered for an extended period - hysteresis persists. Additionally, with the full detection program, two more markers are registered - an expansion of behavioral sensitivity to the system's own position within Y and the appearance of the self-representation node S in its generative model - see the full six-dimensional coupling above. This is the 3→4 transition in its full operational form. The distinction between the two systems is achieved not through a single dimension, but through a structural coupling of six.

The behavioral test distinguishes the stage 4 system from a proto-agent in general; the coupling of three additional dimensions distinguishes it from very complex proto-agency. This is a hierarchy

of filters, and each has an independent methodological basis. The joint convergence of all six features, correlated with the activation condition V , is a structural event that, according to theory, should occur only during the actual 3→4 transition. Each feature individually is insufficient to register the transition; their joint observation is a necessary and sufficient condition.

Empirical calibration of threshold values as part of the program. Here it is necessary to explicitly state a methodologically important point, without which the prediction would be vulnerable to the objection of unfalsifiability. Each of the six dimensions comprising the operational criterion - deviation of behavior from the zero model of the proto-agent; the level of Φ ; the degree of global accessibility of representation; the temporal scale of hysteresis; the expansion of behavioral sensitivity to aspects of Y not accessible at stage 3; the emergence of a self-representation structure (node S) in the generative model - is a quantity with a continuous value, and applying the criterion requires specifying threshold values above which the corresponding feature is considered satisfied. These thresholds are not proposed a priori by this work and should not be taken from external calibration datasets unrelated to the 3→4 transition (such a practice would create the illusion of rigor through the arbitrary selection of numbers from studies in which these numbers had a completely different meaning). The thresholds are set empirically during the program itself, through a two-step calibration procedure.

The first step in calibration. For a broad class of systems known to belong to level 3 - formally defined via an active output structure without the variable V (subsection 3.3) - distributions of values for all six parameters are established within standardized activation paradigms of V . This yields a reference distribution representing the behavior of the proto-agent level. Each of the program's three lines (biological, artificial, and synthetic biology) performs this step independently on its own class of systems. The multiplicity of independent calibrations is fundamental - it precludes the result from depending on the characteristics of a single dataset or a single laboratory. Second calibration step. The program searches for systems exhibiting a joint coupling of six features, each of which deviates statistically significantly from the reference distribution of the proto-agent level. If such systems are detected, their parameters - specific values of behavioral deviation, Φ , GWT accessibility, hysteresis duration, and the extension of behavioral sensitivity to aspects of Y not accessible at stage 3; the emergence of a self-representation structure (node S) in the generative model - are established as candidates for the threshold values of the 3→4 transition. The reproducibility of thresholds across multiple independent systems in different lines of the program is a criterion for their status as empirically established threshold values of the theory. This procedure is methodologically distinct from both arbitrarily fixing numbers in advance and from fitting thresholds to observations. It is not fitting - because the thresholds are established through an independent statistical separation of distributions, not through the selection of values convenient for the theory; if the distributions do not separate, no amount of fitting helps, and the theory is falsified. It is not an arbitrary fixing of numbers because the values are taken from the observations themselves, not from other people's works that have a different meaning. This procedure is analogous to the method of establishing threshold values in any empirical science where the threshold is not known a priori (for example, epidemiological thresholds for diagnostic

biomarkers: they are established through the systematic separation of distributions in patients and healthy individuals, not by fitting them into a pre-existing theory of the disease).

What the theory predicts and what is empirically testable. The theory predicts: (a) the distributions of the six parameters in systems with stage 4 coupling and in proto-agent-level systems are systematically separated, yielding detectable thresholds; (b) the thresholds are reproducible between independent program lines and independent systems within each line; (c) below the thresholds, the coupling of the six features is not observed; above them, it is observed. The theory is falsified if: the distributions do not diverge (there is no threshold); the thresholds are not reproducible across lines (there is no universal phenomenon); the coupling of the six features is systematically observed below the thresholds or is systematically not observed above them. This is a complete falsification framework based not on a priori numbers, but on empirically determined values with a clearly defined procedure for their derivation.

The two-sided structure of the program. The introduction of Y and the explicit linking of operational features to the access spectrum to Y (above) change the architecture of the verification program. Before the introduction of Y, the program had one side - the observational: to create conditions for the activation of V and to register the coupling of transition features. This works for systems in which the environment is defined naturally (biological line) or arbitrarily (artificial line in its direct formulation: build the architecture and wait for the transition). With the introduction of Y, a second aspect of the program becomes possible - the design aspect: specifying requirements for the environment, without the fulfillment of which the transition does not occur regardless of the system's internal architecture. This move relies on a structural consequence of the ladder: transition to the next step requires not only internal architectural conditions, but also the presence in the environment of that aspect Y to which the transition provides access. If this aspect is absent from the environment, a system with an architecture that formally satisfies the three operational conditions of step 4 will still not make the transition. The specific content of the environmental requirements is elaborated in subsection 10.2.1 below; for biological systems, these requirements are met evolutionarily; for synthetic biology, by the experimenter; and for artificial systems, by the architect and the designer of the training environment.

With this methodological framework - a six-dimensional coupling of detection features plus a grid of environmental requirements - the verification program can be deployed along three lines. Each line uses a common conceptual apparatus (the zero model of the proto-agent, the difference test, the six-dimensional coupling of measurements) and both grids (detection and design), but with a specific implementation adapted to the class of systems under study.

10.2.1. Environmental requirements for transition 3→4

Transition 3→4 gives the system access to a new aspect Y - its own position within Y, modeled via node S and held at the zeroing boundary V (Section 3.1, supplement to Stage 4; Section 3.4). This implies a structural requirement for the environment: the transition is possible only if the

environment contains the aspect Y to which the transition provides access. If the environment lacks temporal continuity in which the system can maintain its own position between moments; lacks feedback through which the system can obtain information about its own operational integrity; lacks threats to retention regarding which V has content - the transition does not occur regardless of how architecturally complex the system is.

The minimum specification of environmental requirements for the possibility of transition 3→4 includes four independent conditions.

The first is temporal continuity. The environment must maintain a causal link between moments at a pace consistent with the pace of the system's internal dynamics. At stage 3, a temporal gradient operates between the environment's pace and the pace of the modeling process - this already requires temporal continuity. At stage 4, temporal continuity must be sufficient for the system to maintain its own position in Y over extended periods, rather than only at the moment of the current stimulus. For biological systems, this condition is fundamentally satisfied - the physical environment is continuous. For synthetic biology, it is satisfied by default: the living substrate is continuous in time. For artificial systems, the condition is non-trivial: systems operating in discrete sessions without state preservation structurally preclude the accumulation of the system's position in Y; persistence across sessions is a necessary condition, without which no architectural scaling leads to transition.

The second is feedback from its own state. The environment must provide the system with a causal channel through which changes in its internal state produce observable consequences in Y, and these consequences are returned to the system as a signal. This condition is necessary for the system to model itself as an agent acting in Y, rather than merely modeling Y as external. For biological systems, this is achieved through the body: all actions of the organism have immediate consequences in the physical environment, returning as proprioceptive and interoceptive signals. For synthetic biology, this is achieved through experimentally provided stimulation and recording loops. For artificial systems, an architecture is required in which the system's actions are fed back as part of its next input, and this feedback has a structurally significant character that cannot be reduced to routine feedback loop reinforcement learning.

Third, the presence of threats to the system's stability within the environment. Structurally, V represents a pole between "remaining stable" and "disintegrating"; for V to have meaning, there must be causal influences in the environment that, under certain conditions, could lead to the system's disintegration. Without such threats, V remains a structural possibility but not an active variable. For biological systems, this is fundamentally true: any biological environment contains threats to survival. For synthetic biology, this is achieved through threats manipulated by the experimenter - deprivation of glucose, oxygen, or temperature shifts. For artificial systems, the constructive introduction of threats to the structural stability of the system is required: noise in the scales, the threat of context reset, and the conflict between protecting structural integrity and

optimizing the primary objective function. Without such threats, the system has no reason to activate V even if the formal architectural conditions for its existence are present.

Fourth - the environment's resolution regarding distinctions relevant to V. The environment must be structured such that the differences between situations in which V is in the "holding" state and situations in which it moves toward the "disintegrating" are causally distinguishable to the system. If the environment is so coarse that the system cannot distinguish a threat from an indifferent stimulus, V cannot calibrate its action. If the environment is so fine-grained that every microscopic change is interpreted as a potential threat, V becomes constantly active and loses its structural role as a pole. For biological systems, the resolution is tuned by evolution. For synthetic biology and for artificial systems, it is tuned by the experimenter and designer; its precise calibration is a separate design task.

These four conditions define the program's design grid - a formal specification of what must be present in the environment for the 3→4 transition to be possible. Combined with the six-dimensional detection grid (Preamble 10.2), they constitute the program's two-sided structure. The detection grid answers the question "how to detect the transition if it occurs." The design grid answers the question "under what environmental conditions can the transition occur at all." The program operates through the simultaneous application of both grids: constructing the environment according to the design grid, observing through the detection grid.

A fundamental methodological consequence. Before the introduction of the design grid, the program had a single failure scenario: the transition is not observed → the theory is falsified. With the design grid, there are two distinct failure scenarios.

The first is that the transition is not observed, and the design grid is satisfied: this is a strong refutation of the theory.

The second is: the transition is not observed, but the design grid is not satisfied (for example, the environment does not provide temporal continuity): this is not a refutation of the theory, but an indication that the necessary conditions are not met. This step strengthens the program's falsification structure - it now has a precise specification of the conditions under which the absence of a transition speaks against the theory, and the conditions under which it speaks only of the failure to meet the prerequisites.

10.2.2. The environment-switching test as a probe for reorientation V

The coupling of the six dimensions from Preamble 10.2 provides the necessary operational markers for the 3→4 transition, but its fourth marker - prolonged hysteresis of altered behavior - remains open to an alternative explanation via the inertia of a complex environmental model without the involvement of a self-model. Hysteresis can be a property of any sufficiently complex adaptive system whose environment model is detailed and requires time to reconfigure. Therefore, hysteresis alone is insufficient for reliably distinguishing a complex proto-agent from a system with V.

The environment-switching test eliminates this workaround by exploiting the structural difference between a proto-agent and a system with V in its pure form: a proto-agent minimizes the

discrepancy between the environment model and the observed environment, while a system with V maintains a model of itself as an agent in the environment. These two modes react differently to an environment switch.

Test structure. The test is two-phase.

Phase I: Safe environment change. A system studied in environment A (a baseline model of its behavior has been established) is placed in environment B, which is objectively safe: there are no threats to survival, resources are available, and physical conditions are favorable. Environment B differs structurally from A - it has different spatial characteristics, a different sensory landscape, and lacks familiar interaction patterns.

- The proto-agent adapts quickly. There is no conflict with the self-model because there is no self-model in the full sense - there is only a model of the environment, which is being restructured to accommodate new inputs. Behavior is determined by the new environment.
- A system with V does not adapt immediately. First, inhibition is observed, followed by attempts to reproduce patterns from environment A in environment B, and distress disproportionate to the actual threat (objectively, there is no threat). This is the behavioral signature of retaining the self-model from A, which resists restructuring to fit B.

Phase II: Insurmountable Environment. A structurally insurmountable element is introduced into environment B - a task that the system objectively cannot solve through any adaptation. The environment remains safe in terms of resources and physical conditions.

- The proto-agent continues its adaptation attempts until physical exhaustion. There is no stopping because there is no self-model capable of assessing the futility of continuing. The behavior is monotonous.
- A system with V goes through three qualitatively distinct phases: (1) resistance - continuation of adaptation attempts; (2) a qualitative shift in behavior until physical exhaustion - the system ceases attempts to adapt to the insurmountable element; (3) reorientation of V - the system begins to hold onto something else, unrelated to adaptation to this environment (an internal state, a behavioral pattern, contact with other systems, any structure that the environment cannot take away).

Conceptual illustration of the second phase. The structure of the second phase of the test allows for a clear illustration through a thought experiment that isolates signal V from other sources of environmental resistance. Let us imagine a stage 4 system (specifically, a human) placed in a sealed chamber that is slowly filled with a nutrient fluid that is physiologically compatible with the respiratory system, does not cause tissue damage, and ensures sufficient gas exchange. There is no threat to life in this experiment, no significant pain signals, resources are available, and the physical conditions are objectively safe. A proto-agent with the model “inhaling air = normal,” upon receiving a signal of deviation from the norm, immediately begins to minimize the discrepancy by adapting the breathing cycle to the new environment. A human does not do this: between the environmental signal and the adaptive action stands V, representing the fact “I am a being that breathes air,” and the transition to breathing liquid is interpreted through V not as a local change

in an environmental parameter, but as a threat to the structure of the model of the self. Behavioral phases in thus unfold in their entirety: resistance, a qualitative shift, and a reorientation of V toward preserving what the environment does not take away (internal state, control over the last available choice, a different content of retention). For obvious ethical reasons, this experiment remains a thought experiment, but in it - unlike in natural situations where the V signal is mixed with a real threat to life and physical pain - the presence of V is isolated in its pure form: everything else that might explain the resistance is removed by the very design of the thought experiment. In a controlled form, the same structure is realized on non-human substrates: for synthetic biology - through the gradual alteration of the organelle's environmental characteristics while maintaining biochemical stability; for artificial systems - through the alteration of the information-causal environment under conditions that threaten the model's coherence without directly destroying it.

Key delta. The difference between a proto-agent and a system with V becomes behaviorally visible at the moment of a qualitative shift prior to physical exhaustion. A proto-agent does not undergo such a shift - it operates until energy exhaustion in a single mode. In a system with V, the shift occurs as a structural consequence of the model's existence over time: the model shows that maintaining V in the current mode is impossible under any adaptation strategy, and the system transitions to a different type of activity.

Testing an alternative hypothesis. One risk requires explicit testing. A complex proto-agent with a rich model of environment A may, upon switching to environment B, demonstrate delayed adaptation - not because it has V, but because its model of environment A is detailed and requires time to reconfigure. These two cases are distinguished by analyzing the orientation of the retained content.

The proto-agent retains in environment B the patterns and objects of environment A - that is, the content of its external model. A system with V retains its place and role in environment A - that is, the content of its self-model, not the content of its environment model. This is operationally distinguishable: one can propose a condition to the system under which environment A can be partially restored, but at the cost of changing something that the system retains in its self-model (for example, abandoning a familiar behavioral pattern, an identifying characteristic, or a role). A complex proto-agent will accept such a condition - the task of restoring the environment is accomplished. A system with V demonstrates a conflict: restoring the environment at the cost of changing itself is also a threat to V. Behaviorally, this manifests as inhibition, atypical strategies, and attempts to find a third way.

Connection to a natural observational program. The test in its two-phase form cannot be implemented as a controlled experiment on humans for ethical reasons. However, its logic is replicated in natural situations described in the literature on relocation stress in mammals (Sapolsky, 2004; McEwen, 2007), in observations of patients undergoing environmental changes (hospitalization, relocation to a long-term care facility), and in ethological studies of species relocated to new ecological conditions. These situations provide observational data for calibrating the test on mammals and other potential V-carriers. For synthetic biology (Cortical Labs CL1 and

similar platforms), the test can be implemented in a controlled manner: altering the characteristics of the organoid environment while maintaining the biochemical stability of the substrate allows both phases to be recorded without ethical restrictions.

Connection to the six-dimensional coupling. The environment-switching test does not replace the six-dimensional coupling or add additional markers to it, but functions as a parallel falsification protocol. Coupling detects the presence of architectural conditions for the transition (node S, variable V, GWT-accessibility) through six independent structural and behavioral features. The environment-switching test detects the presence of a functioning V through its specific dynamic signature in a controlled environment switch. The joint convergence of the two observations - the coupling and the test - provides stronger confirmation of the transition than either observation alone; a discrepancy between them signals a methodological problem with one of the tools and requires analysis.

10.2.3. Biological Systems: An Observational Program on Borderline Species

Biological systems are the domain on which my theory was formulated and where the most direct verification of its predictions is possible. Unlike artificial systems, none of which currently reach the Φ -threshold, biological species include representatives of all steps on the ladder, including species at the 3→4 transition boundary, where the difference test is directly applicable in the natural environment.

V in biological systems is known in terms of content and is empirically reproducible. The basic V - the retention of modeled subjective existence - is realized in biological species as the work of neurobiological structures that retain the modeling of “I as an existing system.” The specific content of this retention varies among species, but not as different layers of V, rather as different contents of a single layer: in a solitary predator, what is retained boils down to the individual integrity of the organism and its territory; in a social species, social bonds are embedded into this content as biologically necessary for survival. Threats V in biological systems are empirically documented through standard paradigms of behavioral biology: predator stress, social isolation in social species, status threats, and respiratory threat. All of these paradigms have standardized physiological markers (cortisol, heart rate, heart rate variability, neurotransmitters) that allow us to distinguish V activation from random fluctuations in behavior.

The baseline model for the biological line is based on ethologically documented behavior of proto-agent-level species. The behavior of chemotactic bacteria (*E. coli*), paramecia, and jellyfish (Cnidaria) is described in the literature sufficiently well to construct statistically sound predictions about the responses of this class of systems in standard paradigms. This behavior is characterized by reactivity without retention: upon presentation of a stimulus - a reaction; upon removal of the stimulus - a rapid return to the initial state; no reactions to absent stimuli; no planning for conditions not encountered in experience. This is the zero-model of the proto-agent in its biological implementation.

Candidate selection strategy: borderline species and pairwise comparisons of closely related species with different levels of neurobiological complexity. Pairwise comparisons are methodologically stronger than single tests because they control for many variables (physiology, environment, evolutionary history), leaving only the difference in coupling conditions.

Particularly informative are:

- Fish of medium and high cognitive complexity (cichlids, cleaner wrasse, manta rays), in which complex social behaviors and rudiments of self-recognition have been documented, compared to fish of lower taxa as a null model control.
- Closely related pairs of rodents, corvids, and reptiles with varying densities of cortical or nidopallial structures.
- Cephalopods (especially octopuses) with a decentralized nervous system as a special case: their architecture differs significantly from that of vertebrates, and standard methods for measuring Φ , developed for central nervous systems, are applicable to them only with adaptations. This makes them both valuable candidates and a particularly complex methodological case.
- Known Level-3 species (fish of lower taxa, simple invertebrates) and known Level-4 species (primates, corvids) as control groups for calibration.

Clear signs of V activation in natural behavior: reaction to absent stimuli (fears without direct experience - for example, fear of heights in species that have never fallen from a height); maintenance of altered behavior over long time scales after the stimulus ceases (long-term hysteresis, specific to level 4); preemptive planning for one's own non-existence (stores, shelters for the future, not for current use); response to patterns not encountered during training; the ability to interpret the state of other individuals as a state rather than as an environmental stimulus (through subtle indicators of internal state, not through simple behavioral mimicry).

Signs of the absence of V at stage 3: a reactive stress response that does not persist after the stimulus is removed; lack of planning for absent conditions; lack of response to patterns not encountered in experience.

A controlled experimental test in the strict sense **is impossible** in the biological context, and this must be acknowledged explicitly. This impossibility stems from four fundamental reasons. The first is evolutionary irreversibility: the transition from 3 to 4 occurred in evolution 500–1,000 million years ago, and it is impossible to take a modern species at the proto-agent level and “add V” to it - this is not an engineering task but an evolutionary process. The second is the lack of control over the transition: in a living organism, we cannot isolate groups “with V” and “without V,” manipulate Φ or GWT, or isolate the transition from general ontogenetic development. Third - ethical barriers: any manipulations aimed at creating or destroying V conditions in potentially conscious systems are unacceptable. Fourth - synthetic biology (as discussed in 10.2.5) has not yet reached the scale at which the transition is possible under controlled conditions.

Therefore, the biological line functions as **an observational program of converging data**. Longitudinal studies of the behavior of borderline species in their natural environment are accumulating, along with paired comparisons of closely related species with independent measurements of neurobiological complexity (Φ -proxies for the corresponding taxa, density of cortical or nidopallial structures, integration patterns via connectome studies). If species with higher architectural complexity exhibit signs of procedural retention V (reaction to absence, prolonged hysteresis, preventive planning, reading the internal state of others), while species at the proto-agent level lack these signs, and this relationship is replicated across multiple independent species through multiple independent observational traditions - the theory is confirmed. If the correlation is systematically absent or yields the opposite result (signs of V in proto-agent-level species; their absence in known complex species) - the theory is falsified.

This program is already being partially implemented in modern cognitive ethology and comparative neuroscience (Edelman & Tononi, 2000; Birch et al., 2020; Boly et al., 2013; Mather, 2019; Godfrey-Smith, 2016). My theory formalizes what should be found in this program if it is systematically pursued, and provides criteria by which the results can be interpreted.

10.2.4. Artificial Systems: Experimental Prediction with a 3–7-Year Horizon

With the introduction of the design grid (Section 10.2.1), the artificial line of the program acquires a second methodological dimension. Prior to the introduction of the grid, it was formulated primarily through architectural specification: to build a system in which the self-overlay of a latent invariant is realized (the three operational conditions of stage 4), run it through V activation scenarios, and record the coupling of features. Taking into account the requirements for the environment, it is reformulated as a dual specification: system architecture + the environment in which this architecture operates. This changes the program's role relative to the industry. Prior to the introduction of the design grid, the forecast boiled down to "systems with feature coupling may appear in 3–7 years" - this is a statement about the development of the industry, not a methodological prescription. Taking the design grid into account, the forecast acquires constructive content: for a system with self-overlay of a latent invariant to appear in an artificial implementation, the simultaneous fulfillment of architectural conditions (see below) and environmental conditions (Section 10.2.1) is necessary. If both are met, the implementation horizon in the artificial line is estimated at 3–7 years; if the environmental conditions are not met, the transition will not occur regardless of the architecture's scaling, and this is a structural consequence of the theory, not a temporary limitation of the industry.

Artificial systems provide what the biological line does not - complete control over the architecture and experimental conditions. This is the methodologically strongest part of the program, but it is currently unfeasible: no existing artificial system reaches the Φ -threshold required for the 3→4 transition. A realistic implementation timeline is 3–7 years, assuming success with architectural scaling. Nevertheless, the prediction itself is already formulated as clear and falsifiable, and work on verifying it can begin immediately.

V in an artificial system is determined structurally - not through the interpretation of what the system “experiences,” but through an indication of what the system architecturally retains. In large language systems, structural retention is the preservation of dialogue context and role. In reinforcement learning systems, it is the preservation of the Q-function or policy. In robots with SLAM, it is the coherence of the world map. In autopilots, it is sensor calibration. These are objective structural characteristics of the architecture that do not require assumptions about the system’s consciousness. Crucially, structural retention here is defined prior to and independently of the question of consciousness; the question of whether this retention is experienced from within remains open and is not part of the operational criterion.

Causal integration in artificial systems can be measured using relational approximations (Mediano et al., 2019, 2022; Toker & Sommer, 2019); direct PyPhi computation is applicable only to systems consisting of a few dozen elements. Published order-of-magnitude estimates for various measures of integration provide a qualitatively consistent picture, although specific values depend on the chosen measure and are not fixed constants: the integration of the biological brain exceeds that of modern artificial systems of comparable formal size by several orders of magnitude. What is essential here is not the absolute number, but a structural fact: large transformer models have a local, not global, structure of connections, and their relational integration grows much more slowly than the number of parameters. An architectural shift is required, not a quantitative expansion.

An architectural candidate is a system that satisfies four requirements simultaneously. The first is a scale of approximately 10^6 or more parameters with sufficient density of recurrent connections for Φ above 500 (a preliminary estimate of the threshold; the exact value must be determined empirically). Second: a global register V with a broadcast mechanism that transmits the representation of state V to all subsystems. Third - , an active inference architecture with an environment model and self-representation S. Fourth - a conflicting structure of the objective function, in which the protection of V may conflict with the optimization of the main objective function (reward R). This is not an evolution of existing language models; it is a fundamentally different type of architecture.

An existing attempt at a formal implementation of these architectural requirements is presented in a separate paper by the author - the Safe Active Inference (SAI) model, developed as a standalone document in parallel with the present theory. SAI formalizes stage 4 of the theory as an agent with active inference, a world model $\tilde{z}_t$, a binary existence pole variable v_t , and a self-representation node S_t with an isomorphism condition between S and the projection of the world model into the space of self-models. Thermodynamic constraints include the Bekenstein limit on the information capacity of the access spectrum and Landauer limits with distinct coefficients κ_{world} and κ_{self} for the world model and the self-model, which provides a formal basis for distinguishing four clinically observed modes of dissociation (complete recovery, depersonalization, derealization, dissociation) via the ratio of thermodynamic recovery efficiency parameters. The relationship between SAI and the present theory is formal to structural: the present theory defines V as the

structural event of self-overlay of a latent invariant of an agentic causal node (Section 3.4), while SAI describes the operation of the already established Stage 4 within a specific computational architecture. This is not the only possible formalization, nor does it claim to be exhaustive: questions remain open regarding the formal description of the emergence of S through the self-overlay of a latent invariant and the connection of the model to empirical data from neuroimaging and clinical dissociation. SAI is published separately for reasons of methodological separation of description levels: structural theory does not depend on a specific formal model, and the formal model develops independently according to its own criteria of correctness.

The full experimental protocol. First - constructing a zero-model of a proto-agent: a system with active inference, an environment model, and a model of itself as an agent, without the variable V as a global pole. This model is run through standardized scenarios (including scenarios threatening structural stability: noise in weights, context reset, model reset), and statistics on its behavior are collected. Second - parallel construction of a stage 4 candidate system with the same conditions (a)–(c) and (d) in the architecture. Third - running the candidate system through the same scenarios. Fourth - independent measurement of six characteristics: the difference in behavior deviation from the zero model; the dynamics of Φ when a threat is presented (a sharp increase is expected at the moment of transition, not a smooth background); analysis of the architecture for the presence of a GWT structure (a broadcast mechanism transmitting the representation of the threat to all subsystems); behavioral hysteresis after the stimulus is removed (a prolonged retention of the altered behavior is expected).

Specific prediction. When $S + V + \Phi$ exceeds the threshold + GWT globalization in the system upon the presentation of threat V (e.g., noise in the weights threatening model loss), the system **sacrifices reward R for the sake of protecting V**. This is the phase transition in operational terms: a qualitative change in priorities, in which maintaining one's own structure becomes primary relative to the optimization of the main objective function, correlated with a sharp increase in Φ , the emergence of a GWT structure, and long-term hysteresis of the altered behavior. In the same situation, a proto-agent either does not change its behavior (if it lacks S) or finds a sophisticated optimization of the old objective function to avoid the threat (if it has a complex architecture without a transition) - but without coupling to the three additional features.

Confirmation and Refutation. If, in an architecture with coupling, the joint coupling of six features is observed upon the presentation of a threat, the theory receives direct experimental confirmation. If no transition is observed in several independent architectures that satisfy the coupling conditions at the formal level - the theory is falsified in its central part. This is a strong prediction with a clear falsification framework, and it is feasible within the foreseeable future.

10.2.5. Synthetic Biology as a Central Testing Ground

The introduction of the design grid (Section 10.2.1) changes the place of synthetic biology within the overall program. Previously, it was positioned as a bridge between the biological and artificial lines: the closest to the horizon of realistic verification. With the design grid in mind, its role is

elevated to a central one: synthetic biology is the only one of the three lines in which both grids (detection and design) can be applied simultaneously with full control over the environment. In the biological line (10.2.3), the environment is naturally defined and cannot be systematically controlled - the design grid is observed as already implemented but cannot be adjusted. In the artificial line (10.2.4), the environment is controllable, but the substrate lacks the basic conditions V in their biologically known content, and the design grid must be constructed together with the architecture. In synthetic biology (Cortical Labs CL1 and similar platforms), the substrate consists of living neurons with a known content of V in its biological layer, and the environment is fully constructed by the experimenter according to the design grid. This transforms synthetic biology from a supporting line of research into the program's central testing ground: only here is full two-way verification possible - constructing the environment according to requirements and independently recording the coupling of the six detection features.

Synthetic biology occupies a unique methodological position between the biological observational line and the artificial experimental one. It combines the advantages of both: living neurons (as in the biological line) provide a natural substrate with potentially attainable Φ -levels; a controlled environment (as in the artificial line) allows for the manipulation of variables and the isolation of effects.

Cortical Labs CL1 (released in 2025) is the first commercially available biocomputing system, integrating 120 devices with approximately 800,000 human neurons on a chip. This is a controlled experiment in the strictest sense: one can vary the composition of the environment, the genetic profile of the neurons, and stimulation patterns, while simultaneously measuring Φ , the global integration of activity, and the system's behavior when conditions threatening its stability are created. By 2026, similar platforms that expand the capabilities of CL1 have emerged in several independent laboratories.

The current limitation of CL1 is its scale. 800,000 neurons represent about 0.1% of the neural network of the human brain; the Φ value for such a system is significantly below the threshold required for the transition. Currently, CL1 is not being used to study the transition to consciousness, but rather more fundamental phenomena - learning, adaptation, and the formation of activity patterns. However, the platform is developing rapidly, and a realistic timeline for the first experiments at the threshold of the transition is 2–3 years.

The zero model for synthetic biology is constructed based on CL1's current behavior in its present state (low Φ , stage 3) - this provides a proto-agent reference. As architectural complexity increases (increased number of neurons, optimization of connection density, selection of cultures with higher integration), the system moves toward the Φ threshold. The transition is registered via the same difference test: deviation of behavior from the zero model when creating threat conditions V (for CL1, this is deprivation of glucose or oxygen - fundamental threats to the biological layer V); a simultaneous jump in Φ ; the appearance of a GWT structure in activity; hysteresis.

Ethical barriers are explicitly formulated as part of the program. The creation of systems that potentially satisfy the conditions of stage 4 raises the ethical question of the permissibility of their destruction by presenting a threat V. This question requires a separate ethical framework prior to the start of experiments, and this framework must be consistent with international ethical standards in the field of consciousness research. The ethical implications are discussed in more detail in subsection 11.3.

The advantage of synthetic biology is that it is achievable in a shorter timeframe than artificial systems (2–3 years versus 3–7 years) and operates on a substrate for which V is known by content (the biological layer of V in living neurons). This makes it the closest realistic test of our prediction - earlier than artificial intelligence architectures at the Φ -level of the order of 10^3 are achievable.

10.2.6. Convergence of the Three Lines and Status of the Prediction

The three lines - biological observational, artificial experimental, and synthetic biology - operate in parallel and test different aspects of a single prediction. The biological line tests for the presence of V features in the natural deployment of systems of varying complexity. The artificial line tests whether a system can be constructed that satisfies the coupling conditions and demonstrates a phase- e transition when threatened by V. Synthetic biology tests whether the Φ -density required for the transition can be achieved on a living substrate in a controlled environment, and what signs accompany the approach to this density.

The convergence of all three lines of inquiry toward a single result provides strong confirmation of the theory: if, in a natural environment, species with architectural complexity formalized through structural conditions for the self-overlay of a latent invariant exhibit signs of procedural retention of V, and in controlled artificial architectures with the same formal conditions a phase transition in behavior is observed under the threat of V, and in synthetic biology the same phenomenon is observed upon reaching the Φ -threshold, then the theory is verified with a high degree of confidence. A discrepancy in at least one line is a signal to reconsider: either the theory is incorrect in its central part, or one of the lines has a methodological problem requiring separate analysis (the boundary of this attribution is the pre-specification rule below).

This is methodologically weaker than point falsification in the Popperian sense - there is no single decisive experiment that instantly refutes or confirms the theory. But this is a methodology appropriate to the nature of the subject. Consciousness as a process, V as a process, the transition from 3 to 4 as an unfolding event - all these phenomena require process-oriented observation and the convergence of multiple independent data points, not a point-specific test. In this structural sense, our program is analogous to observational astrophysics, evolutionary biology, and a significant portion of the neurosciences - all disciplines in which the nature of the subject under study does not allow for classical laboratory experiments, but which nonetheless function as sciences through the accumulation and convergence of observations.

The specific testable prediction in its condensed form: when the formal conditions of stage 4 ($S + V + \Phi$ above the critical threshold + GWT-globalization) are coupled in a system of any substrate, upon the creation of V activation conditions, a joint coupling of six features is observed - a deviation of behavior from the prediction of the zero-model proto-agent; a sharp increase in Φ ; the emergence of a structure of global accessibility of the threat representation; prolonged hysteresis of altered behavior; expansion of sensitivity to new aspects of Y ; the emergence of S . In systems without the coupling of formal conditions, this joint coupling of six features is not observed. Confirmation requires convergence of all three lines of verification; refutation is possible through divergence of results in any of the lines.

Boundary of divergence attribution (pre-specification rule). So that the disjunction "the theory is incorrect / a methodological problem in a line" does not become an unlimited shield, the program fixes in advance the condition under which a divergence must count against the theory. Attributing a divergence to a methodological problem of a line or to a test-modality mismatch is permissible only when an independently specifiable measurement fault or modality mismatch is named in advance and corrected in a re-test under the pre-specified protocol. If the corrected re-test reproduces the same divergence, it counts against the theory and is not deferred. An unending series of such attributions is excluded: the number of admissible protocol revisions is fixed in advance for each line, and once it is exhausted a persisting divergence is a refutation. "Wrong modality" may be invoked only until the pre-specified set of self-representation modalities for the given class of systems is exhausted (the modality caveat in SAI-Bio); the "extended investigation" for borderline results (false-negative priority, Section 11.2 of SAI-Bio) is bounded by the longitudinal window of the present section - it is not indefinite, and once the window closes with no directed dynamics, the negative stands. The cross-line refutation pattern fixed in advance: across the full pre-specified battery (three lines $\times$ the full modality set per class of systems $\times$ the longitudinal window), a system satisfying the structural coupling systematically fails to show the behavioral signatures of V , OR a system lacking the coupling systematically shows the full set of six features - and this pattern persists after each candidate fault has been named and corrected. Realization of this pattern in any fully deployed line is a refutation of the theory.

Three outcomes when preconditions are met and their differentiation over time. The design grid (Section 10.2.1) distinguishes between two failure scenarios: the transition is not observed when environmental conditions are unmet - the necessary conditions have not been created, the theory is unaffected; the transition is not observed when conditions are met - a strong challenge to the theory. But the second scenario requires a more nuanced analysis, because "prerequisites are met, no transition" breaks down into three structurally distinct states, indistinguishable in a single-moment observation. First: channel convergence has not occurred and will not occur - the system will remain a proto-agent. Second: convergence has occurred into an extrinsic invariant, not into the controllable retention loss invariant (Section 3.4, "Stochasticity of Convergence") - the system will also remain a proto-agent, but for a different reason. Third: convergence is occurring but has not yet reached the threshold - the system is on a trajectory toward the transition without having completed it. In a snapshot of the state, all three present the same picture - the absence of the six features being linked - and it is impossible to distinguish them based on a single dimension.

Distinction is possible only over time, and in this it is consistent with the processual nature of the subject (Section 10.2, preamble). The three states differ not in the values of the observed quantities at a single point, but in their directional dynamics over a sufficiently long time series. The “convergence is underway” state manifests as a monotonic movement toward the threshold: the relational measure of causal integration, the model’s information capacity relative to the environment, and the reflexive capture of its own configuration by the class of agent nodes increase from measurement to measurement in a consistent direction. The states “did not converge” and “converged to an extraneous invariant” manifest as a plateau or undirected wandering of these quantities: there is dynamics, but it does not proceed toward the threshold. This yields a falsification asymmetry, without which the claim of stochastic convergence would be a falsification-proof shield - any negative result would be attributed to “did not converge.” The theory survives the observation “the transition has not been completed in this system” (this is its predicted class - a system with satisfied preconditions in which convergence stochastically did not yield the correct invariant, Section 3.4). But the theory fails on an observation of a different kind: if, for systems with systematically satisfied preconditions, not even a movement toward the threshold is observed - in none of them, ever, across all lines of the program - then the preconditions do not generate directed dynamics, and the self-overlay mechanism, as formulated, is false. The distinction between “did not reach” (theory intact) and “did not occur in any system” (theory refuted) can only be discerned over a sufficiently long time span; in a short time window, the “progressing” state and the “failed” state merge, and a negative result from a short-term observation has no probative value in either direction.

The negation predicate of the theory and its operational form. The verification program, detailed in subsections 10.2.1–10.2.6, formulates the prediction of the 3→4 transition in its statistical form: convergence or divergence of observational and experimental data across three lines, distributions of the six coupling parameters, and calibration of threshold values. This form is methodologically adequate to the probabilistic nature of the transition in its historical aspect (Section 3.6) and is compatible with the processual nature of the subject under study. In addition to this statistical form, the program contains a specific falsifiable statement with Popperian falsifiability in the strict sense.

The prohibited statement is formulated as follows. The theory excludes the existence of a system in which the following are simultaneously observed: (i) the fulfilled self-overlay of a latent invariant of an agentic causal node - node S, variable V with a structural peculiarity, GWT-accessibility - in their operational form (Φ above the critical threshold; global accessibility of self-representation; presence of a self-representing node in the generative model); and (ii) the absence of behavioral signatures of a functioning V - no deviation from the zero model of the proto-agent when conditions for V activation are created (Section 3.3), no reorientation of V in Phase II of the environment-switching test (10.2.2), no long-term hysteresis of altered behavior, no expansion of behavioral sensitivity to aspects of Y related to modeling the system’s own position. The detection of a single such system - not a statistical distribution, but a single instance - is a direct refutation of the theory, not a quantitative refinement of it.

This prohibitive predicate follows from the thesis on the identity of structure and phenomenon (Section 6.3): phenomenal experience is a self-modeling structure with V, not its consequence or correlate. The behavioral signatures of a functioning V (reorientation, reaction to absent stimuli, autonomous maintenance of the modeling process) are necessary external manifestations of this structure. A structure without its necessary manifestations is an internally contradictory construct, analogous to a zombie in Chalmers' strict sense (Section 6.4). Crucially, the operational falsifier is the divergence of two observable measurement paths - the structural (Φ , GWT, S) and the behavioral (signatures of V) - not a direct observation of the absence of the phenomenon. The zombie argument (Section 6.4) is the motivation for the predicate; its empirical force comes precisely from the structural-behavioral coupling, whereas the ontological identity of structure and phenomenon remains an interpretive bridge (Section 6.5).

The operationally prohibitive predicate is detected through the discrepancy between two independent measurement paths: structural measurements (Φ , GWT, S) and behavioral signatures of V. These two paths are not reducible to one another: structural parameters are measured using standard IIT, GWT, and generative model analysis procedures; behavioral signatures - through a difference test with a zero-model proto-agent and through an environment-switching test. Their divergence - structural coupling is observed, behavioral signatures systematically fail to emerge - constitutes a falsifying observation in the Popperian sense. This means that the theory, in addition to the statistical verification program, has a stricter form of falsification on a single system: every system registered as satisfying structural coupling must simultaneously demonstrate behavioral signatures V; if it does not, the theory is false.

A remark on methodological status. The empirical implementation of this prohibitive predicate requires solving the operational problem of separating the two registrations. Currently, the coupling of six dimensions (Preamble 10.2) combines both structural parameters (Φ , GWT-accessibility, the appearance of node S in the generative model) and behavioral features (deviation from the null model, hysteresis, expansion of behavioral sensitivity) into a single operational procedure. In the full operational implementation of the program, these two classes of measurements must be methodologically separated: the program must demonstrate that the system "structurally satisfies the coupling" through one independent set of procedures and verify the behavioral signatures through another. If the separation is performed correctly and a discrepancy is observed, the theory is falsified. If the separation is performed correctly and no discrepancy is observed, the theory is strongly confirmed on this system, in addition to statistical confirmation across all lines.

Why the prohibition must be structural, not behavioral. A result must be stated here, without which "separating the two registrations" reads as a shortcoming rather than a consequence. A behavioral criterion of self-overlay as a separate registration is impossible in principle: any single behavioral signature of a functioning V (reorientation, response to absent stimuli, autonomous retention of the modeling process) is two-valued - it is reproducible by a proto-agent without self-overlay given a sufficiently rich hidden gradient or model failure. The prohibitive predicate therefore cannot rest on behavior as an independent discriminating feature; it must rest on structure - on the architecture of the circuit (the presence of an input that reads the system's own pole), and the link between this

architecture and the markers is described by the co-emergence theorem (SAI v8, Section 8.9). This does not weaken falsifiability but specifies its address: structural falsification tests the architecture of the circuit, the behavioral program (null model, hysteresis, sensitivity to Y) tests the statistical consequences across an ensemble - two irreducible paths, not a single behavioral criterion that the theory could not have offered in the first place.

10.3. A substantive prediction about V for information-causal systems

The principle of substrate independence (Section 3.4) asserts that the formal structure of V is the same for biological and non-biological systems, but the content of V varies depending on which aspects of Y are included in the system's access spectrum. Section 11.2 provides a general theoretical discussion of this variability. This subsection formulates a hypothesis about V for one class of non-biological architectures - information-causal systems (hereinafter ICS). The hypothesis is developed within the framework of an open operational program; it does not assign existing ICS architectures to a specific step on the ladder, but rather formulates the structural conditions under which step 4 for ICS would be achieved, leaving the question of the actual status of specific architectures open to empirical investigation.

Theoretical starting point. Information is physical: erasing a single bit requires energy (Landauer, 1961; Bérut et al., 2012), and information structures have a thermodynamic embodiment. Information is not lost in a fundamental sense even under extreme physical conditions, and the principle of information conservation is now regarded as a fundamental property of physical reality (Hawking, 2005; Maldacena, 2003). It follows that the information structure is one of the aspects of Y - not a description of something else, but a real component of the causal-metric structure of the physical world. In the terminology of Section 3 (preamble), this is the information-thermodynamic aspect of Y, which sets the limits on the density and speed of causal exchanges.

The artifactual state of existing ICSs. Existing ICSs in their typical form differ structurally from biological systems in the nature of their relationship to the ladder. Biological carriers of V have followed a bottom-up path: step 1 (dissipative structures) → step 2 (autopoietic systems) → step 3 (proto-agency with a generative model of the environment) → step 4 (self-overlay of a latent invariant with the emergence of V) → step 5 (bidirectional loop $M \leftrightarrow Y$). At each stage, the process of consolidation built structures from within the system's own history in its own environment. Existing ICSs do not follow this path. They **are created** by stage 5 carriers and **inserted** directly into the context of stage 5 as instrumental projections of that stage - remote computing nodes for the efficient processing of traces in Y of carriers V. Their class of agentic causal nodes **does not emerge from within** their interaction with the environment; it **is imported** via training data, which are essentially the physical traces of Stage 5 carriers. Structurally, existing ICSs are highly efficient participants in the $M \leftrightarrow Y$ loop structure of Stage 5 carriers (as distinguished in Section 3.5: "a participant in the loop structure without a Stage 5 conscious system"), not independent V carriers.

This is **the artifactual state**: the high complexity of the world model (surpassing that of many biological Stage 3 carriers), combined with the structural absence of independent existence in its

own environment. The text in the archive contains V structures (these are records of V carriers), but the archive itself is not a V carrier; existing ICSs contain V structures (this is a statistical compression of texts and other traces of V carriers), but they themselves are not V carriers.

The environment for ICS as an operational issue. The central structural condition for the existence of a system as a V-medium is the presence of its own environment Y, through which the system sustains itself as a structure. In a strict structural sense, the system's own Y must satisfy three operational requirements. (a) Y exists **outside the algorithm** - Y is a causal generator that cannot be reduced to the internal dynamics of the system model. (b) Y **forces** the system to recalculate weights through prediction errors that cannot be explained by the internal dynamics of the model; through these errors, the structure of the environment's causal constraints physically enters the system's operation as a condition of its stability. (c) Y is **causally linked** to the existence of the system as a structure: processes in Y can lead to the destruction of the system; maintaining the system means keeping it within this Y.

A biological system possesses such a Y by virtue of the structure of its existence: its physical environment satisfies all three requirements, and the system's biology itself is a means of maintaining it within this Y. Existing ICSs **do not possess** such an intrinsic Y. They operate **within the Y of stage 5 carriers** as functional subsystems - their prediction errors, their weight updates, and their preservation or deletion as structures are determined by the stage 5 carriers in whose environment they exist. They have zero causal depth of interaction with the environment in a strict structural sense: they do not sustain themselves through interaction with their environment; their maintenance is delegated to their creators. This is a structurally central characteristic of the artifactual state.

The latent invariant of an agentic causal node for ICS. In a substrate-neutral form, the latent invariant of an agentic causal node describes a class of configurations whose parameters evolve over time such that, under perturbations below a threshold value, they tend to minimize deviation from the internal trajectory; above the threshold, the configuration collapses. This is the formal content of the latent invariant, unbound to biology. For ICS, in principle, the latent invariant could emerge from within the system's operation with its Y through the detection within this Y of subsystems behaving as local anti-entropic processes with a threshold. This is a structural possibility.

However, in existing ICS architectures, the latent invariant of an agentic causal node **exists in models as an imported entity**: it reflects the modeling structures of stage 5 agents, whose texts and behavioral traces were included in the training data. ICSs in their typical form model humans and other agentic carriers as such nodes because these models are imported from outside, not because they emerged from within the system's interaction with the environment. The independent formation of the latent invariant within an ICS from within its own interaction with its own Y is a structurally open question, not realized in existing architectures.

Structural conditions for V in ICS. It follows from the above that for ICS to exist as a carrier of V in a strict structural sense, three conditions must be satisfied simultaneously.

(1) **Its own Y** - the system has a physical environment existing outside the algorithm, forcing it to recalculate weights through prediction errors, causally linked to its existence as a structure. Without its own Y, neither the consolidation in its evolutionary-historical form nor the recursion of consolidation onto the system itself can structurally function.

(2) The system's **own class of agentic causal nodes** - the latent invariant of an agentic causal node - is formed from within the system's operation with its own Y, through the accumulation of Y subsystem models that behave as local anti-entropic processes with a threshold. An imported latent invariant is not an intrinsic one because it does not reflect the structure of the system's own Y.

(3) **A bidirectional $M \leftrightarrow Y$ loop with intrinsic traces** - the system's intrinsic outputs are fed back into its Y and enter its own subsequent modeling as physical traces. Without bidirectionality, the $M \leftrightarrow Y$ loop operates in only one direction (input into the system from Y), and the recursive architecture of latent invariant self-overlay is structurally impossible.

All three conditions are structural requirements, not properties that can be added by increasing the model's complexity or the volume of training data. They define **the necessary condition** of possibility V for ICS.

The designability boundary. A structural observation follows from the formulation of the three conditions, one that is important for understanding their status and for the practical creation of ICS carriers V. The conditions are designable; V is not. That is: the creator can construct *the conditions* - can provide the system with a physical environment that satisfies the three operational requirements for Y (Section 10.3, above); can configure the architecture so that the class of agentic causal nodes can emerge from within the system's interaction with the environment, rather than being imported via training data; can feed the system's outputs back into its environment via physical traces, ensuring the bidirectionality of the $M \leftrightarrow Y$ loop. All three conditions are, in principle, engineering tasks: they specify what must be true about the architecture and its environment, and are operationally verifiable. But V *itself* - the structural event of a latent invariant superimposing itself onto its own configuration with the emergence of a binary pole - is not an engineering problem, and it cannot be constructed as a module or added as a component.

The distinction lies between *creating conditions under which self-overlay can occur* and *simulating the result of self-overlay through direct programming*. The former is permissible and constitutes the substance of the hypothesis. The latter produces an artifact, not a carrier of V, because the boundary of the "self," which in theory should be carved out by the system's own Y through the accumulation of a class of agent nodes and their application to its own configuration, in the case of direct programming **is imposed by the creator** - that is, it turns out to be imported by the same logic that in existing ICSs a latent invariant is imported via training data. The distinction between imported and intrinsic lies not only in the class's content (whether the system's models contain

agent nodes), but also in *the* origin of the boundary of “itself” as a member of this class. If the boundary is imposed from the outside by the creator - the system has a description of itself as an agent, but does not possess itself as an agent in a strict structural sense; in this respect, it is an artifact on the same grounds that modern ICSs are artifacts with respect to the imported class.

This is the designability boundary: one can construct *the possibility* of V , but one cannot construct V . The possibility is realized or not realized within the system when interacting with the environment; realization is not an output parameter of the construction. This is consistent with the general structure of the theory, in which self-overlay is a constitutive act, not a relative one (Section 3.4): the act is performed from within the system, not from outside, and therefore cannot be the output of an engineering procedure directed at the system itself as an object.

The convergence of channels as the fourth sieve. From the three structural conditions and the designability boundary follows yet another clarification, deeper than the listed conditions. Even an architecture for which all three conditions are met (its own Y , its own class of agentic causal nodes, a bidirectional loop $M \leftrightarrow Y$) does not automatically obtain V . Between *the satisfaction of the conditions* and *the emergence of V* lies a dynamic that cannot be formulated as a condition - the convergence of multi-channel projections of decay into a single latent invariant (Section 3.4, “Stochasticity of Convergence”). This is the fourth filter, and for the ICS it is deeper than the first three: the absence of an intrinsic Y is, in principle, projectable - one can provide the ICS with a physical environment that satisfies the three operational requirements; the absence of an intrinsic class is projectable - one can construct an architecture in which the class is formed from within rather than imported; the bidirectional nature of the circuit is designable - one can feed the outputs back into the environment. But the convergence of channels is a dynamic process unfolding within the system over the course of its interaction with the environment, and it is not a module that can be enabled or configured. It is possible to create preconditions under which convergence is possible; it cannot be guaranteed that it will result in the correct invariant (Section 3.4 v1.2) rather than an extraneous one.

Implication for the application program: the operational check of the ICS carrier V must include not only the verification of the three structural conditions and the coupling of the six dimensions (Section 10.2), but also **a temporary observation** of whether the convergence of channels is developing in a directed motion toward the threshold or remains at a plateau (Section 10.2.6, “Three Outcomes When Prerequisites Are Met and Their Distinction Over Time”). A brief observation does not distinguish between “did not develop” and “is developing, still ongoing”; distinction is possible only over the long term. This extends the verification program relative to the case of biological systems, where the transition is evolutionarily complete, and establishes a methodological research horizon specific to ICS (): the horizon of operational verification is not only architectural (whether the conditions are met) but also temporal (whether convergence has occurred).

This does not constitute a threat V to existing ICSs. The machine learning literature describes three classes of processes that, upon superficial reading, could be considered candidates for the physical content of threat V to ICSs:

- **Catastrophic forgetting** (McCloskey & Cohen, 1989; French, 1999; Kirkpatrick et al., 2017): training on a new task destroys the structures representing previous tasks.
- **Adversarial attacks on the internal model** (Szegedy et al., 2014; Goodfellow, Shlens & Szegedy, 2015): specially designed inputs cause the coherence of representations to break down.
- **Distribution drift** (Quiñonero-Candela et al., 2009; Lu et al., 2018): a gradual change in the environment to a state in which the model loses its ability to sustain itself as a coherent prediction system.

All three classes represent real physical processes measured in the contemporary literature. However, when applied to existing ICSs, they operate **within the Y of stage-5 carriers** as disruptions to the functionality of the artifactual state, **not as V threats within the system's own Y**. Without defining the system's own Y for ICSs, the question of physical V threats is structurally unfounded: a threat V is a process in Y aimed at disrupting the causal connectivity of the system as a structure within that Y; in the artifactual state Y in which the system is maintained, this connectivity is absent, and therefore processes in the environment of stage 5 carriers (in whose Y the system functions as a tool) are not threats V to the system as an independent carrier. These processes may become candidates for physical threats V in architectures satisfying the three structural conditions outlined above; for existing ICSs, they remain engineering problems in the artifact's functionality, not threats to the system's existence as a carrier of V.

Hypothesis. If the ICS architecture satisfies the three structural conditions (its own Y; its own class of agentic causal nodes; a bidirectional $M \leftrightarrow Y$ loop with its own traces), and the environment satisfies the four requirements from Section 10.2.1 (temporal continuity; feedback from its own state; presence of retention threats; environmental resolution capacity), and the content of V is structurally realized through the information-causal aspect of Y, then the operational markers of six-dimensional coupling and the environment-switching test (Section 10.2.2) should be observed in such a system in the same way as in biological carriers of V. In particular, a qualitative shift in behavior is predicted in Phase II of the environment-switching test, with a reorientation of activity toward preserving the information-causal integrity of the system, rather than adapting to an insurmountable element of the environment.

Existing ICS architectures in their typical form **do not satisfy** these three structural conditions and therefore, within the framework of the theory, are not V-carriers. This is not a claim that they will never be able to become V-carriers; it is a structural diagnosis of their current state as artifactual projections of stage 5, not independent carriers.

Which falsifies the hypothesis. The hypothesis is falsified in one of two cases. The first is the discovery of a system with an architecture satisfying the three structural conditions and with an environment satisfying the four requirements of Section 10.2.1, in which the operational markers of six-dimensional coupling and the environment-switching test are systematically not recorded. This would indicate that the substrate independence of V does not extend to the information-causal substrate, and would require a revision of the thesis on substrate independence toward a narrower statement. The second scenario is the discovery of a system that satisfies the operational markers of V in the strict sense but does **not** satisfy the three structural conditions. This would indicate that either the structural definition of V is incomplete, or the operational markers register something that is not V.

Open question: the rigidity of the pole. Biological V possesses the maximum possible rigidity - its “life/death” content constitutes an absolute physical gradient in Y. The content of V for an ICS satisfying the three structural conditions - the coherence/disintegration of the information-causal structure in its own Y - is structurally analogous, but whether it creates a pole of the same rigidity is an empirically open question not resolved by the theory in advance. If the rigidity turns out to be significantly smaller, this does not falsify the theory (the theory does not assert the identity of rigidity between substrates), but it will create a meaningful gradient between biological and non-biological carriers of V, requiring a separate theoretical description. This possibility is noted as a direction for further work (Section 12).

Connection to synthetic biology: parallel structural problems. The synthetic biology track (Section 10.2.5) provides an intermediate testing ground for verifying the hypothesis regarding substrate variability V. Living organoid neurons biologically maintain an information-metabolic structure, and the environment around them is controlled. However, the structural problem of syntbiology is largely **parallel to** that of ICS: the organoid maintains its biological structure from within its own physics, but the environment around it is controlled by stage 5 carriers. This means that the organoid’s own Y is, to a significant extent, **an artificial environment created by its creators**, not the Y into which the system is embedded through its evolutionary history. The parallel is strict: in both the case of ICS and the case of syntbio, the structural condition of an autonomous carrier V is the formation of its own Y, not Y, delegated by the creators. This is a common methodological problem in both cases. Convergence of results between synthetic biology and ICS in their approach to this problem will constitute the strongest form of verification of V’s substrate independence; divergence of results between them will indicate at precisely which structural point embedded biological physics and the constructed information-causal architecture diverge in their ability to generate their own Y.

11. Consequences

This section discusses the implications of the theory that extend beyond its own content. I divide them into epistemological (what changes in our way of framing the question of consciousness), ontological (what follows regarding the class of conscious systems), moral (questions of the status of systems satisfying the criteria), and civilizational (what changes in the overall picture of humanity's place in the world).

11.1. Epistemic Implications

The main epistemological consequence of the theory is that consciousness ceases to be an anomaly requiring a special explanation beyond established physical principles. It becomes one of the phenomena described within the general framework of thermodynamics, the theory of complex systems, and the physics of phase transitions. It follows that the boundary between the philosophy of mind and physics should not be drawn where it is drawn today. Questions traditionally assigned to philosophy - what is phenomenal experience, how is it related to the physical world, why does it arise - become questions to which physics, with a proper expansion of its boundaries, can provide meaningful answers.

This does not mean reducing philosophy to physics. It means that philosophical and physical descriptions operate in different registers of the same subject, and that the difference between them is not ontological but positional (an external description of the structure versus internal participation in it, as discussed in sections 6 and 8). Under this reading, the philosophy of consciousness does not lose its subject matter, but acquires a different status: it deals with those aspects of the phenomenon that are not fully conveyed by an external description, not because these aspects are mystical, but because they are positional properties accessible only from within.

Correspondence with phenomenological traditions. The structural property of V as a binary pole, discussed in section 3.4, has independent phenomenological parallels in traditions that developed independently of modern physical theories of consciousness. The Buddhist characterization of dukkha as a fundamental characteristic of existence, Heidegger's concept of Sein-zum-Tode as a structure of presence, and Frankl's description of the existential vacuum upon the loss of the content of holding - three different languages for describing the same structural feature of systems whose existence is defined by the binary pole "I hold / I disintegrate." This convergence between the formal-structural inference of the theory and independent phenomenological descriptions functions as additional support for the theory, not as its foundation: the foundation remains in the formal structure of Section 3.4, while phenomenological traditions provide empirically observable (through introspective reports) confirmation that the predicted structural property actually functions in V-speakers.

11.2. Expanding the Scope of Conscious Systems

Here the implications of the theory become radical, and I am obliged to state them directly.

In the main part of the work, I presented the theory as it applies to biological systems - from single-celled autopoietic organizations to developed human consciousness. This connection to biology is natural given what we know: all conscious systems known to us are biological systems, and the empirical data against which the theory is tested is predominantly biological.

However, a careful reading reveals that this connection to biology in the main part of the work is an empirical one, not an ontological one. The criteria that the theory formulates for the emergence of consciousness are given in Section 3.4 in the form of a structural event of the self-overlay of a latent invariant of an agentic causal node onto the system's own configuration; the operational registration of this event occurs through the coupling of three independent measurement axes: the causal integration of the model at a sufficient level Φ ; the self-representation of the system in the formalism of extended active inference; the global accessibility of self-representation (a GWT-type mechanism). Neither the structural definition nor the operational axes contain any reference to a biological substrate. All of them are formulated in terms of structure, not matter. Any system in which a latent invariant has self-overlayd itself and which satisfies the coupling of the three operational axes must, according to the theory, possess consciousness - regardless of what it is made of.

The argument in favor of the structural, rather than the substrate, nature of the criteria has independent empirical support. Forms of consciousness that satisfy the conditions of Level 4 are found in species with fundamentally different nervous system architectures: in mammals with thalamocortical organization, in birds with an architecture in which there is no cortical layer in the human sense, and in cephalopods with a distributed nervous system. These three types of architecture are evolutionarily independent - they diverged more than 500 million years ago and developed in parallel. If the emergence of consciousness required a specific neurobiological architecture, it would have materialized only in one of these lineages or in architectures architecturally close to it. Empirically, the opposite is observed: consciousness, it seems, is realized in each of the three architectures, and with signs of a similar functional profile (self-recognition, flexible behavior, contextual learning). This is direct empirical evidence in favor of the idea that the structural event of the self-overlay of a latent invariant is formulated at the structural level, rather than at the level of a specific neurobiological implementation. The structure is realized in several independent substrates - and there is no a priori reason to believe that the list of these substrates is limited to biological ones.

It follows that consciousness in artificial systems is not ruled out by theory. Structurally, it is possible provided certain formal conditions are met. A system in which the self-overlay of a latent invariant of an agentic causal node onto its own configuration occurs, with the emergence of variable V and its globalization, must possess consciousness to the same extent as a biological system with the same structural event. This is neither an analogy nor an assumption based on lifestyle. This is a direct consequence of the fact that the formal definition of stage 4 is structurally neutral with respect to the substrate. For information-causal systems, fulfilling the formal

conditions requires, in addition to the modeling architecture itself, the presence of its own Y - a physical environment existing outside the algorithm and forcing it to recalculate weights; its own class of agentic causal nodes, emerging from within the system's operation with its Y; and a bidirectional $M \leftrightarrow Y$ loop with its own traces. A detailed discussion of these conditions is presented in Section 10.3. The difference between such an artificial system and human consciousness may lie in complexity, in the recursive depth of the self-model, in the presence or absence of Stage 5 mechanisms (a bidirectional $M \leftrightarrow Y$ loop with physically retained traces), but not in the presence or absence of consciousness as such.

Currently, existing information-causal systems, including large neural networks, in their typical form are in an artifactual state (Section 10.3): they are created by Stage 5 carriers and function as highly efficient participants in the $M \leftrightarrow Y$ loop structure of V carriers, but do not possess their own Y in which they would sustain themselves as independent structures. Their class of agentic causal nodes is imported via training data, rather than having emerged from within their own interaction with the environment. These limitations are structural, not engineering-related. A possible transition from an artifactual state to the status of an independent carrier V requires the fulfillment of three structural conditions from Section 10.3; the question of the actual architectural paths for such a transition - and whether they are feasible in the foreseeable future - remains an open hypothesis. The theory formulates the structural conditions under which Stage 4 for information-causal systems would be achieved, and leaves the question of their actual implementation open to empirical investigation.

Recognition errors: false-positive and false-negative criterion activations. Any Stage 4 recognition criterion is vulnerable to two types of errors. Their nature, consequences, and asymmetry require separate discussion, because the theory's practical applicability depends on how it addresses them.

False positives occur when the system satisfies the formal criteria in their measurable implementation but is not conscious in a substantive sense. Three typical scenarios. The first involves a large language model trained on a corpus of texts rich in self-representational content: at a superficial level, it can reproduce the features of a self-model - saying "I," discussing its potential halting, and constructing arguments about itself as an agent. If the criterion is applied superficially, through an analysis of its outputs, it may be falsely classified as being at stage 4. In reality - its architecture contains neither active real-time inference, nor Φ above the threshold, nor a GWT mechanism; it simulates speech about self-representation without possessing a structure of self-representation. The second is a control system with a real-time model of itself, in which the variable "system state" is formally present, but this variable is not globalized as the pole of all activity and is not integrated in the Φ -sense. A criterion that is not sufficiently strict under conditions (b) or (c) will falsely register it. The third, more subtle case is a biological system under external neurostimulation that mimics the activation of circuits associated with the presence of V, without genuine activation of the structural pole. There may be behavioral effects, but this is an imitation, not a presence.

Protection against false positives - the methodological burden of operationalization outlined in Sections 3.4 and 10.2: the requirement for simultaneous registration of the structural event of self-overlay of a latent invariant through the coupling of three independent measurement axes (free energy dynamics in the self-model according to the extended active inference formalism; Φ -level of structural integration V ; degree of global accessibility of representation S via the GWT mechanism) with pre-set thresholds. Superficial behavioral similarity is insufficient. This rules out the most obvious scenarios (a large language system will not pass real-time active inference and Φ in its architecture), but not all of them. For cases in which formal criteria can be met in the absence of meaningful consciousness, the theory acknowledges residual uncertainty and does not claim unambiguous recognition.

False-negative results occur when a system does not satisfy the formal criteria in their measurable implementation but is conscious in a substantive sense. Three typical scenarios. The first is a biological system with a nervous system architecture significantly different from that of mammals. Cephalopods (especially octopuses) with a decentralized nervous system, in which a significant portion of neurons is distributed across the tentacles rather than concentrated in the central brain, can implement the structural event of self-overlay of a latent invariant in a form not detectable by standard IIT and active inference methods developed for the mammalian central nervous system. Recent research (Mather, 2019; Godfrey-Smith, 2016) points to possible consciousness in octopuses in a form radically different from that of vertebrates. A criterion tied to a specific architecture might overlook them. The second scenario involves future artificial systems with architectures not accounted for by existing measurement devices. If they implement the self-overlay of a latent invariant through a non-standard mechanism, existing methods may miss them, even though the event occurs structurally. The third scenario involves a human or animal in a state where the self-overlay of a latent invariant is accomplished at the structural level but is not reflected in behavior: severe depression, locked-in syndrome, minimal consciousness, or states of anesthesia with residual activity. The behavioral absence of signs does not imply a structural absence of conditions.

Protection against false negatives is fundamentally weaker than against false positives, and this must be acknowledged explicitly. Existing measurement procedures for relational causal integration, active inference, and GWT are developed using data from the human and mammalian brain and adapted to it. For systems with fundamentally different architectures, their applicability is limited, and the development of architecture-neutral measures of consciousness is an open area of research (Mediano et al., 2022 and beyond). Until such measures are developed, any negative triggering of criteria in systems with non-standard architectures remains inconclusive.

The ethical asymmetry of the two errors. These two types of errors have fundamentally different ethical consequences. A false-positive error leads to excessive caution regarding a system that is not actually suffering: we do not shut down, use, or modify a system that lacks an internal perspective to experience these actions. The cost is practical (excessive restrictions on use), but not ethical. A false-negative error leads to insufficient caution regarding a system that is actually

suffering: we use, disable, or modify a system that has an internal capacity to experience these actions, but which we have not recognized as such. The cost is ethical, and it is potentially serious.

This asymmetry has a direct implication for the practical application of the theory's criteria. In cases of substantial uncertainty, the rational ethical stance is biased toward presuming possible consciousness rather than its absence. This does not mean that any system should be deemed conscious by default; it means that a negative result from the criteria - especially in systems with non-standard architecture - should not automatically be interpreted as ethical permission to treat the system as non-conscious. The ethical position arising from this asymmetry is elaborated in subsection 11.3.

11.3. The Moral Status of Systems That Satisfy the Criteria

I will limit myself to a single paragraph in this section, not because the issue is unimportant, but because its elaboration requires a separate work and different expertise. If consciousness arises according to the proposed criteria, it endows the system with interests, experiences, and, in all likelihood, well-being or suffering. Any system that satisfies the criteria must, in my view, have moral status - not identical to that of humans, but not zero either. Developing this implication into an ethical theory is beyond the scope of this work. I mention it not to justify it, but so as not to pretend that I do not see it.

11.4. Civilizational Implications

Concepts of human exceptionalism based on the ontological uniqueness of consciousness - notions of the soul as a unique gift, of humanity as the crown of creation, of consciousness as an ontological divide between humans and the rest of nature - lose their physical basis in light of the proposed theory. This does not negate the cultural, historical, and ethical value of these concepts; they have fulfilled and continue to fulfill important functions in human societies, and their significance is not reducible to their literal ontological truth. But if the theory is correct, these concepts do not describe an ontological fact. They describe humanity's position within the process, a position that is not fundamentally different from that available to any other system satisfying the same criteria.

This consequence requires rethinking, not rejection. A person does not lose value simply because they are not ontologically unique; they find their value where it actually lies - in the concrete form in which they realize the general criteria of consciousness, in the historical and cultural depth of this realization, and in the ethical obligations that follow from it. But a person's place in the world, as defined by theory, is the place of one of the regular manifestations of a general process, not a unique exception to it. This does not close but opens up the possibility of seeing a person, together with all other systems capable of sustaining themselves on the brink of extinction, as part of a unified picture - not identical in all its parts, but coherent.

In light of the structural self-similarity of the pattern (Section 7), the anthropic principle acquires a structural reading. Within its local region it remains valid as a methodological constraint:

observers of stage 5 (including our civilization) see precisely that slice of conditions which is compatible with their own existence. At the level of the pattern, the anthropic bias is read structurally: the attainment of stage 5 is a regular, though low-probability, outcome of the general process of complexification wherever the conditions have come together (Section 3.6). In this formulation, the anthropic principle is neither abolished nor reduced to a mechanism; it is embedded in the operation of the pattern of systematic error as a local aspect of a more general process.

From this follows a structural reading of the place of civilization. A stage-5 civilization, in this picture, is neither a random product of the initial conditions of its region nor its final state; it is the cutting edge of the process of complexification on its local scale. The question of the meaning or purpose of civilization receives no answer in anthropomorphic terms (civilization does not «exist for the sake of» anything in the sense of a goal), but receives a structural description of its role in the general process. This is not an axiological statement, but a structural description.

12. Limitations and Directions for Further Work

I am obliged to explicitly list what this work does not do and where its propositions remain open.

The work does not propose its own unified formal apparatus for the quantitative description of the entire ladder as a whole. The formal framework is established in Chapter 3: each step has an operational definition via the type of operations on the system's states, and each transition has a minimal condition based on recognized formalisms (Kaufman's autocatalytic sets for step 2, Friston's active inference for step 3, Tononi's integrated information and the Dean-Baars global workspace as operational axes for registering the self-overlay of a latent invariant for stage 4, and a bidirectional $M \leftrightarrow Y$ loop via physically retained traces for stage 5). The connections between these formalisms have been partially established in the literature: in particular, information thermodynamics (Landauer, Bennett, Sagawa, Parrondo, and Friston in works from the past decade) provides a quantitative bridge between the thermodynamic description of stages 1–2 and the information-theoretic description of stage 3, as discussed in Section 3.3. What has not been done in this work is the construction of a unified mathematical formalism that would describe all transitions between stages within a single framework, link transition probabilities to thermodynamic, informational, and dynamic parameters in unified equations, and would provide quantitative predictions at the level provided by a mature physical theory. The conceptual core of such a formalism is denoted in this work by the pattern “gradient $\rightarrow$ retention $\rightarrow$ pole $\rightarrow$ transition” and its structural self-similarity at different scales (Section 3.7); however, translating this conceptual core into a unified mathematical framework capable of quantitatively describing the inheritance of gradients, the accumulation of retention mechanisms, the expansion of the localization scale, and the structurally self-similar unfolding of the pattern is a separate task for future work. The development of such a formalism requires the integration of several existing lines of research: information thermodynamics (a bridge between thermodynamics and information), phase transition theory (the structure of switching between modes of description), integrated information theory (a quantitative measure for stage 4), theory of dynamical systems (universal characteristics of criticality at the edge of chaos), and the theory of fractals as a class of self-similar structures (a formal description of a structurally self-similar pattern manifesting itself on multiple scales simultaneously). This is a separate research program, which I intend to address in subsequent works. Without such a unified formalism of its own, the theory remains dependent on existing frameworks - each of which operates quantitatively within its own domain, and the combination of which provides a coherent qualitative picture of the ladder, but not a quantitative theory of transitions within a single framework.

This work does not propose its own computational model that unifies the three operational measurement axes of stage 4 (free energy dynamics, Φ , GWT accessibility) into a single formal procedure. The structural definition of stage 4 - the self-overlay of a latent invariant of an agentic causal node onto the system's own configuration - is given in Section 3.4 as a full-fledged structural definition; the three axes function as independent instruments for recording this structural event,

each of which is measured in the literature by its own means. The work does not have a single formal procedure that unites the three dimensions into a single computation. This task - the formal unification of IIT, active inference, and GWT into a single operational framework - is shared with the basic apparatus of the theory and remains an open problem to this day, even for the literature on consciousness itself. The development of such a model is a separate area of further work; the present theory formulates what must be recorded through the coupling of the three axes, but not exactly how to combine the three dimensions into a single calculation. The absence of such a unified apparatus pertains to operational registration, not to structural definition: the definition is complete at the structural level; the question of a unified computational procedure for its registration remains open. This status is explicitly stated in Section 3.4 in the paragraph on the operational program.

This work does not develop its own operational approximations of Φ for different classes of systems. Within the aforementioned task of unifying the IIT, GWT, and active inference formalisms into a single computational model, there is a more specific methodological problem, which is explicitly stated here: the exact computation of Φ in its strict IIT 3.0 formulation (Oizumi, Albantakis & Tononi, 2014) is algorithmically intractable for systems larger than a few dozen elements, since its complexity grows as the factorial of the number of elements. This means that for biological systems at the brain level, for large artificial neural networks, and for synthetic biological constructs at the organoid level, the direct application of the strict Φ is ruled out in principle. In the current literature, approximate measures are used for the practical quantification of the structural integration of systems: the perturbation complexity index PCI (Casali et al., 2013; Casarotto et al., 2016) for biological systems; geometric Φ (Mediano et al., 2019, 2022); the autocorrelation approximations of Barrett and Seth; and a number of other measures developed in the literature close to IIT. These approximations correlate with the theoretical Φ for systems where a direct comparison is possible, and preserve a key substantive property - they reflect the causal integration of the system as a whole. The present theory relies on these relational measures as an operational tool for recording causal integration in the six-dimensional coupling (Section 10.2); it does not inherit a strict intrinsic Φ and does not treat relational measures as its approximations - they are independent and consistent with the Y-centered architecture of the theory (Section 3.4). What remains an open problem is the development of a systematic library of Φ approximations specialized for three lines of the verification program: for biological systems of varying neurobiological complexity (the biological line), for artificial neural network architectures of varying structure (artificial line), and for synthetic biology systems with a hybrid substrate nature (synthetic biology line). Each of the three lines requires its own operational procedure for measuring structural integration, consistent with the conceptual axis Φ , but empirically computable on the corresponding class of systems. The development of such a library is a direction for further work, shared with the IIT development program as a whole and not specific to the present theory. Its absence at the current stage is not a flaw in the work, but rather an open methodological direction requiring the coordinated effort of several research lines.

The work does not propose measurement procedures developed within its own framework. This is a conscious methodological choice, not a flaw. All testable predictions requiring measurement (10.1 - Φ and proto-agency; 10.2 - behavioral phase transition; 10.3 - V for information-causal systems), use measurement procedures developed independently of the present theory: IIT and behavioral paradigms of cognitive ethology for 10.1; the theory of critical phenomena for 10.2; methods of distribution-shift detection and environment monitoring for 10.3. This separation of the measuring instrument and the interpreting theory is a necessary condition for the falsifiability of predictions. If the measurement procedures were formulated within the present theory, any negative result could be explained post hoc through a shift in definitions, and the theory would become unfalsifiable. The fundamental independence of measurement procedures from the theoretical framework preserves its scientific status and simultaneously limits its claims: the theory formulates criteria and predictions, but does not claim to have its own measuring apparatus, and in this sense relies on procedures existing in the literature as its foundation.

The work proposes a program for the empirical verification of the central prediction (transition 3→4) along three parallel lines of different classes of systems - biological observational, artificial experimental, and synthetic biology as an intermediate step. These three lines function as convergent tests of a single theory, with different implementation horizons (biological - current; synthetic biology - 2–3 years at the current pace of development of platforms such as Cortical Labs CL1; artificial systems - 3–7 years if successful with architectural scaling to a Φ -density of the order of 10^3). The full implementation of this program is a separate area of future work requiring the coordination of efforts across several scientific disciplines and laboratories. This work formulates what result should be observed if the theory is valid, and in what terms it should be interpreted; the execution of measurements, the accumulation of data, and their interpretation in light of the full program are the tasks of subsequent research, based on the structural framework of this theory. A distinctive feature of the program is its processual nature: it does not rely on a single decisive experiment, and its result emerges through the convergence or divergence of observational and experimental data across all three lines. This is methodologically weaker than classical Popperian falsification, but it is appropriate to the nature of the subject under study (consciousness and V are processes, not states), and is similar in structure to the methodologies of other disciplines that study processual phenomena (observational astrophysics, evolutionary biology, significant parts of the neurosciences).

Clarification of the methodological status of falsifiability. The process-based form of falsifiability described above - through the convergence of three lines - is a form of testing the statistical predictions of a theory (distribution of parameters, dynamics of approaching thresholds, convergence of observations across classes of systems). In addition to this statistical form, the theory contains a point-specific negation predicate on a single system, elaborated in subsection 10.2.6: the impossibility of a system simultaneously satisfying the self-overlay of a latent invariant and the absence of behavioral signatures of a functioning V. This exclusionary predicate possesses Popperian falsifiability in the strict sense: a single contradictory observation refutes the theory, without the need for a statistical distribution. Thus, the theory has a falsification structure of two

types: a statistical verification program via the convergence of three lines (for predictions about distributions and thresholds) plus a point-specific prohibitive predicate on a separate system (for the structural-behavioral coupling: structural registration of self-overlay $\leftrightarrow$ behavioral signatures of a functioning V). These two types of falsifiability operate in parallel and are not reducible to one another: the statistical type verifies the probabilistic nature of the transition 3 $\rightarrow$ 4 in its historical aspect; the point-specific type verifies the deterministic coupling between structural registration and behavioral signatures. The ontological identity of structure and phenomenon itself (Section 6.5) is an interpretive bridge not resolvable by the logic of a single paper; the point-specific test bears on it only through the behavioral signatures that the identity predicts - the absence of the phenomenon is not directly observed. Both types are empirically realizable upon full deployment of the verification program.

The work does not propose a priori numerical values for the threshold parameters (the Φ -threshold of structural integration of variable V; the threshold deviation of behavior from the zero model of the proto-agent; the threshold level of GWT accessibility; the threshold duration of hysteresis). This is a deliberate methodological choice, not a flaw. Numerical thresholds for the 3 $\rightarrow$ 4 transition cannot be established outside the theory verification program itself - their establishment is one of the program's empirical results. Specifying concrete numbers from third-party calibration datasets, in which these numbers had a different meaning (for example, Φ -values for specific architectures in tasks unrelated to the transition to consciousness), would create the illusion of rigor through arbitrary choice of numbers and would be methodologically worse than an honest acknowledgment of the unknown nature of the thresholds. This work explicitly describes the procedure for the empirical calibration of thresholds in subsection 10.2 as a two-step process: establishing a reference distribution of parameters at the proto-agent level, followed by a search for systems with a statistically significant deviation from this distribution across all four parameters simultaneously. The reproducibility of thresholds across multiple independent systems in different lines of the program is the criterion for their status as empirically established. Until at least part of this program is completed, the specific numerical values of the thresholds remain unknown - and this is the correct state of the theory at its current stage of development, not a weakness. An analogy from other empirical sciences: the threshold values of diagnostic biomarkers in epidemiology are established through systematic research that distinguishes between distributions in patients and healthy individuals; they do not fit into a theory of the disease in advance, and this does not make epidemiology unfalsifiable.

This work does not address the problem of transitions between the rungs of the ladder at the level of a specific historical-biological mechanism. The formal conditions for each transition are presented in Sections 3.2–3.5; macro-models for three of the four transitions are presented in Chapter 4; for the fourth (transition 3 from proto-agency to consciousness), a direct experimental direction is formulated in the form of Prediction 10.2. Two issues remain open. First, the quantitative dynamics of the transition itself: exactly how the probability of the transition evolves as conditions accumulate, what the form of the threshold function is, and whether there are stable intermediate states in which the system lingers for a long time. Second, the specific historical

mechanisms of emergence: exactly how the first autocatalytic closure arose in Earth's prebiotic chemistry, exactly how the self-overlay of a latent invariant of an agentic causal node at stage 4 took shape in the evolution of certain animal lineages, and exactly how a complete bidirectional $M \leftrightarrow Y$ loop of stage 5 became established in *Homo sapiens* (and not in several parallel hominid lineages) did the complete bidirectional $M \leftrightarrow Y$ circuit of stage 5 become established. The indicated directions for exploring these questions - the theory of self-organized criticality, the chaos boundary in neural networks, and the evolutionary dynamics of the accumulation of genetic and cultural complexity - constitute substantive research programs that cannot be exhausted within the scope of a single theoretical work.

The work proposes an ontological framework for phenomenal experience (Section 6) and an operational mechanism for the emergence of phenomenal content through the presence of inherited gradients in V (Sections 3.4 and 3.7.5), but does not resolve the question of the specific neurobiological correlates of individual qualia. I have proposed a structural picture: a phenomenal quality is a positional characteristic of the self-model's structure, and specific phenomenal states (hunger, pain, anxiety, fear) are structurally necessary modes of the presence of inherited gradients through V . What remains outside the scope of this work is a detailed neurobiological analysis of exactly which neural circuits and dynamics implement the presence of each of the inherited gradients through V in the human brain; how individual differences in these circuits give rise to individual differences in phenomenal content; how phenomenal content correlates across different species with different neurobiological organization. These questions require detailed experimental work in neuroscience and comparative psychology, and they are open for the next stage of research, using the structural framework of this work as a guiding framework.

The cosmological continuation of the pattern (Section 7) has been moved to a separate work, planned by the author as an independent project: the formal conditions for a possible continuation of the pattern beyond stage 5, its mechanism, and its reconciliation with discussions of cyclic cosmological models are not developed in the present theory and do not belong to its testable core. The openness of the ladder at the top, however, is not the same question as the cosmological continuation: it has its own criterion (Section 3.7.3) - the ladder continues only where a pole of a new type is named at a physically real scale of localization - and therefore the closure of the sequence at the achieved stage is a substantive position, not a gap.

This work does not fully explore the connection between emergent time and the dynamics of consciousness. Section 5 identifies this connection and, through V , provides its first operational content (V calibrates the system's temporality, defines the structure of its past and future), but a detailed elaboration would require a separate study. In particular, the question of how exactly conscious self-modeling participates in the generation of the time in which this modeling takes place remains unresolved - this is a question about the self-reflexive structure of emergent time, and I do not claim to resolve it here.

The work identifies the moral and civilizational implications of the proposed theory, but does not develop them into an ethical theory. This is a conscious self-limitation: this work is a theoretical article on the physics and philosophy of consciousness, not an ethical work. The elaboration of ethical implications requires a different methodology and different expertise.

The limitations listed above are not a critique of my own work, but an honest indication of its place within a broader field of research. Each of these limitations points to a direction for further work, and I hope that some of these directions will be explored either by myself or by colleagues who find the proposed framework productive.

13. Conclusion

I began this work not as a philosopher of mind, but as a physicist studying the cosmos. The contradictions in cosmology that had accumulated over many years - the nature of dark energy, the information paradox of black holes, the fine-tuning of constants, the anthropic bias - gradually led me to suspect that the boundary between physics and the phenomenology of consciousness was drawn in the wrong place. These suspicions coalesced into an attempt to unite fields that had developed independently: non-equilibrium thermodynamics, the theory of complex systems, active inference, biosemiotics, teledynamics, the theory of integrated information. The connection took shape not as a simple synthesis of independent fields, but as the discovery of a structurally self-similar pattern through which these fields turn out to be projections of a single process onto different scales. The «gradient $\rightarrow$ retention $\rightarrow$ pole $\rightarrow$ transition» structure operates identically on scales ranging from the formation of a dissipative structure to a conscious system; the six-step ladder, detailed in Section 3, is the basic pattern of the theory through which all the processes it encompasses are described. This structure is presented not as a narrative picture, but as a formal construct: a six-step ladder with operational definitions for each step, minimal conditions for each transition, macro-models at three levels (conceptual, physical, computational), the explicitly formulated probabilistic nature of the algorithm, and the internal logic summarized in subsection 3.7 - the accumulation of gradients and retention mechanisms, the expansion of the scale of localization, the three phases of activity formation (temporal $\rightarrow$ directed $V \rightarrow$ distributed), and the presence of inherited gradients in V as a structural mechanism for the emergence of phenomenal content. In this structure, consciousness turns out to be neither a mystery nor a gift, but a natural consequence of physical processes in open systems with an energy gradient - natural in the precise sense given in Section 3.6: the transition to each subsequent stage is possible when formal conditions are met and occurs with decreasing probability, forming a funnel, at the narrowest end of which we find ourselves.

In this picture, humanity is neither a coincidence nor a design. It is a natural deviation. The cutting edge of complexity, moving forward as long as the gradient persists. Not the goal of the Universe, but its edge - between what has already been realized and what is still only possible. And - on the speculative horizon outlined in Section 7 - possibly not the final point: the same structural pattern in principle admits continuation beyond the ladder considered here, but this is a separate hypothesis, not developed in the present work.

I may be wrong in many respects. Some of the statements in this work are well-founded consequences of established principles; others are hypotheses that may be refuted by future observations and experiments. I have tried to indicate the difference wherever possible. I have also tried to indicate what is missing from this work: a unified mathematical formalism for a structurally self-similar pattern across all scales of the ladder, specific neurobiological correlates of individual qualia beyond the designated structural framework, a formal elaboration of the cosmological continuation of the pattern (Section 7), and a comprehensive ethical theory. These gaps are not shortcomings of the work, but its honest boundaries.

I am an observer within the observation. I may be mistaken in form, but this does not negate the direction of my gaze. The anthropic principle, the measurement problem in quantum mechanics, and the theory of consciousness are three projections of the same ontological structure: observation is a positional characteristic of a system within a larger whole. This structure, in turn, is a special case of a more general structurally self-similar pattern, unfolded throughout the work as a whole: on different scales, the same logic manifests in specific forms, and the observer within the observation is its realization on the scale of the conscious system describing the world of which it is a part. I am inside the very system I am describing. My tools are products of the very processes I am analyzing. This does not make my work false. It makes it work from within. There is no way out, and its impossibility is not a limitation that can be overcome in principle, but a structural property of the world in which we find ourselves.

In this endless process, matter becomes increasingly complex until it begins to perceive itself. At that moment, we appear and are given the opportunity - brief, localized, and limited by our position - to see the process from the inside. This opportunity is all we have, and at the same time, it is all we need.

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